

Quantitative-Analytic Reports on Chess
Berliner's games,
5th World Ch. ICCF 1965-1968

Generated by Stockfish 17.1

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Chapter 1

Abramson Moisey vs Berliner Hans J (USA)

1.1 Game Information

Event	Ch World (corr.)
Site	corr.
Date	1965.??.??
Round	5
White	Abramson Moisey
Black	Berliner Hans J (USA)
Result	0-1
Opening	A41 – Unknown

1.2 Player Statistics

1.2.1 Abramson Moisey (White)

- Total moves: 36
- Accuracy: 85.8%
- Average centipawn loss: 30.7
- Best/Excellent moves: 6 / 7
- Good moves: 11
- Inaccuracies: 6
- Mistakes: 5
- Blunders: 1

1.2.2 Berliner Hans J (USA) (Black)

- Total moves: 36
- Accuracy: 93.6%

- Average centipawn loss: 13.3
- Best/Excellent moves: 15 / 7
- Good moves: 10
- Inaccuracies: 2
- Mistakes: 2
- Blunders: 0

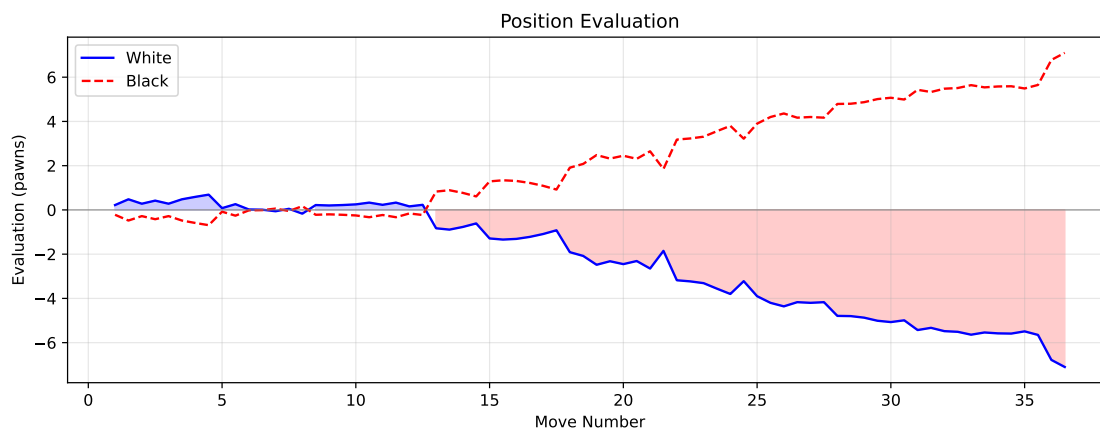
1.3 Positional Analysis

This section shows how key positional factors evolved throughout the game.

1.3.1 Position Evaluation Over Time

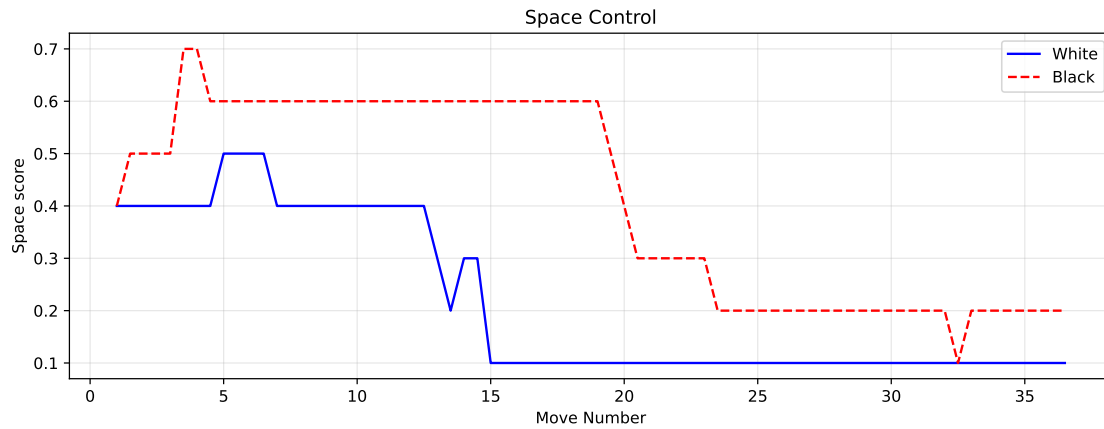
Game Character Classification. Evaluation range: minimum $m_1 = -7.10$, maximum $m_2 = +0.69$, spread $d = m_2 - m_1 = 7.79$.

This game is classified as **chaotic** (normal). A chaotic game with moderate swings.



1.3.2 Space Control

- White average: 0.22 (range: 0.10 to 0.50)
- Black average: 0.41 (range: 0.10 to 0.70)



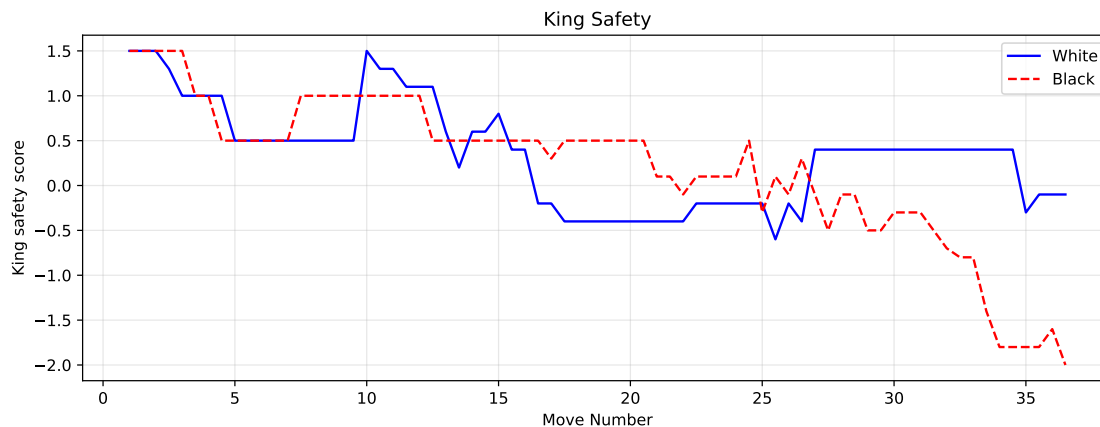
1.3.3 Mobility (Piece Activity)

- White average: 1.93 (range: 0.90 to 2.65)
- Black average: 1.84 (range: 1.00 to 2.60)



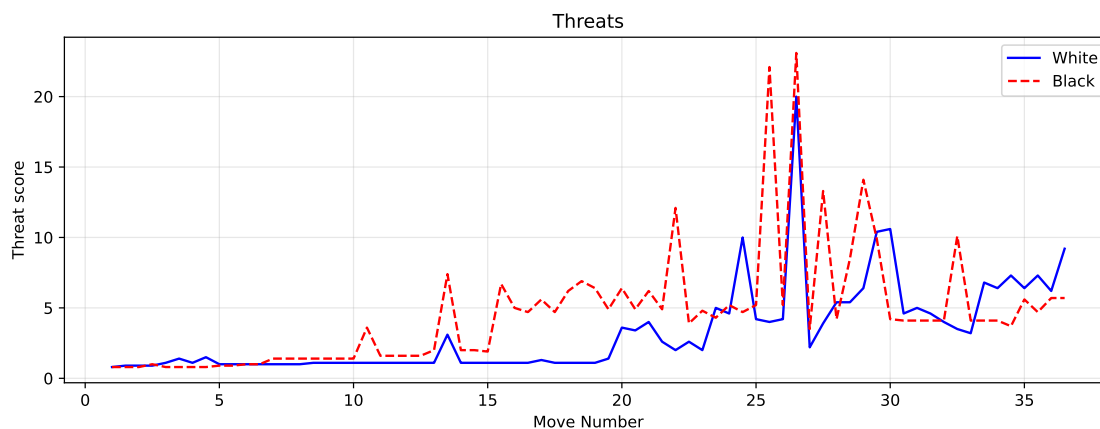
1.3.4 King Safety

- White average: 0.34
- Black average: 0.20



1.3.5 Threats

- White average: 3.26 (range: 0.80 to 20.00)
- Black average: 4.60 (range: 0.80 to 23.10)
- Average advantage: -1.35 (Black maintained more threats)



1.4 Annotated Game

1. Nf3 g6 2. c4 Bg7 3. d4 d6 4. Nc3 e5 5. e3? *White loses 63cp. Consider instead: dxe5, e4, g3.*

5...Nd7 6. a3?! Ngf6 7. b4 O-O 8. Bb2 c6?! 9. Be2 Re8 10. O-O e4 11. Nd2 Nf8 12. Re1 h5 13. f4? *White loses 100cp. Consider instead: a4, d5, b5.*

13...exf3! 14. Bxf3! N8h7?! 15. e4? *White loses 70cp. Consider instead: d5, Qb3.*

15...Ng4! 16. Nf1 Qh4 17. Qd2! Ng5 18. Re2? *White loses 96cp. Consider instead: Bxg4.*

18...Be6! 19. Qd3?! c5 20. bxc5 dxc5 21. Nd5 cxd4? *Black loses 73cp. Consider instead: b5.*

22. Bxg4?? *White loses 136cp. Consider instead: Bxd4.*

22...Qxg4! 23. Rae1 b5! 24. c5 Bxd5? *Black loses 69cp. Consider instead: Rac8.*

25. exd5?! Rxe2! 26. Qxe2 Qxe2 27. Rxe2 d3! 28. Rd2?! Ne4 29. Bxg7 Nxd2! 30. Bd4! Ne4 31. c6?! Kf8! 32. Be5 f6 33. Bf4?! Ke7 34. Ne3! d2! 35. Kf1 Nc3 36. d6+?
White loses 117cp. Consider instead: Bc7, Nd1, Kf2.

36...Ke6!

0-1

1.5 Critical Positions

This section highlights critical moments where the evaluation shifted substantially—either due to errors or missed opportunities. Each diagram shows the position *after* the move was played.

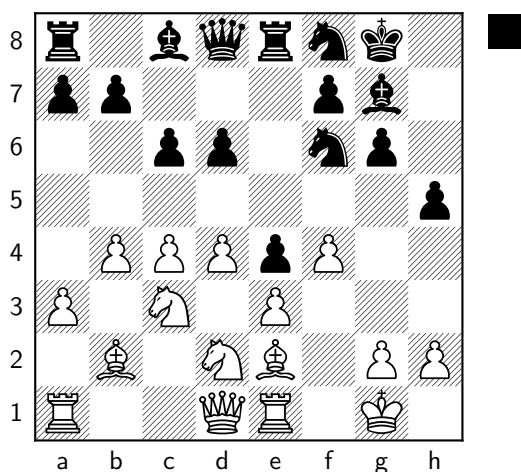
- **Instead of [move]:** The move(s) that should have been played instead.
- **Best continuation:** The optimal sequence of moves from the diagrammed position.

Positions marked with ★ represent the biggest evaluation swings in the game. The positional metrics table summarizes the spatial control, piece activity, king safety, and tactical threats for each side.

Position 1: After 13. f4

Critical moment

Evaluation: -0.83



Metric	White	Black
Space	0.30	0.60
Mobility	2.15	2.25
King Safety	0.60	0.50
Threats	1.10	2.00

Instead of f4: a4, d5, b5

Best continuation: exf3 Bxf3 Be6

Metric	White	Black
Space	0.10	0.30
Mobility	2.45	2.10
King Safety	-0.40	-0.10
Threats	2.00	12.10

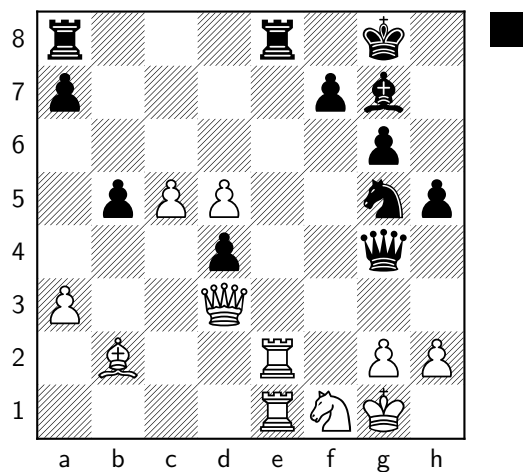
Instead of Bxg4: Bxd4

Best continuation: Qxg4 Ree1 b5

Position 4: After 25. exd5 ★

Biggest evaluation swing(s) (68cp)

Evaluation: -3.90



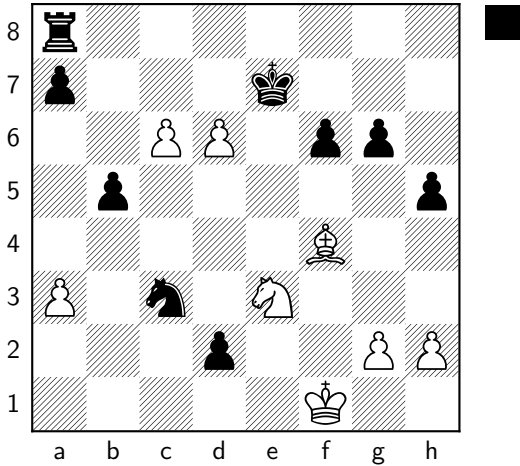
Metric	White	Black
Space	0.10	0.20
Mobility	2.10	2.20
King Safety	-0.20	-0.30
Threats	4.20	5.20

Best continuation: Rxe2 Qxe2 Qxe2

Position 5: After 36. d6+

Critical moment

Evaluation: -6.78



Metric	White	Black
Space	0.10	0.20
Mobility	0.90	1.30
King Safety	-0.10	-1.60
Threats	6.20	5.70

Instead of d6+: Bc7, Nd1, Kf2

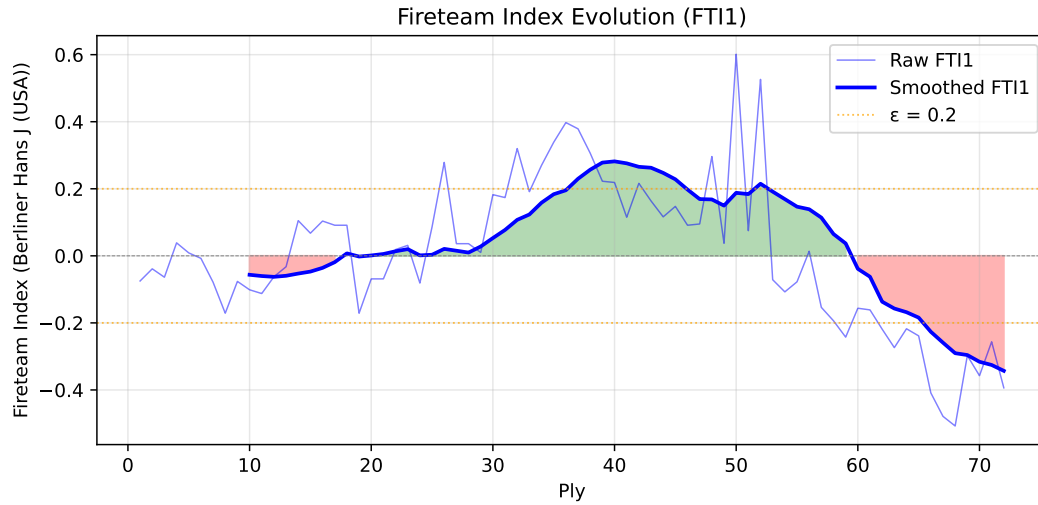
Best continuation: Ke6 Bg3 h4

1.6 Fireteam Index Prediction

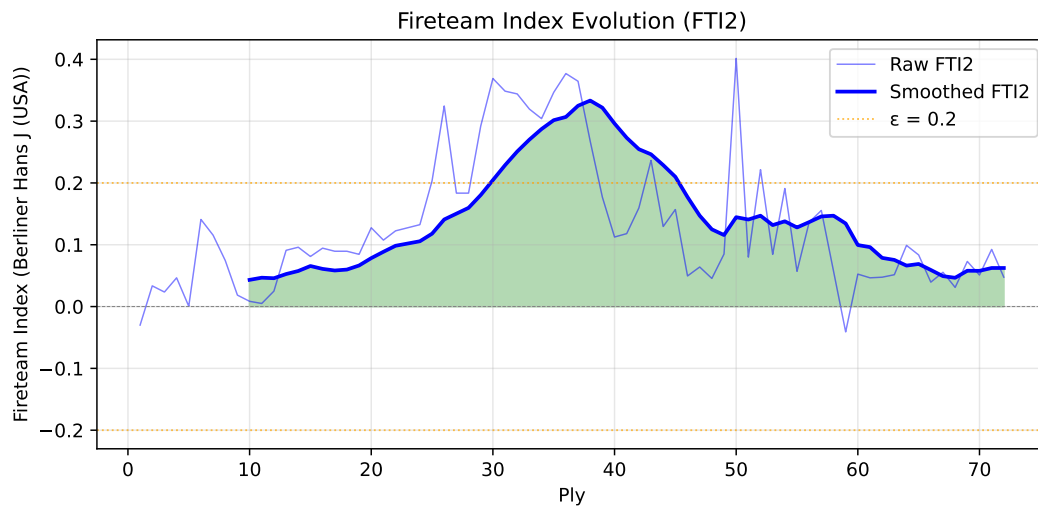
Tracking: **Berliner Hans J (USA)** (Black)

FTI1: Harmonious ($\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}$)

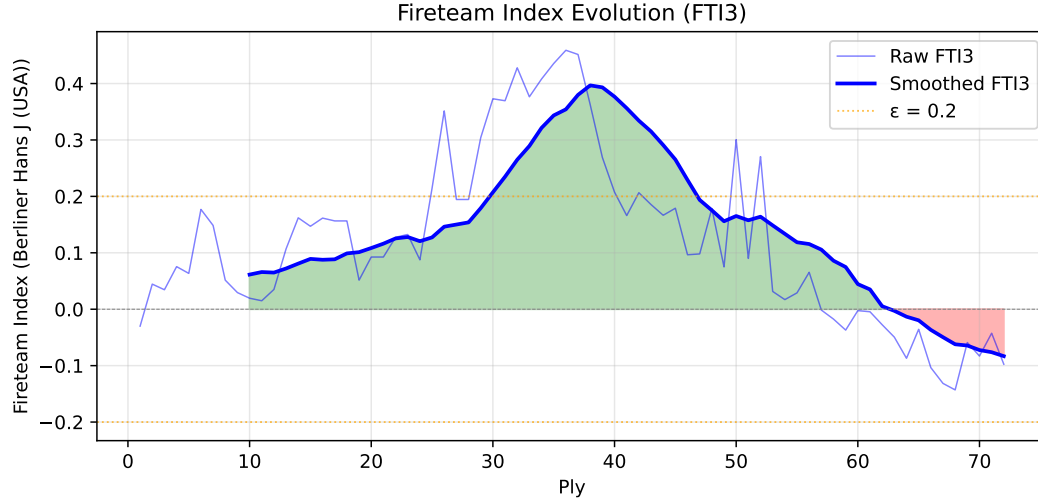
Algorithm	Prediction	Max Streak / Confidence
Per-ply	DRAW	5 plies
Windowed	DRAW	6 plies
Balance	DRAW	$B_n=+0.02$, $C=0.02$

**FTI2: Tactical** (0.6, 0.1, 0, 0.3)

Algorithm	Prediction	Max Streak / Confidence
Per-ply	WIN	10 plies
Windowed	WIN	11 plies
Balance	WIN	$B_n=+0.27$, $C=0.27$

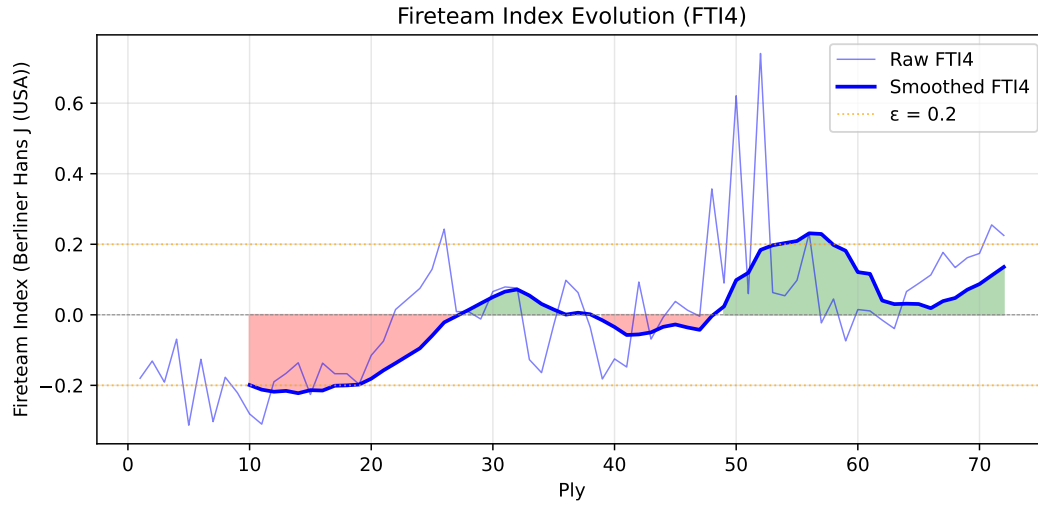
**FTI3: Strategic** (0.7, 0.1, 0.1, 0.1)

Algorithm	Prediction	Max Streak / Confidence
Per-ply	WIN	11 plies
Windowed	WIN	14 plies
Balance	WIN	$B_n=+0.30$, $C=0.30$



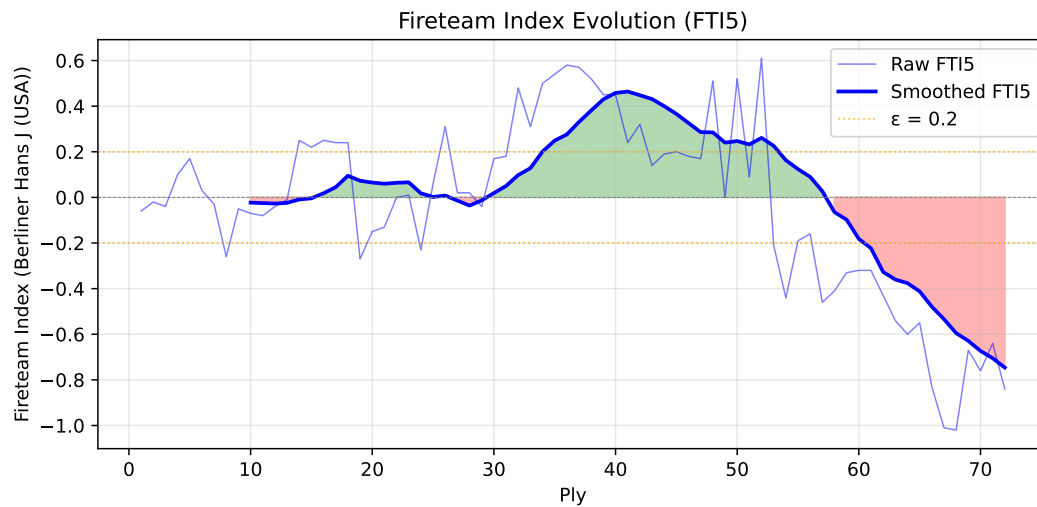
FTI4: Dynamic (0.2, 0.6, 0, 0.2)

Algorithm	Prediction	Max Streak / Confidence
Per-ply	DRAW	1 plies
Windowed	DRAW	0 plies
Balance	DRAW	$B_n = +0.12$, $C = 0.12$



FTI5: Solid (0.3, 0.2, 0.5, 0)

Algorithm	Prediction	Max Streak / Confidence
Per-ply	DRAW	9 plies
Windowed	WIN	13 plies
Balance	DRAW	$B_n = -0.05$, $C = 0.04$



Actual result: 0-1 (Berliner Hans J (USA)'s outcome: WIN)

FTI1: Per-ply	×	Windowed	×	Balance	×
FTI2: Per-ply	✓	Windowed	✓	Balance	✓
FTI3: Per-ply	✓	Windowed	✓	Balance	✓
FTI4: Per-ply	×	Windowed	×	Balance	×
FTI5: Per-ply	×	Windowed	✓	Balance	×

1.7 Analysis Information

- Engine: Stockfish 17.1
- Depth: 22
- Analysis time: 500.4 seconds

Chapter 2

Roman Zinovievich Altshuler vs Hans Berliner

2.1 Game Information

Event	5th Correspondence World Championship
Site	ICCF
Date	1965.07.01
Round	10
White	Roman Zinovievich Altshuler
Black	Hans Berliner
Result	0-1
Opening	B04 – Unknown

2.2 Player Statistics

2.2.1 Roman Zinovievich Altshuler (White)

- Total moves: 25
- Accuracy: 87.9%
- Average centipawn loss: 25.8
- Best/Excellent moves: 9 / 2
- Good moves: 8
- Inaccuracies: 2
- Mistakes: 3
- Blunders: 1

2.2.2 Hans Berliner (Black)

- Total moves: 25
- Accuracy: 92.7%

- Average centipawn loss: 15.2
- Best/Excellent moves: 10 / 4
- Good moves: 6
- Inaccuracies: 3
- Mistakes: 2
- Blunders: 0

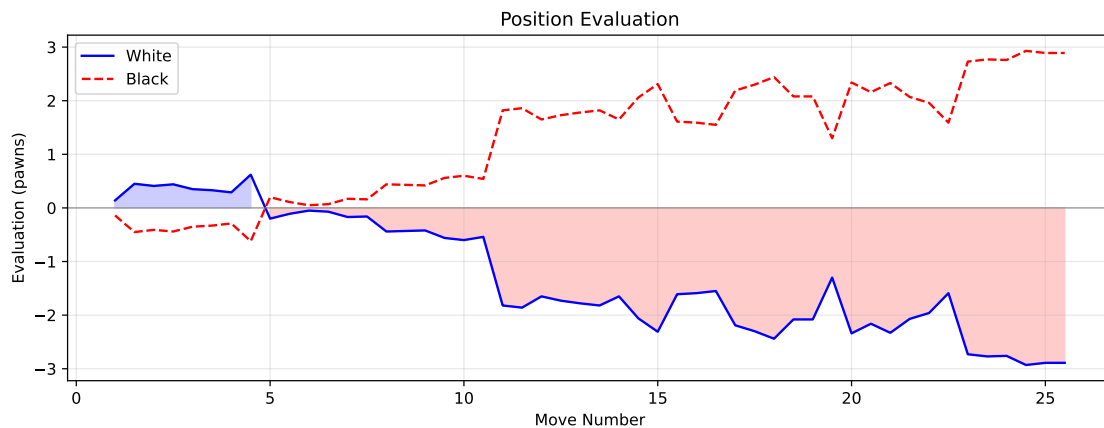
2.3 Positional Analysis

This section shows how key positional factors evolved throughout the game.

2.3.1 Position Evaluation Over Time

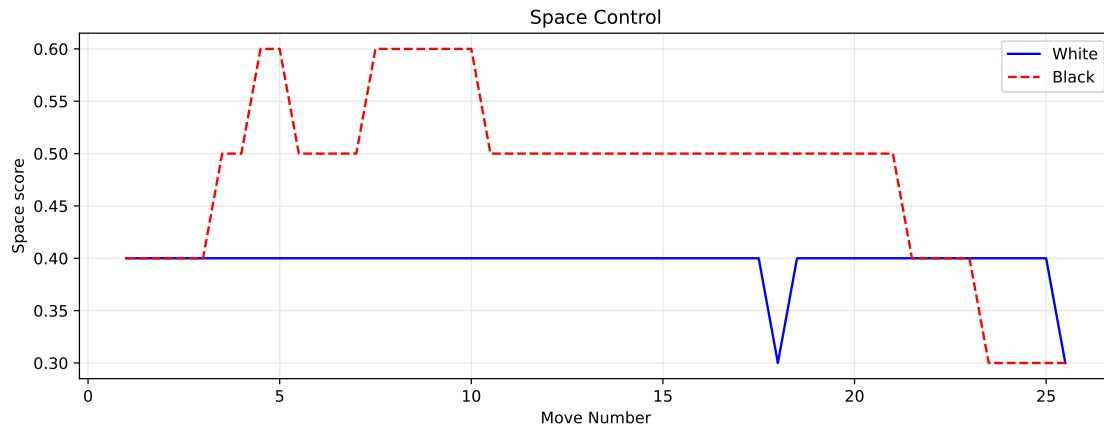
Game Character Classification. Evaluation range: minimum $m_1 = -2.93$, maximum $m_2 = +0.62$, spread $d = m_2 - m_1 = 3.55$.

This game is classified as **tactical** (normal). A tactical game with moderate swings.



2.3.2 Space Control

- White average: 0.40 (range: 0.30 to 0.40)
- Black average: 0.48 (range: 0.30 to 0.60)



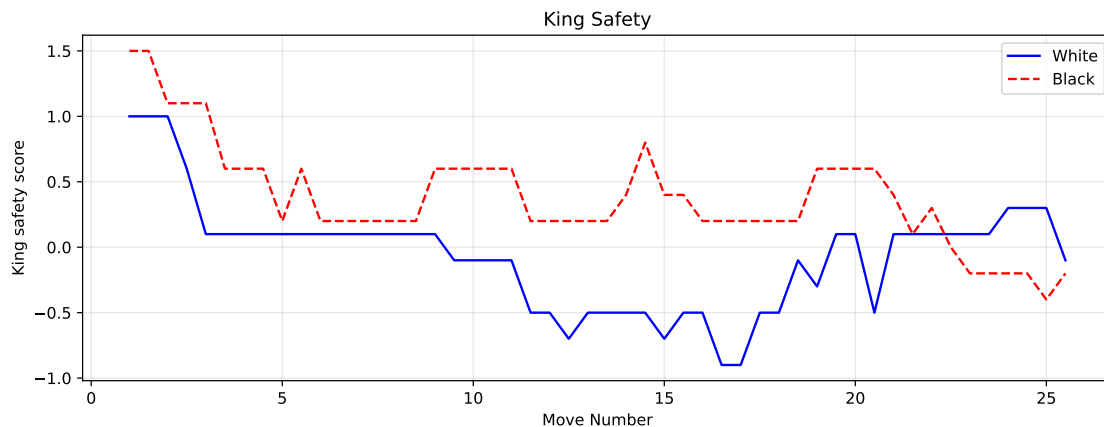
2.3.3 Mobility (Piece Activity)

- White average: 2.13 (range: 1.80 to 2.50)
- Black average: 2.33 (range: 1.60 to 3.00)



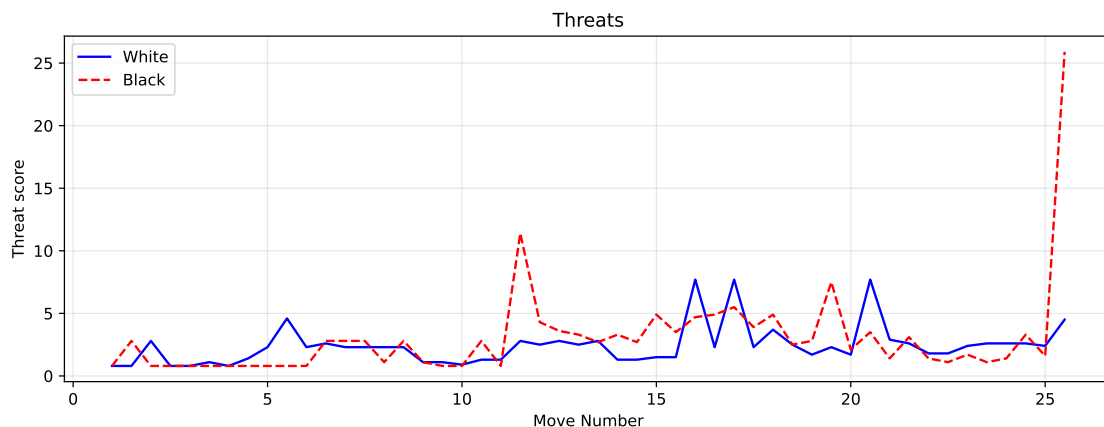
2.3.4 King Safety

- White average: -0.06
- Black average: 0.38



2.3.5 Threats

- White average: 2.42 (range: 0.80 to 7.70)
- Black average: 3.07 (range: 0.80 to 25.90)
- Average advantage: -0.65 (Black maintained more threats)



2.4 Annotated Game

1. e4 Nf6?! 2. e5 Nd5 3. d4 d6 4. Nf3 g6 5. Ng5? *White loses 77cp. Consider instead: Bc4, c4, h3.*

5...dxe5 6. dxe5 Bg7 7. Bc4 c6! 8. Qe2?! h6 9. Nf3! Bg4! 10. Nbd2 b5! 11. Bb3?? *White loses 133cp. Consider instead: Bxd5.*

11...Nf4! 12. Qf1! Qa5! 13. c3! Nd7! 14. Bc2 Nxe5! 15. Nd4 Qc7? *Black loses 68cp. Consider instead: Qb6.*

16. Ne4! Nc4 17. h3?! Bd7! 18. g3 Ne6?! 19. Nf3! Ne5? *Black loses 89cp. Consider instead: Nd6, c5, O-O.*

20. Qe2? *White loses 119cp. Consider instead: Nxe5.*

20...Nxf3+ 21. Qxf3 f5 22. Nd2! O-O?! 23. Nb3? *White loses 115cp. Consider instead: Qe2.*

23...c5 24. O-O! c4! 25. Nd4! Nxd4!
0-1

2.5 Critical Positions

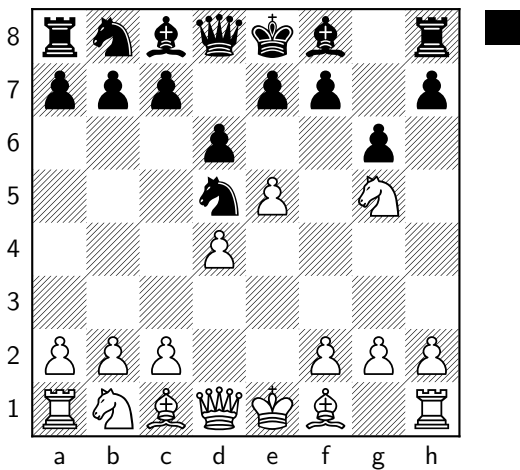
This section highlights critical moments where the evaluation shifted substantially—either due to errors or missed opportunities. Each diagram shows the position *after* the move was played.

- **Instead of [move]:** The move(s) that should have been played instead.
- **Best continuation:** The optimal sequence of moves from the diagrammed position.

Positions marked with ★ represent the biggest evaluation swings in the game. The positional metrics table summarizes the spatial control, piece activity, king safety, and tactical threats for each side.

Position 1: After 5. Ng5 ★

Biggest evaluation swing(s) (82cp)
Evaluation: -0.20



Metric	White	Black
Space	0.40	0.60
Mobility	2.10	1.75
King Safety	0.10	0.20
Threats	2.30	0.80

Instead of Ng5:

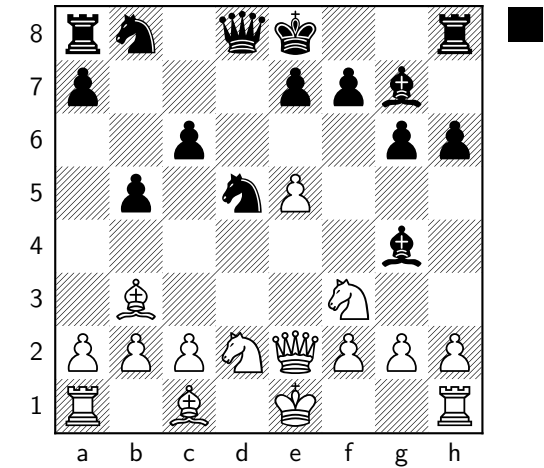
Bc4, c4, h3

Best continuation:

dxe5 dxe5 Bg7

Position 2: After 11. Bb3

White blunders (-128cp)
Evaluation: -1.82



Metric	White	Black
Space	0.40	0.50
Mobility	2.15	2.15
King Safety	-0.10	0.60
Threats	1.30	0.80

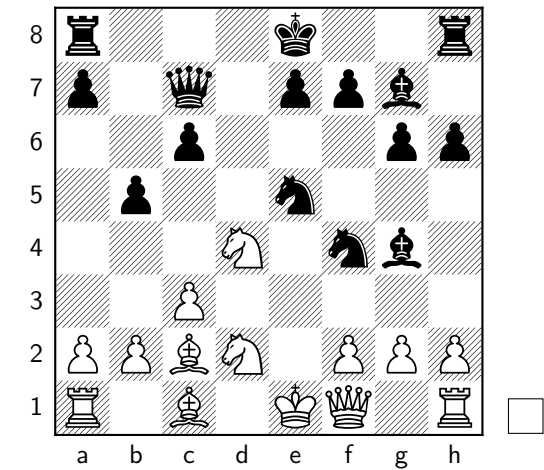
Instead of Bb3: Bxd5

Best continuation: Nf4 Qf1 Qa5

Position 3: After 15...Qc7 ★

Biggest evaluation swing(s) (70cp)

Evaluation: -1.61



Metric	White	Black
Space	0.40	0.50
Mobility	2.00	3.00
King Safety	-0.50	0.40
Threats	1.50	3.50

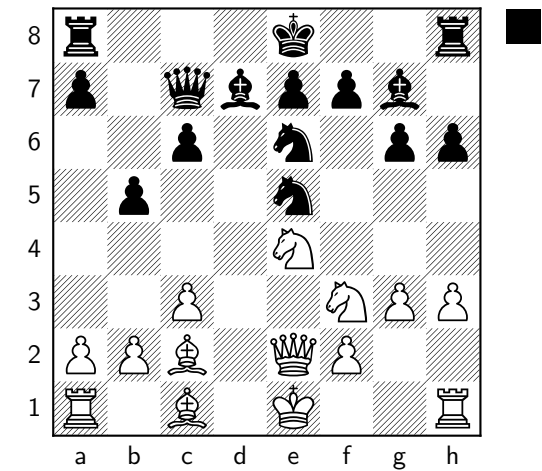
Instead of Qc7: Qb6

Best continuation: Ne4 Nc4 f3

Position 4: After 20. Qe2

Critical moment

Evaluation: -2.34



Metric	White	Black
Space	0.40	0.50
Mobility	2.50	2.75
King Safety	0.10	0.60
Threats	1.70	2.10

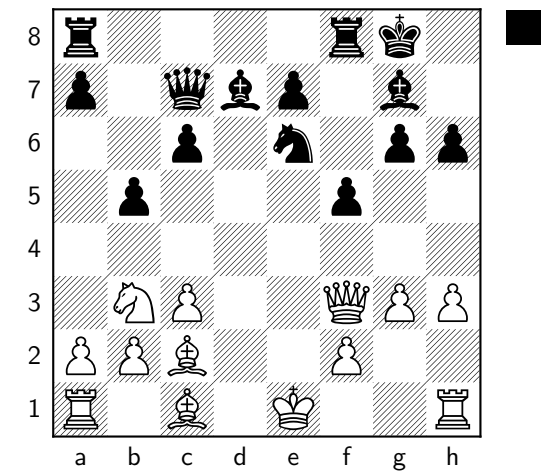
Instead of Qe2: Nxe5

Best continuation: Nxf3+ Qxf3 O-O

Position 5: After 23. Nb3

Critical moment

Evaluation: -2.73



Metric	White	Black
Space	0.40	0.40
Mobility	2.15	2.60
King Safety	0.10	-0.20
Threats	2.40	1.70

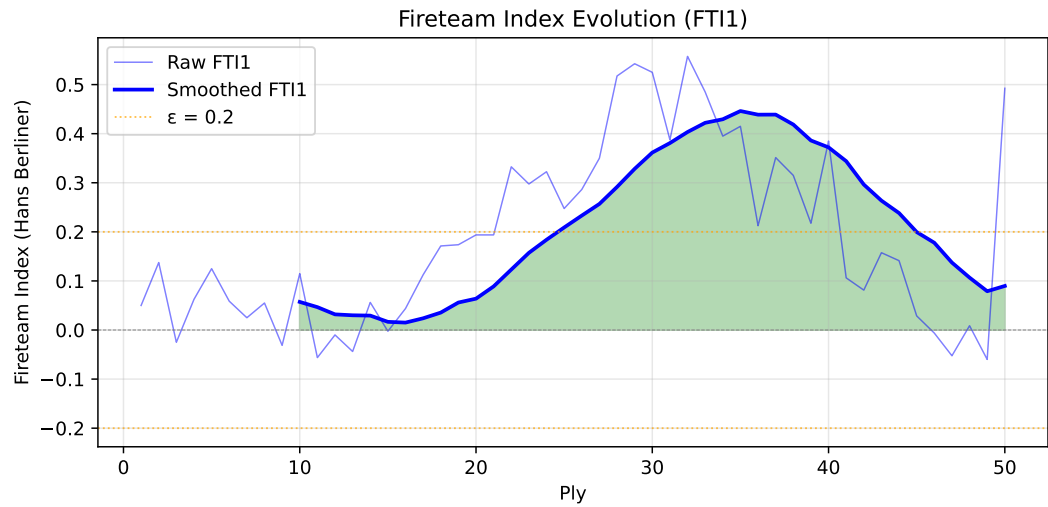
Instead of Nb3: Qe2 Best continuation: c5 O-O c4

2.6 Fireteam Index Prediction

Tracking: **Hans Berliner** (Black)

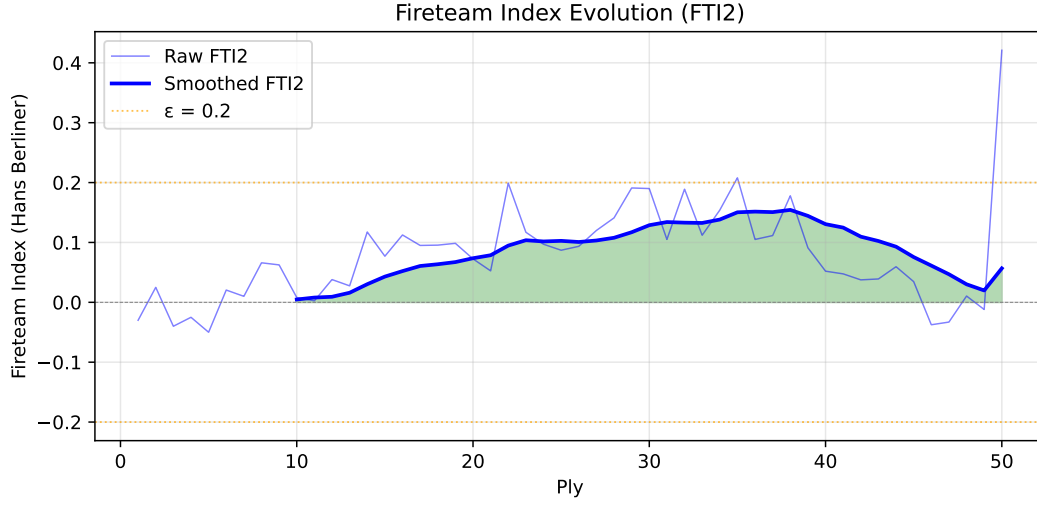
FTI1: Harmonious ($\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}$)

Algorithm	Prediction	Max Streak / Confidence
Per-ply	WIN	10 plies
Windowed	WIN	17 plies
Balance	WIN	$B_n=+0.59$, $C=0.59$

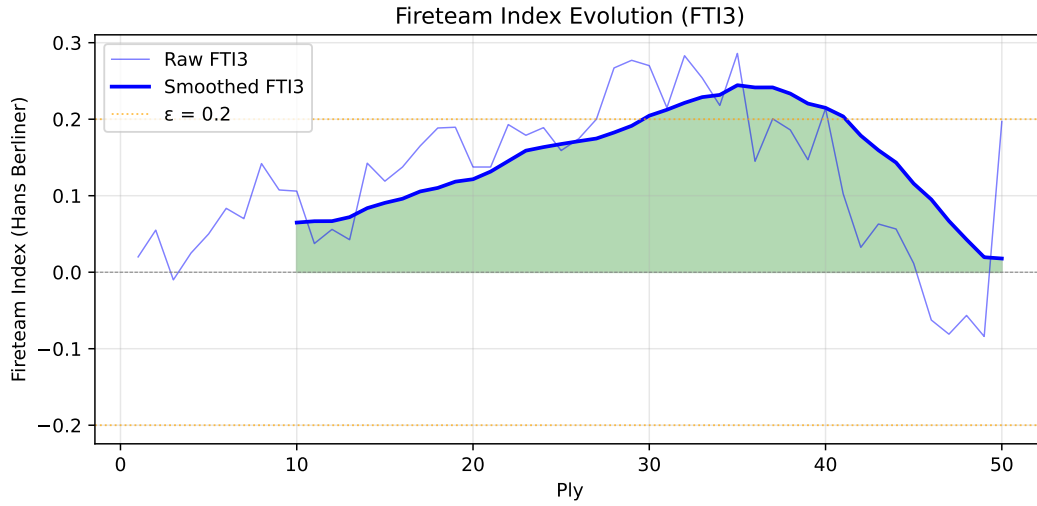


FTI2: Tactical (0.6, 0.1, 0, 0.3)

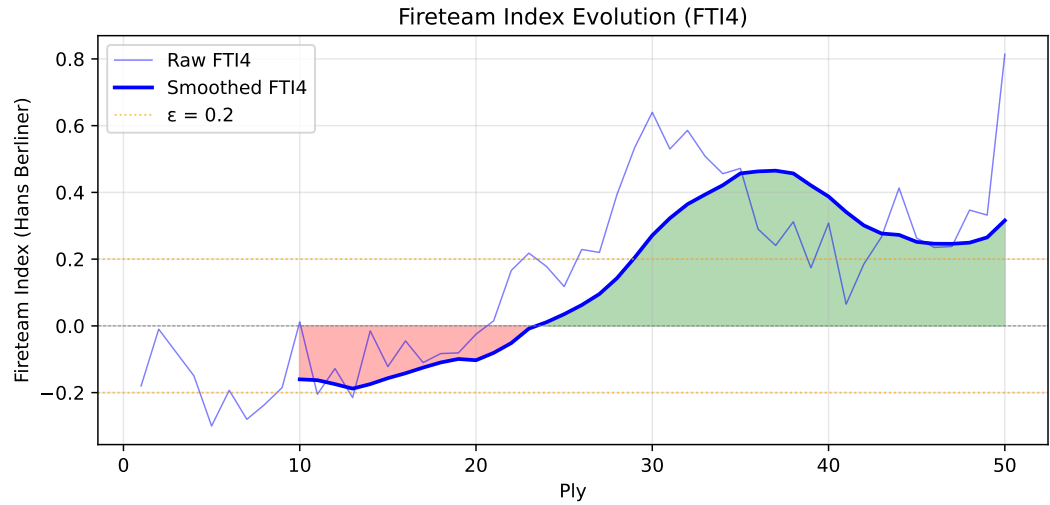
Algorithm	Prediction	Max Streak / Confidence
Per-ply	DRAW	1 plies
Windowed	DRAW	0 plies
Balance	DRAW	$B_n=+0.06$, $C=0.06$

**FTI3: Strategic (0.7, 0.1, 0.1, 0.1)**

Algorithm	Prediction	Max Streak / Confidence
Per-ply	DRAW	3 plies
Windowed	DRAW	0 plies
Balance	WIN	$B_n=+0.29$, $C=0.29$

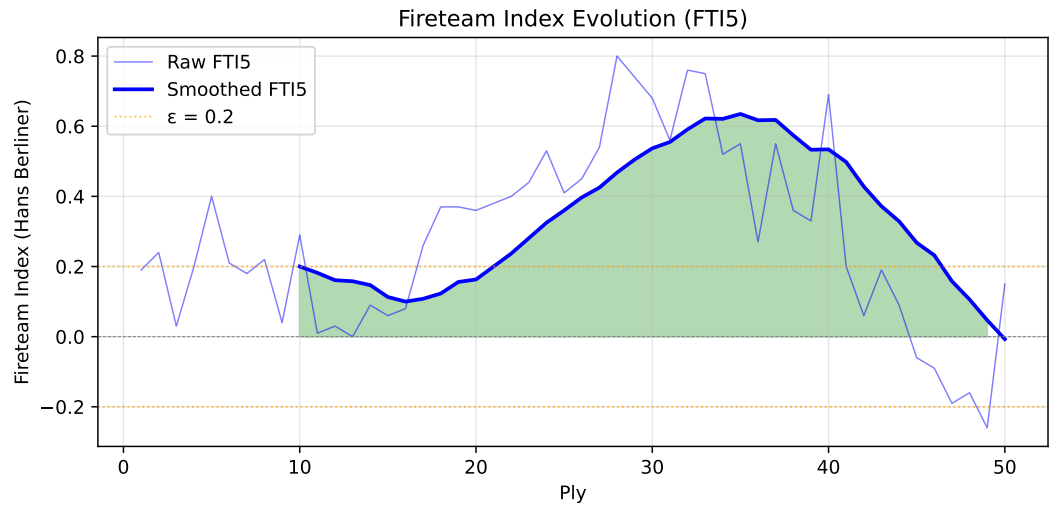
**FTI4: Dynamic (0.2, 0.6, 0, 0.2)**

Algorithm	Prediction	Max Streak / Confidence
Per-ply	DRAW	9 plies
Windowed	WIN	16 plies
Balance	WIN	$B_n=+0.68$, $C=0.68$



FTI5: Solid (0.3,0.2,0.5,0)

Algorithm	Prediction	Max Streak / Confidence
Per-ply	WIN	24 plies
Windowed	WIN	23 plies
Balance	WIN	$B_n=+0.68$, $C=0.58$



Actual result: 0-1 (Hans Berliner's outcome: WIN)

FTI1:	Per-ply ✓	Windowed ✓	Balance ✓
FTI2:	Per-ply ✗	Windowed ✗	Balance ✗
FTI3:	Per-ply ✗	Windowed ✗	Balance ✓
FTI4:	Per-ply ✗	Windowed ✓	Balance ✓
FTI5:	Per-ply ✓	Windowed ✓	Balance ✓

2.7 Analysis Information

- Engine: Stockfish 17.1
- Depth: 22
- Analysis time: 319.9 seconds

Chapter 3

Berliner Hans J (USA) vs Borisenko Georgy K (UZB)

3.1 Game Information

Event	Ch World (corr.)
Site	corr.
Date	1965.??.??
Round	16
White	Berliner Hans J (USA)
Black	Borisenko Georgy K (UZB) (2354)
Result	1-0
Opening	D86 – Unknown

3.2 Player Statistics

3.2.1 Berliner Hans J (USA) (White)

- Total moves: 17
- Accuracy: 96.4%
- Average centipawn loss: 7.4
- Best/Excellent moves: 7 / 0
- Good moves: 8
- Inaccuracies: 2
- Mistakes: 0
- Blunders: 0

3.2.2 Borisenko Georgy K (UZB) (Black)

- Total moves: 16
- Accuracy: 90.3%

- Average centipawn loss: 20.5
- Best/Excellent moves: 4 / 1
- Good moves: 6
- Inaccuracies: 3
- Mistakes: 2
- Blunders: 0

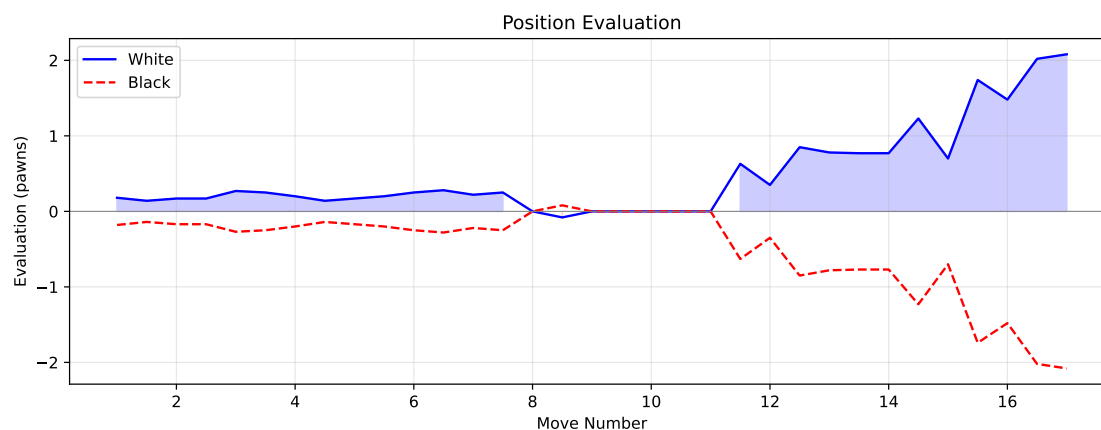
3.3 Positional Analysis

This section shows how key positional factors evolved throughout the game.

3.3.1 Position Evaluation Over Time

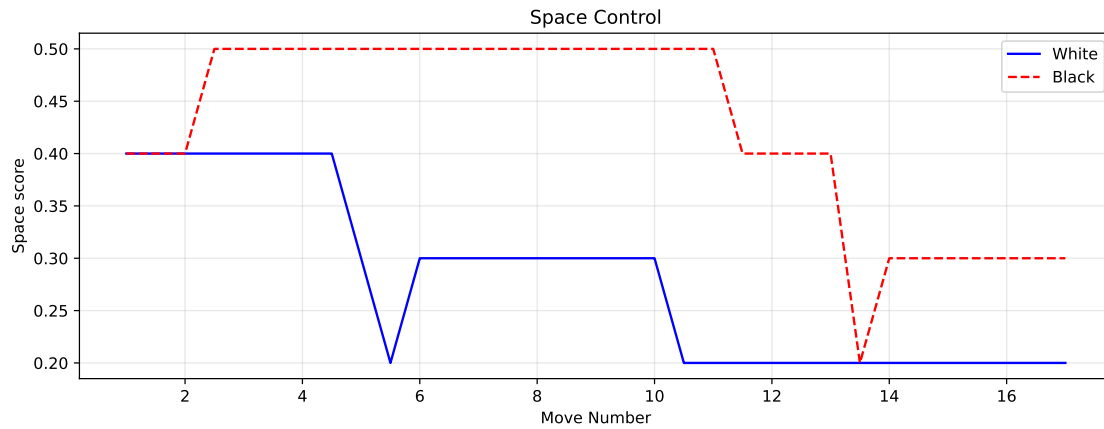
Game Character Classification. Evaluation range: minimum $m_1 = -0.08$, maximum $m_2 = +2.08$, spread $d = m_2 - m_1 = 2.16$.

This game is classified as **tense** (one-sided). A tense, one-sided game. White maintained the advantage throughout.



3.3.2 Space Control

- White average: 0.28 (range: 0.20 to 0.40)
- Black average: 0.43 (range: 0.20 to 0.50)



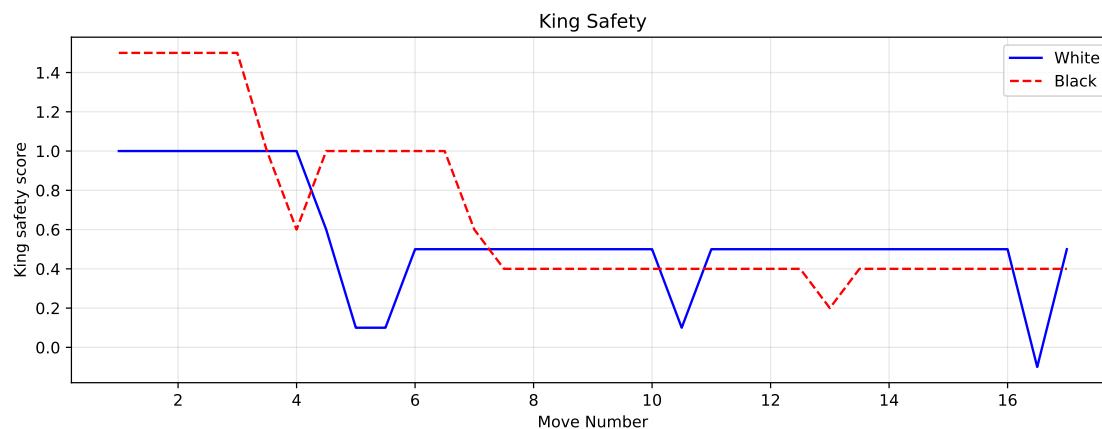
3.3.3 Mobility (Piece Activity)

- White average: 2.26 (range: 1.90 to 2.55)
- Black average: 1.93 (range: 1.60 to 2.20)



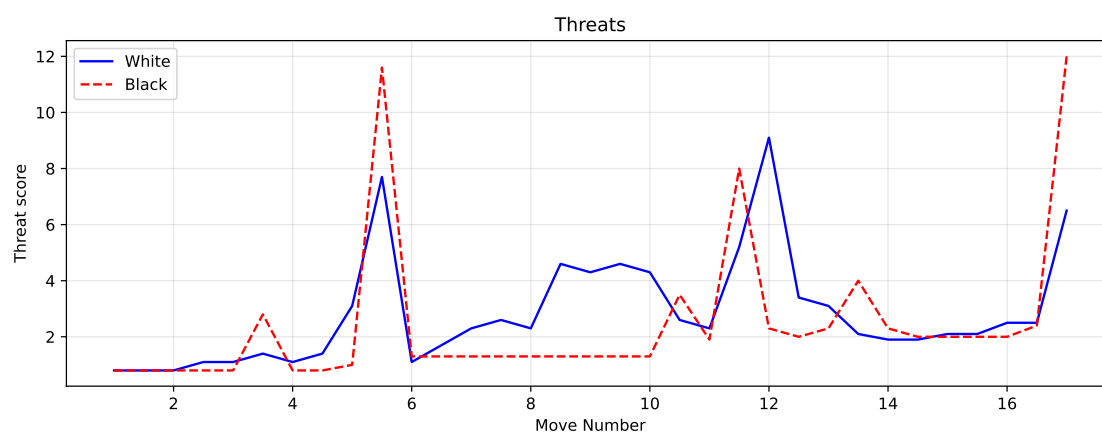
3.3.4 King Safety

- White average: 0.55
- Black average: 0.68



3.3.5 Threats

- White average: 2.86 (range: 0.80 to 9.10)
- Black average: 2.47 (range: 0.80 to 12.00)
- Average advantage: 0.39 (White maintained more threats)



3.4 Annotated Game

1. d4 Nf6 2. c4 g6 3. Nc3 d5 4. cxd5 Nxd5 5. e4 Nxc3 6. bxc3 Bg7 7. Bc4 O-O 8. h4 c5! 9. Ne2! Nc6! 10. Be3! cxd4! 11. cxd4! b5? *Black loses 63cp. Consider instead: Na5, Qa5+, h5.*

12. Bd5 Bd7?! 13. h5! e6! 14. Bb3! Na5?! 15. Qd2?! Nb7? *Black loses 108cp. Consider instead: Nxb3, Bc6, Rc8.*

16. Rc1?! Rc8?! 17. Rxc8!

1-0

3.5 Critical Positions

This section highlights critical moments where the evaluation shifted substantially—either due to errors or missed opportunities. Each diagram shows the position *after* the move was played.

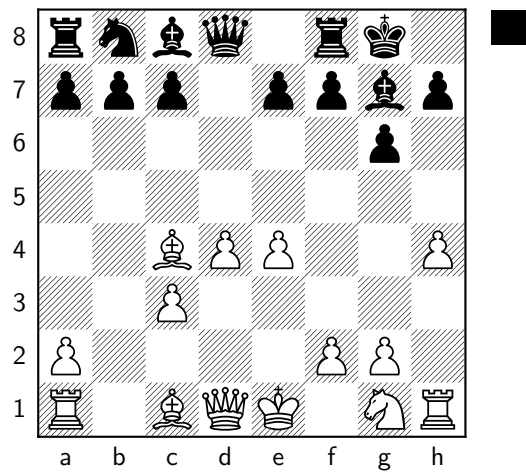
- **Instead of [move]:** The move(s) that should have been played instead.
- **Best continuation:** The optimal sequence of moves from the diagrammed position.

Positions marked with ★ represent the biggest evaluation swings in the game. The positional metrics table summarizes the spatial control, piece activity, king safety, and tactical threats for each side.

Position 1: After 8. h4 ★

Biggest evaluation swing(s) (25cp)

Evaluation: +0.00



Metric	White	Black
Space	0.30	0.50
Mobility	2.35	1.75
King Safety	0.50	0.40
Threats	2.30	1.30

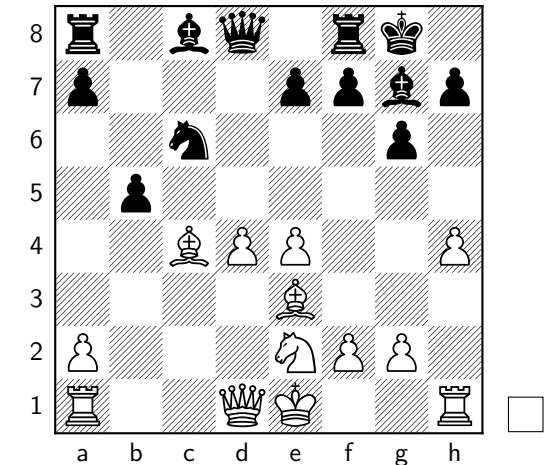
Instead of h4: Ne2, Be3, Ba3

Best continuation: c5 Ne2 cxd4

Position 2: After 11...b5 ★

Biggest evaluation swing(s) (63cp)

Evaluation: +0.63



Metric	White	Black
Space	0.20	0.40
Mobility	2.45	2.15
King Safety	0.50	0.40
Threats	5.20	8.00

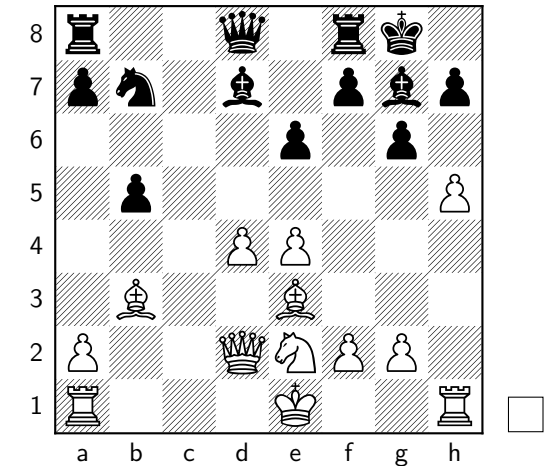
Instead of b5: Na5, Qa5+, h5

Best continuation: Bb3 a5 Rc1

Position 3: After 15...Nb7

Critical moment

Evaluation: +1.74



Metric	White	Black
Space	0.20	0.30
Mobility	2.30	1.95
King Safety	0.50	0.40
Threats	2.10	2.00

Instead of Nb7: Nxb3, Bc6, Rc8

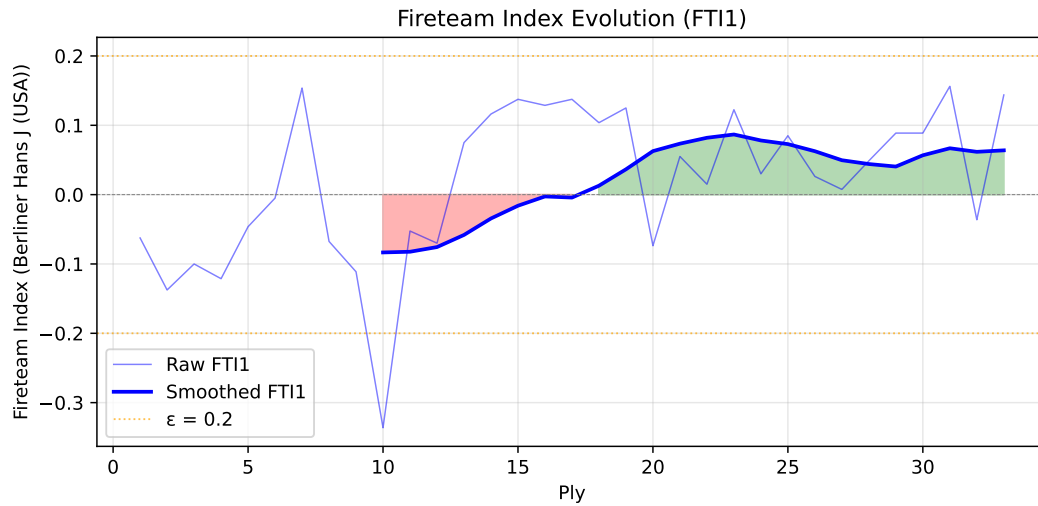
Best continuation: h6 Bh8 Rc1

3.6 Fireteam Index Prediction

Tracking: **Berliner Hans J (USA)** (White)

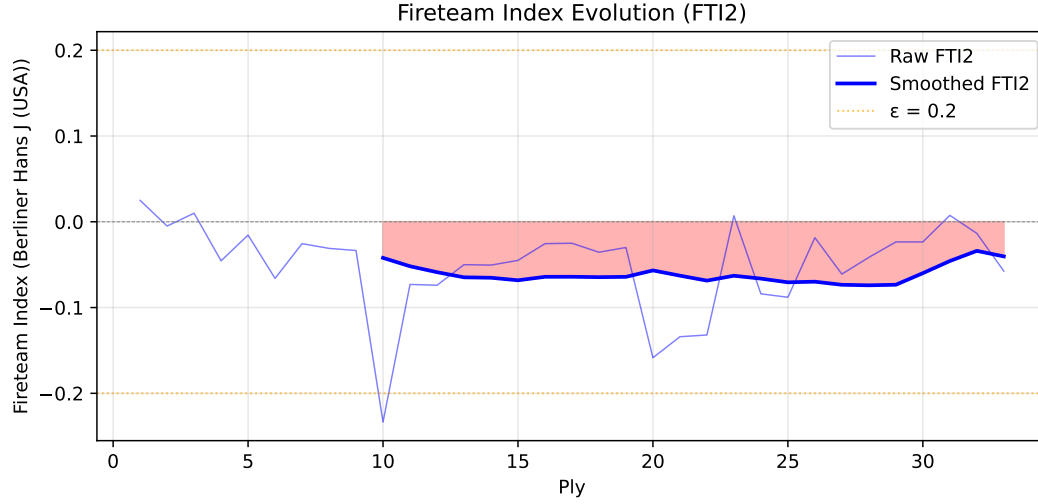
FTI1: Harmonious ($\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}$)

Algorithm	Prediction	Max Streak / Confidence
Per-ply	DRAW	0 plies
Windowed	DRAW	0 plies
Balance	DRAW	$B_n=+0.00$, $C=0.00$

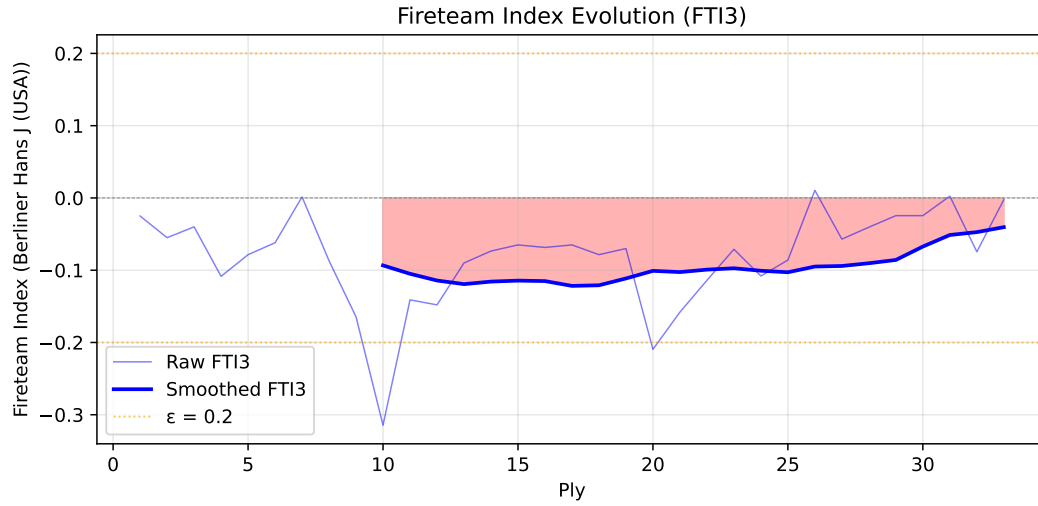


FTI2: Tactical (0.6, 0.1, 0, 0.3)

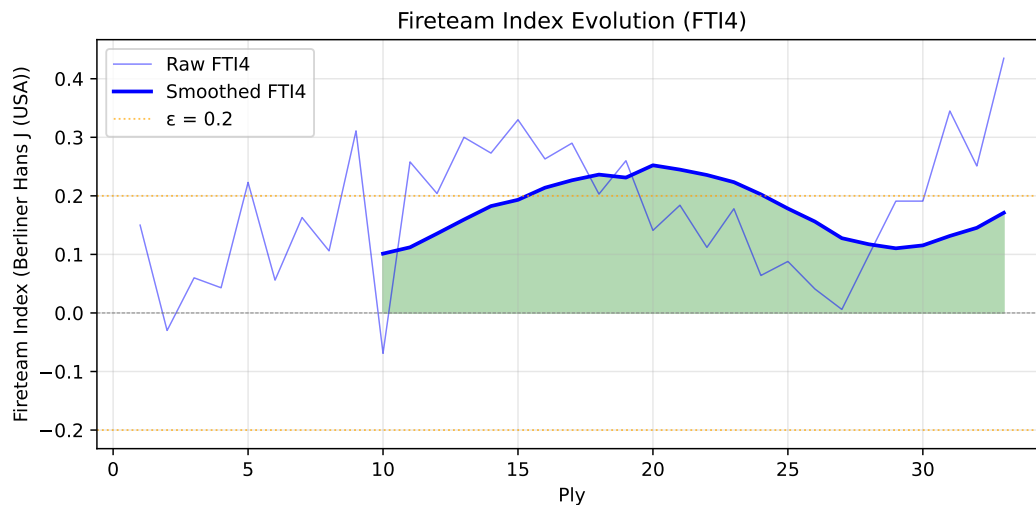
Algorithm	Prediction	Max Streak / Confidence
Per-ply	DRAW	0 plies
Windowed	DRAW	0 plies
Balance	DRAW	$B_n=+0.00$, $C=0.00$

**FTI3: Strategic (0.7, 0.1, 0.1)**

Algorithm	Prediction	Max Streak / Confidence
Per-ply	DRAW	0 plies
Windowed	DRAW	0 plies
Balance	DRAW	$B_n = -0.06$, $C = 0.06$

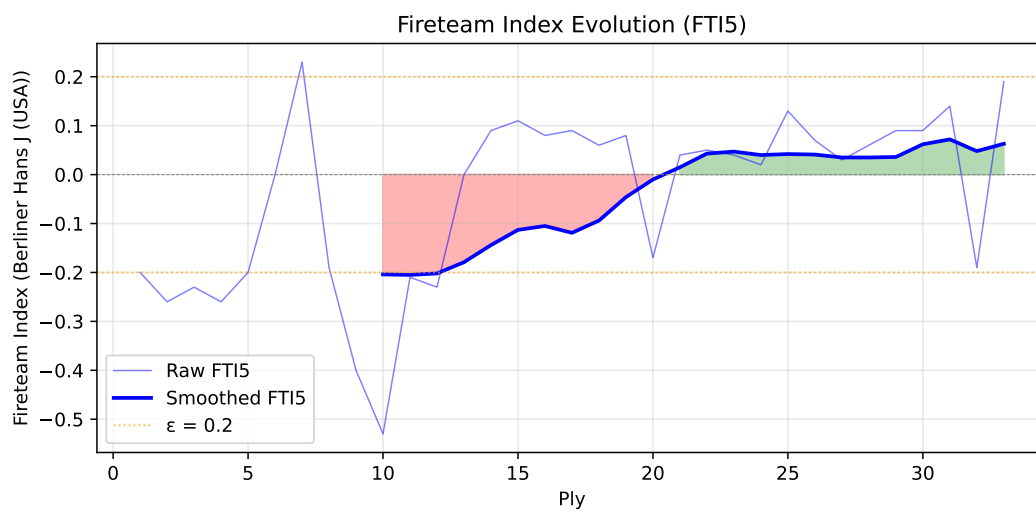
**FTI4: Dynamic (0.2, 0.6, 0, 0.2)**

Algorithm	Prediction	Max Streak / Confidence
Per-ply	DRAW	3 plies
Windowed	DRAW	1 plies
Balance	WIN	$B_n = +0.35$, $C = 0.35$



FTI5: Solid (0.3, 0.2, 0.5, 0)

Algorithm	Prediction	Max Streak / Confidence
Per-ply	DRAW	0 plies
Windowed	DRAW	0 plies
Balance	DRAW	$B_n = +0.00$, $C = 0.00$



Actual result: 1-0 (Berliner Hans J (USA)'s outcome: WIN)

FTI1: Per-ply ✗ Windowed ✗ Balance ✗

FTI2: Per-ply ✗ Windowed ✗ Balance ✗

FTI3: Per-ply ✗ Windowed ✗ Balance ✗

FTI4: Per-ply ✗ Windowed ✗ Balance ✓

FTI5: Per-ply ✗ Windowed ✗ Balance ✗

3.7 Analysis Information

- Engine: Stockfish 17.1
- Depth: 22
- Analysis time: 160.6 seconds

Chapter 4

Heinz De Carbonnel vs Hans Berliner

4.1 Game Information

Event	5th Correspondence World Championship
Site	ICCF
Date	1965.07.01
Round	4
White	Heinz De Carbonnel
Black	Hans Berliner
Result	0-1
Opening	E80 – Unknown

4.2 Player Statistics

4.2.1 Heinz De Carbonnel (White)

- Total moves: 46
- Accuracy: 85.7%
- Average centipawn loss: 30.9
- Best/Excellent moves: 14 / 8
- Good moves: 10
- Inaccuracies: 4
- Mistakes: 8
- Blunders: 2

4.2.2 Hans Berliner (Black)

- Total moves: 46
- Accuracy: 93.0%
- Average centipawn loss: 14.4

- Best/Excellent moves: 22 / 8
- Good moves: 7
- Inaccuracies: 6
- Mistakes: 2
- Blunders: 1

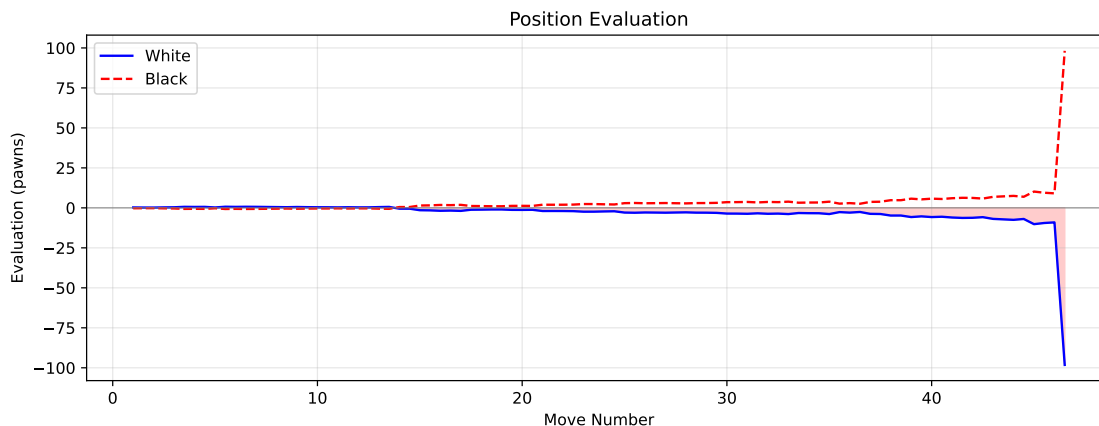
4.3 Positional Analysis

This section shows how key positional factors evolved throughout the game.

4.3.1 Position Evaluation Over Time

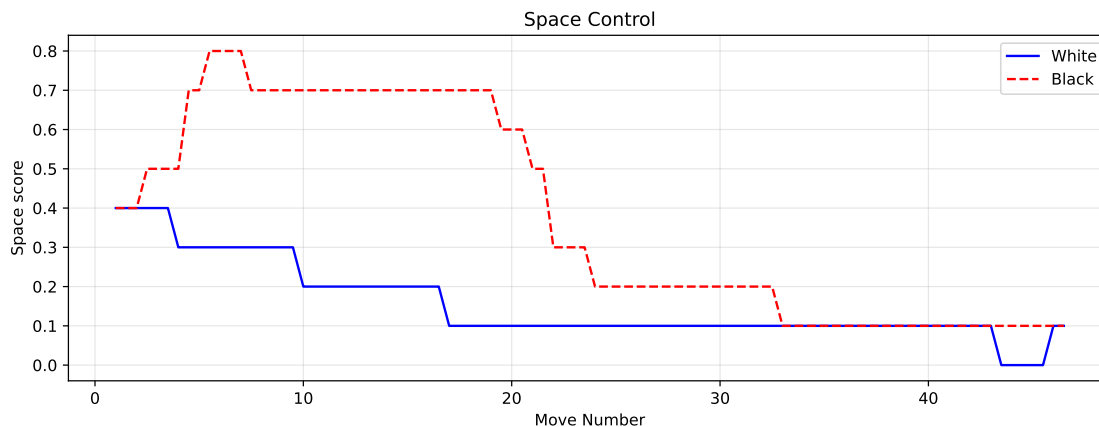
Game Character Classification. Evaluation range: minimum $m_1 = -15.00$, maximum $m_2 = +0.71$, spread $d = m_2 - m_1 = 15.71$.

This game is classified as **chaotic** (normal). A chaotic game with moderate swings.



4.3.2 Space Control

- White average: 0.16 (range: 0.00 to 0.40)
- Black average: 0.38 (range: 0.10 to 0.80)



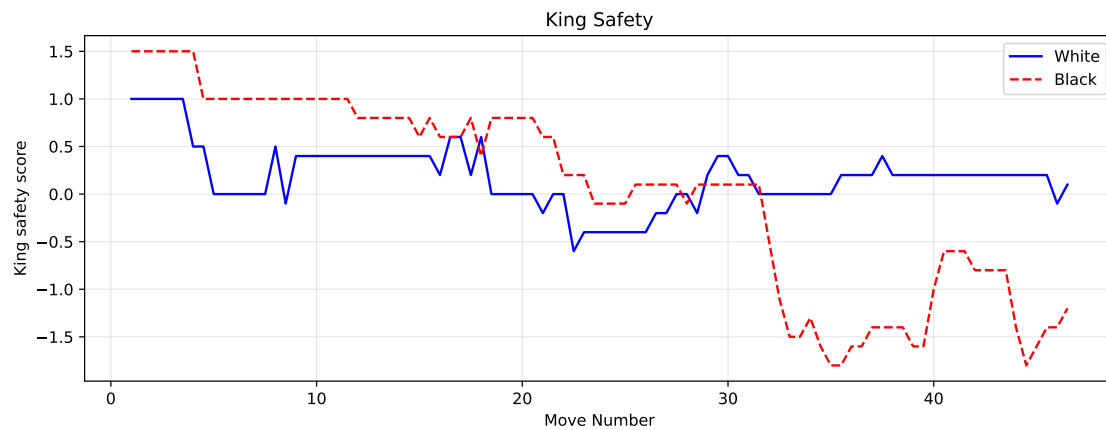
4.3.3 Mobility (Piece Activity)

- White average: 2.23 (range: 1.65 to 2.95)
- Black average: 2.12 (range: 1.55 to 2.95)



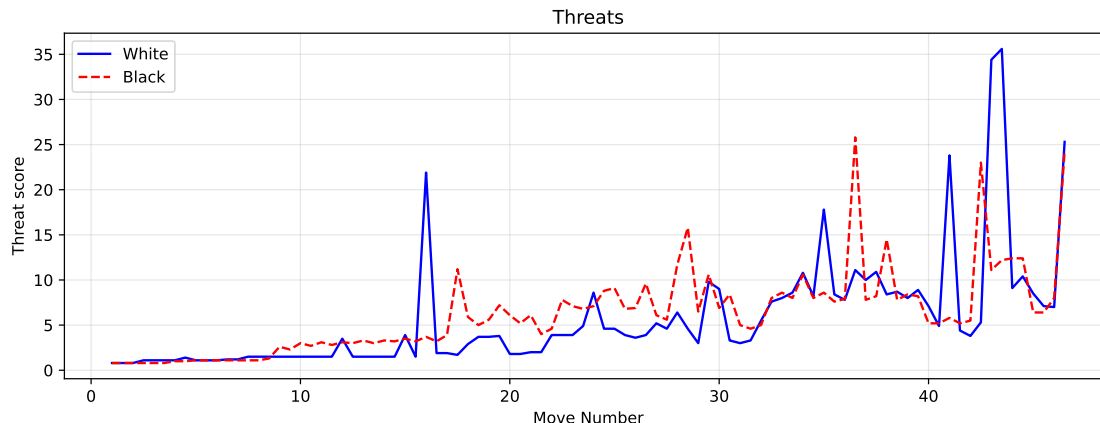
4.3.4 King Safety

- White average: 0.18
- Black average: 0.04



4.3.5 Threats

- White average: 5.72 (range: 0.80 to 35.60)
- Black average: 6.27 (range: 0.80 to 25.80)
- Average advantage: -0.55 (Black maintained more threats)



4.4 Annotated Game

1. d4 Nf6 2. c4 g6 3. Nc3 Bg7?! 4. e4 d6 5. f3?! c6?! 6. Be3 a6 7. Qd2 b5! 8. O-O-O Qa5! 9. Kb1! Nbd7! 10. g4 Rb8! 11. h4 h5! 12. g5! Nh7! 13. Rh2! Nh8! 14. Rc1?? *White loses 123cp. Consider instead: Rf2, e5.*

14...Nb6! 15. cxb5? *White loses 109cp. Consider instead: d5, c5, Nd1.*

15...axb5! 16. Nd1?! Qa8 17. f4 Nc4? *Black loses 70cp. Consider instead: Nfd7, Be6, d5.*

18. Bxc4! bxc4 19. Rf2 c5 20. Nc3! Ne6! 21. dxc5? *White loses 77cp. Consider instead: Nge2, Nf3.*

21...Bb7! 22. cxd6! Bxe4+ 23. Ka1! O-O! 24. dxe7! Rfe8 25. Qd1? *White loses 88cp. Consider instead: Qd7, Qd6.*

25...Rxe7! 26. Nf3 Rd8! 27. Nd2! Nd4 28. Bxd4 Bxd4! 29. Ndx4 Bxf2 30. Qc2?! Be3! 31. Rf1 Rd3?! 32. Nf6+! Kg7 33. f5 Re5 34. fxg6 fxg6! 35. Qh2?! Qb8?? *Black loses 137cp. Consider instead: Bd4.*

36. Rb1 Rb5?! 37. Qh1? *White loses 105cp. Consider instead: Qc2, Qg2.*

37...Bd4! 38. Nxb5? *White loses 96cp. Consider instead: Nfe4.*

38...Qxb5! 39. Qh2? *White loses 79cp. Consider instead: Qe1, Qh2.*

39...Qb7?! 40. Ne8+ Kh8?! 41. Nd6? *White loses 61cp.*

41...Qd5 42. Qf4! Rf3! 43. Qe4? *White loses 60cp. Consider instead: Qxf3, Qe4, Qd2.*

43...c3! 44. Qe8+ Kh7? *Black loses 62cp.*

45. Qe7+?? *White loses 280cp. Consider instead: Re1.*

45...Bg7 46. bxc3! Qxd6!

0-1

4.5 Critical Positions

This section highlights critical moments where the evaluation shifted substantially—either due to errors or missed opportunities. Each diagram shows the position *after* the move was played.

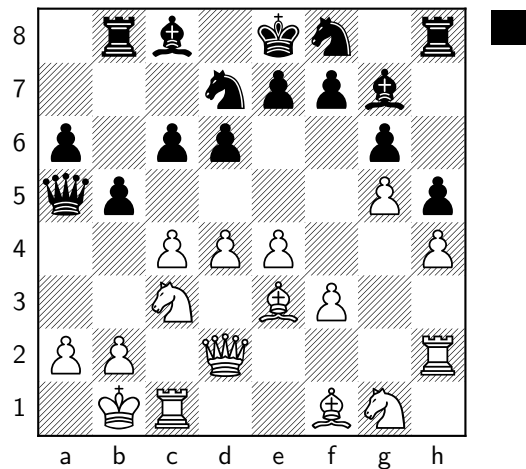
- **Instead of [move]:** The move(s) that should have been played instead.
- **Best continuation:** The optimal sequence of moves from the diagrammed position.

Positions marked with ★ represent the biggest evaluation swings in the game. The positional metrics table summarizes the spatial control, piece activity, king safety, and tactical threats for each side.

Position 1: After 14. Rc1

White blunders (-112cp)

Evaluation: -0.51



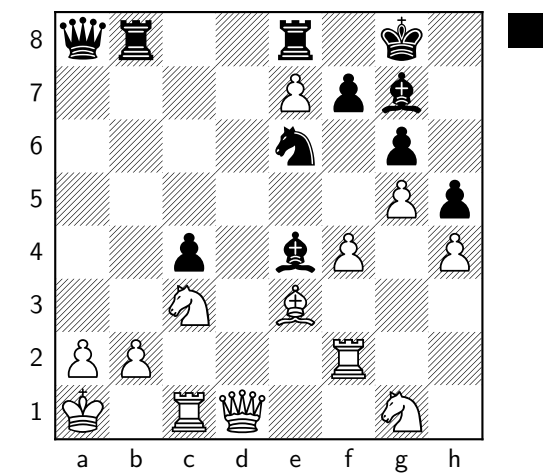
Metric	White	Black
Space	0.20	0.70
Mobility	2.50	1.70
King Safety	0.40	0.80
Threats	1.50	3.30

Instead of Rc1: Rf2, e5 Best continuation: Nb6 d5 c5

Position 2: After 25. Qd1 ★

Biggest evaluation swing(s) (82cp)

Evaluation: -2.94



Metric	White	Black
Space	0.10	0.20
Mobility	2.80	2.35
King Safety	-0.40	-0.10
Threats	4.60	9.10

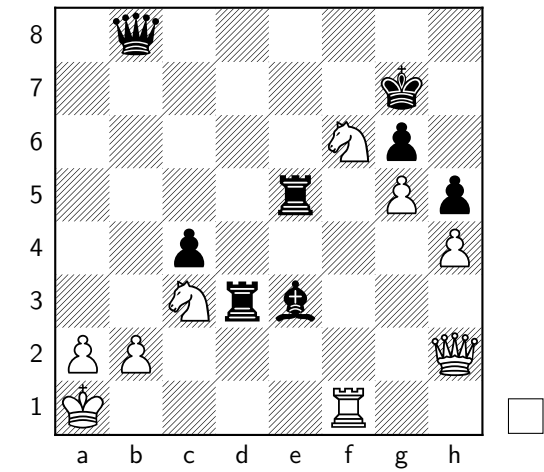
Instead of Qd1: Qd7, Qd6

Best continuation: Rxe7 Nf3 Rd8

Position 3: After 35...Qb8

Black blunders (+122cp)

Evaluation: -2.65



Metric	White	Black
Space	0.10	0.10
Mobility	2.15	2.35
King Safety	0.20	-1.80
Threats	8.40	7.60

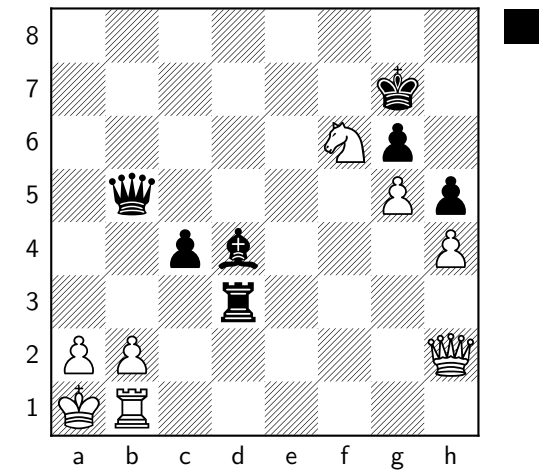
Instead of Qb8: Bd4

Best continuation: Rb1 Qd6 Qg2

Position 4: After 39. Qh2 ★

Biggest evaluation swing(s) (96cp)

Evaluation: -5.75



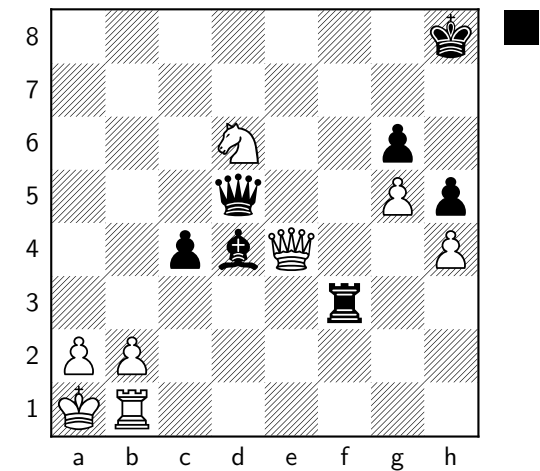
Metric	White	Black
Space	0.10	0.10
Mobility	1.75	1.75
King Safety	0.20	-1.60
Threats	8.00	8.40

Instead of Qh2: Qe1 Best continuation: Be5 Qc2 c3

Position 5: After 43. Qe4

Critical moment

Evaluation: -6.89



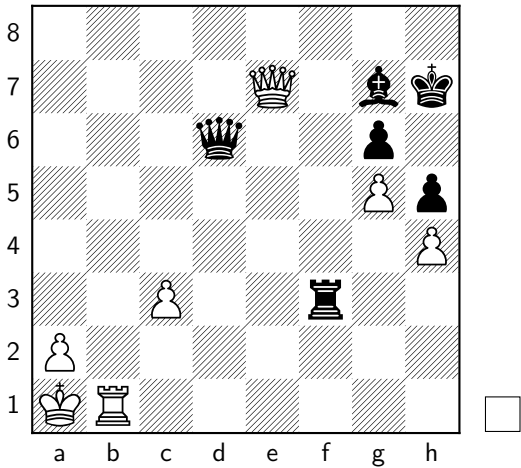
Metric	White	Black
Space	0.10	0.10
Mobility	1.75	1.95
King Safety	0.20	-0.80
Threats	34.40	11.10

Instead of Qe4: Qxf3, Qd2

Best continuation: c3 Re1 Qxe4

Position 6: After 46...Qxd6

Position swings to Black's favor
Evaluation: -98.20



Metric	White	Black
Space	0.10	0.10
Mobility	1.90	1.90
King Safety	0.10	-1.20
Threats	25.30	24.30

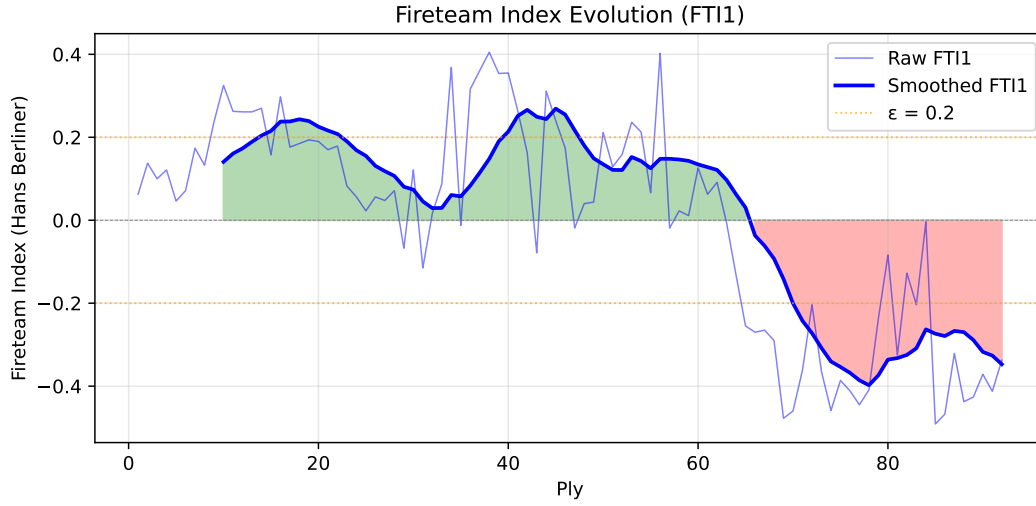
Best continuation: Qb7 Rxc3 Qxg7+

4.6 Fireteam Index Prediction

Tracking: **Hans Berliner** (Black)

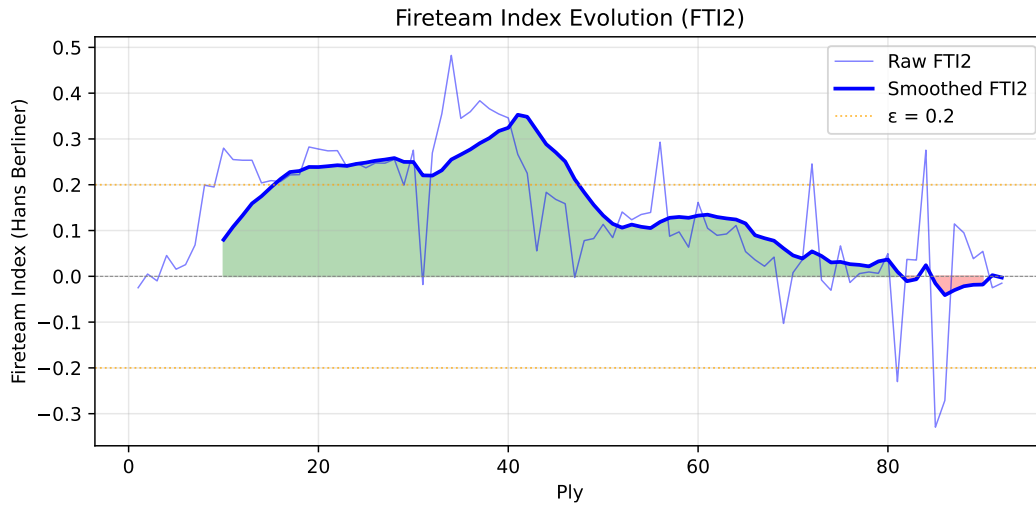
FTI1: Harmonious ($\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}$)

Algorithm	Prediction	Max Streak / Confidence
Per-ply	DRAW	6 plies
Windowed	DRAW	2 plies
Balance	DRAW	$B_n=-0.16$, $C=0.15$



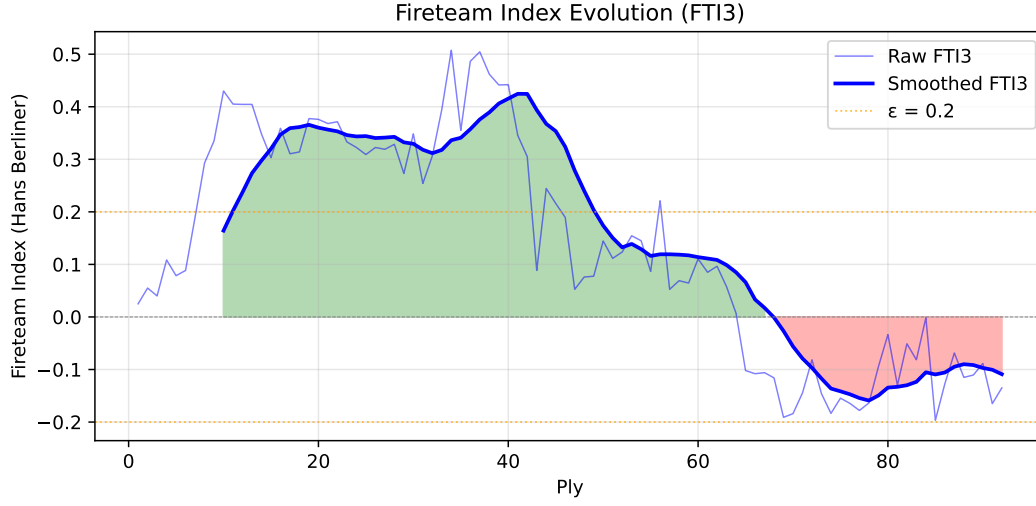
FTI2: Tactical (0.6, 0.1, 0, 0.3)

Algorithm	Prediction	Max Streak / Confidence
Per-ply	WIN	10 plies
Windowed	WIN	13 plies
Balance	WIN	$B_n = +0.32$, $C = 0.25$

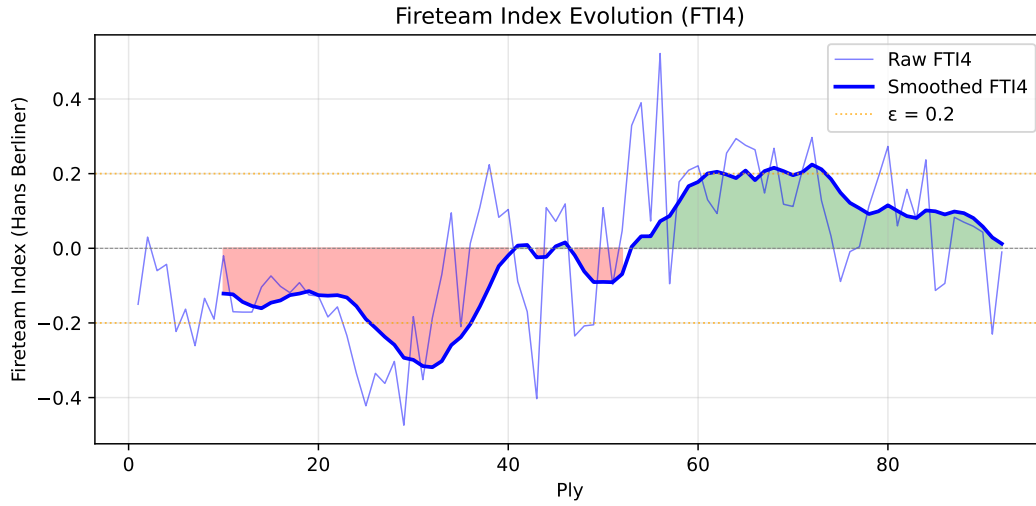


FTI3: Strategic (0.7, 0.1, 0.1, 0.1)

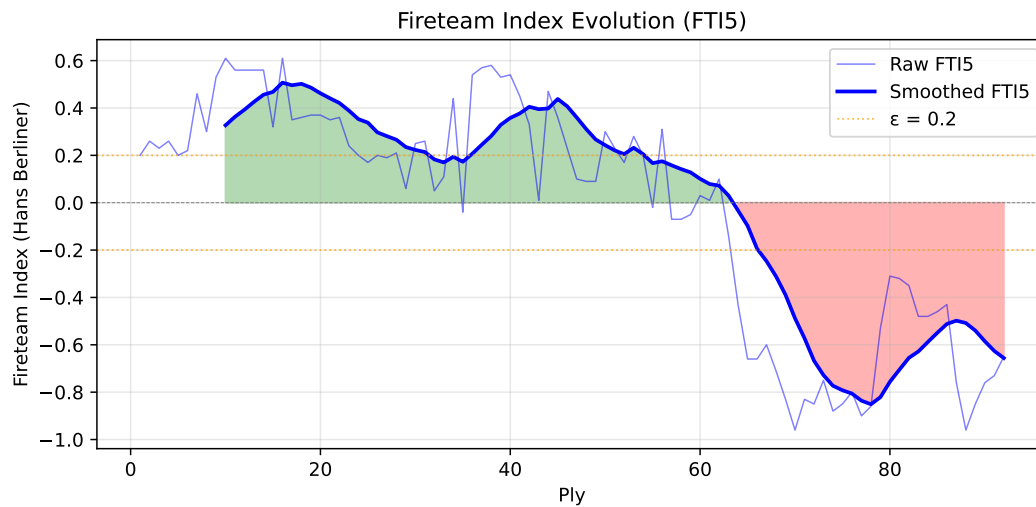
Algorithm	Prediction	Max Streak / Confidence
Per-ply	WIN	26 plies
Windowed	WIN	31 plies
Balance	WIN	$B_n = +0.38$, $C = 0.38$


FTI4: Dynamic (0.2, 0.6, 0, 0.2)

Algorithm	Prediction	Max Streak / Confidence
Per-ply	DRAW	4 plies
Windowed	DRAW	0 plies
Balance	DRAW	$B_n = +0.01$, $C = 0.01$


FTI5: Solid (0.3, 0.2, 0.5, 0)

Algorithm	Prediction	Max Streak / Confidence
Per-ply	DRAW	7 plies
Windowed	WIN	12 plies
Balance	DRAW	$B_n = -0.03$, $C = 0.02$



Actual result: 0-1 (Hans Berliner's outcome: WIN)

FTI1: Per-ply	×	Windowed	×	Balance	×
FTI2: Per-ply	✓	Windowed	✓	Balance	✓
FTI3: Per-ply	✓	Windowed	✓	Balance	✓
FTI4: Per-ply	×	Windowed	×	Balance	×
FTI5: Per-ply	×	Windowed	✓	Balance	×

4.7 Analysis Information

- Engine: Stockfish 17.1
- Depth: 22
- Analysis time: 687.4 seconds

Chapter 5

Lucius Endzelins vs Hans Berliner

5.1 Game Information

Event	5th Correspondence World Championship
Site	ICCF
Date	1965.07.01
Round	6
White	Lucius Endzelins
Black	Hans Berliner
Result	1/2-1/2
Opening	E98 – Unknown

5.2 Player Statistics

5.2.1 Lucius Endzelins (White)

- Total moves: 21
- Accuracy: 95.4%
- Average centipawn loss: 9.3
- Best/Excellent moves: 10 / 3
- Good moves: 6
- Inaccuracies: 1
- Mistakes: 1
- Blunders: 0

5.2.2 Hans Berliner (Black)

- Total moves: 21
- Accuracy: 96.3%
- Average centipawn loss: 7.5

- Best/Excellent moves: 6 / 7
- Good moves: 6
- Inaccuracies: 2
- Mistakes: 0
- Blunders: 0

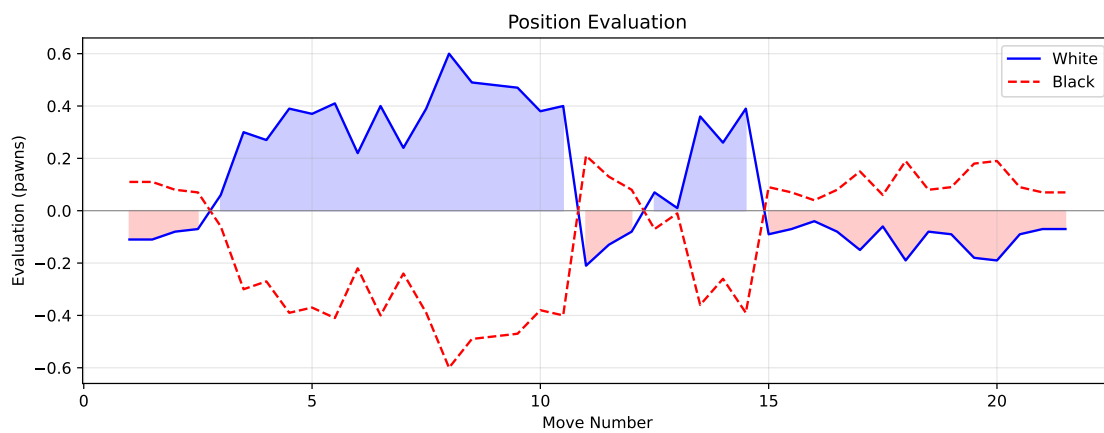
5.3 Positional Analysis

This section shows how key positional factors evolved throughout the game.

5.3.1 Position Evaluation Over Time

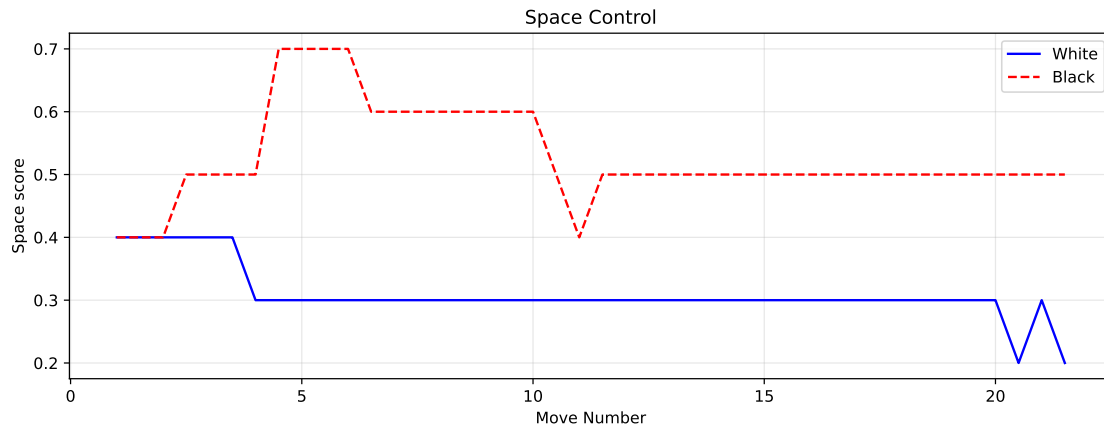
Game Character Classification. Evaluation range: minimum $m_1 = -0.21$, maximum $m_2 = +0.60$, spread $d = m_2 - m_1 = 0.81$.

This game is classified as **balanced** (one-sided). A balanced, one-sided game. White maintained the advantage throughout.



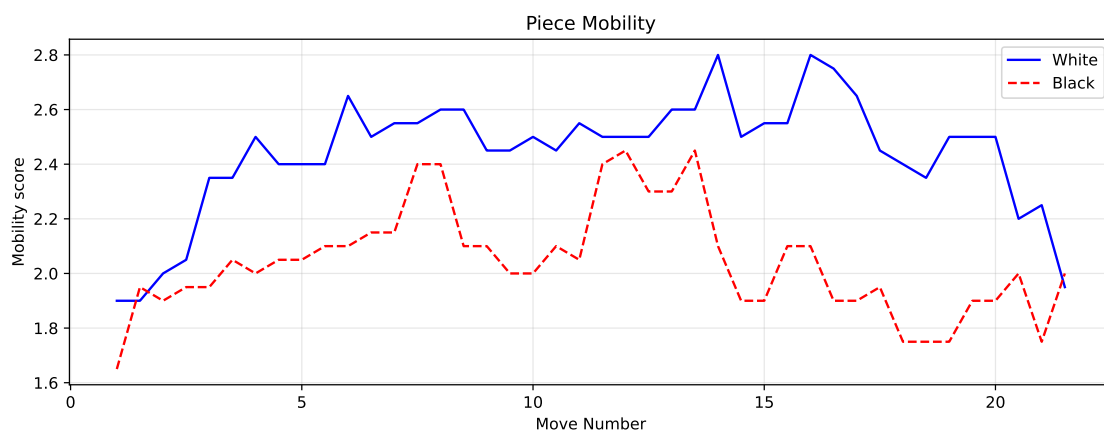
5.3.2 Space Control

- White average: 0.31 (range: 0.20 to 0.40)
- Black average: 0.53 (range: 0.40 to 0.70)



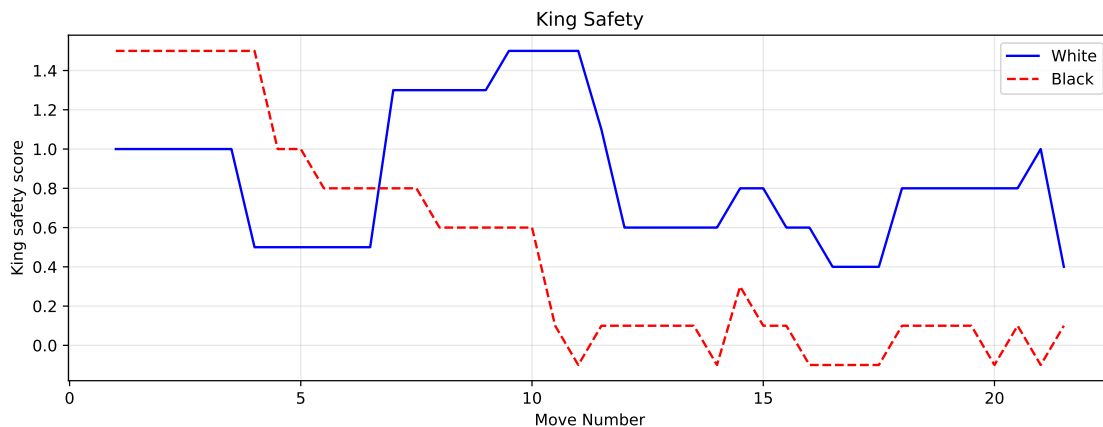
5.3.3 Mobility (Piece Activity)

- White average: 2.44 (range: 1.90 to 2.80)
- Black average: 2.04 (range: 1.65 to 2.45)



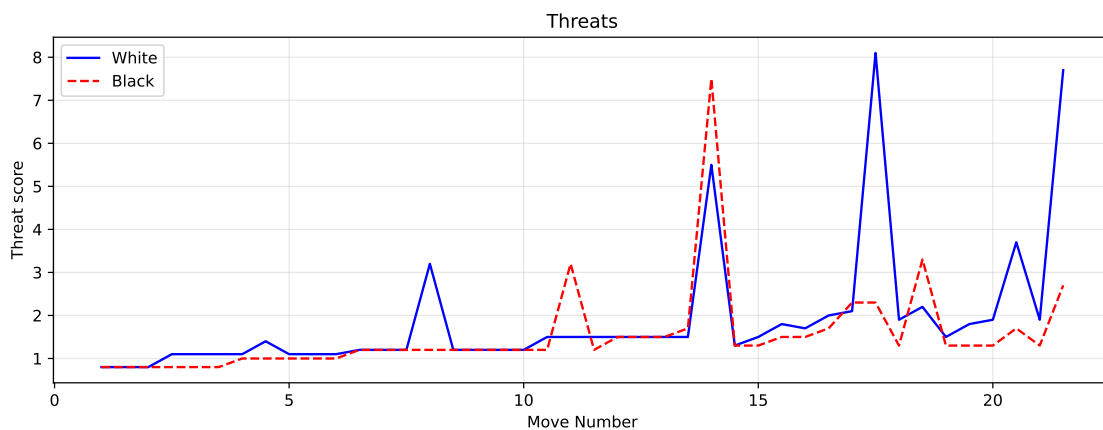
5.3.4 King Safety

- White average: 0.85
- Black average: 0.49



5.3.5 Threats

- White average: 1.91 (range: 0.80 to 8.10)
- Black average: 1.54 (range: 0.80 to 7.50)
- Average advantage: 0.37 (White maintained more threats)



5.4 Annotated Game

1. d4 Nf6 2. c4 g6 3. Nc3 Bg7?! 4. e4 d6 5. Be2 O-O 6. Nf3 e5 7. O-O Nc6 8. d5! Ne7! 9. Ne1 Nd7 10. Nd3 f5! 11. exf5? *White loses 67cp. Consider instead: f3, b4.*

11...Nxf5 12. f3! Nf6 13. Nf2 Bh6?! 14. Bxh6! Nxh6 15. Nfe4?! Nf5 16. Qd2! Nd4! 17. Nb5 Nxe2+ 18. Qxe2 a6 19. Nbc3! Bf5! 20. Qe3 Bxe4 21. Nxe4! Nxe4!

1/2-1/2

5.5 Critical Positions

This section highlights critical moments where the evaluation shifted substantially—either due to errors or missed opportunities. Each diagram shows the position *after* the move was played.

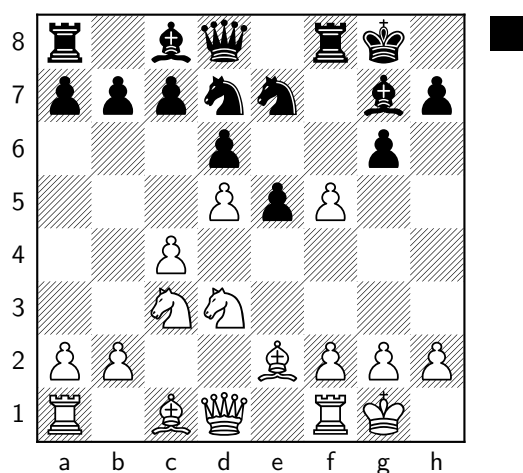
- **Instead of [move]:** The move(s) that should have been played instead.
- **Best continuation:** The optimal sequence of moves from the diagrammed position.

Positions marked with ★ represent the biggest evaluation swings in the game. The positional metrics table summarizes the spatial control, piece activity, king safety, and tactical threats for each side.

Position 1: After 11. exf5 ★

Biggest evaluation swing(s) (61cp)

Evaluation: -0.21



Metric	White	Black
Space	0.30	0.40
Mobility	2.55	2.05
King Safety	1.50	-0.10
Threats	1.50	3.20

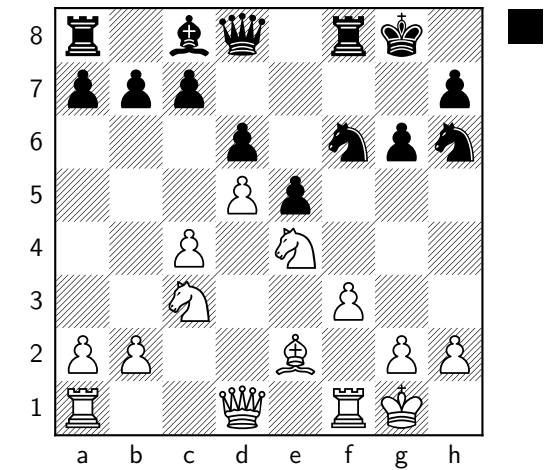
Instead of exf5: f3, b4

Best continuation: gxf5 f4 Ng6

Position 2: After 15. Nfe4 ★

Biggest evaluation swing(s) (48cp)

Evaluation: -0.09



Metric	White	Black
Space	0.30	0.50
Mobility	2.55	1.90
King Safety	0.80	0.10
Threats	1.50	1.30

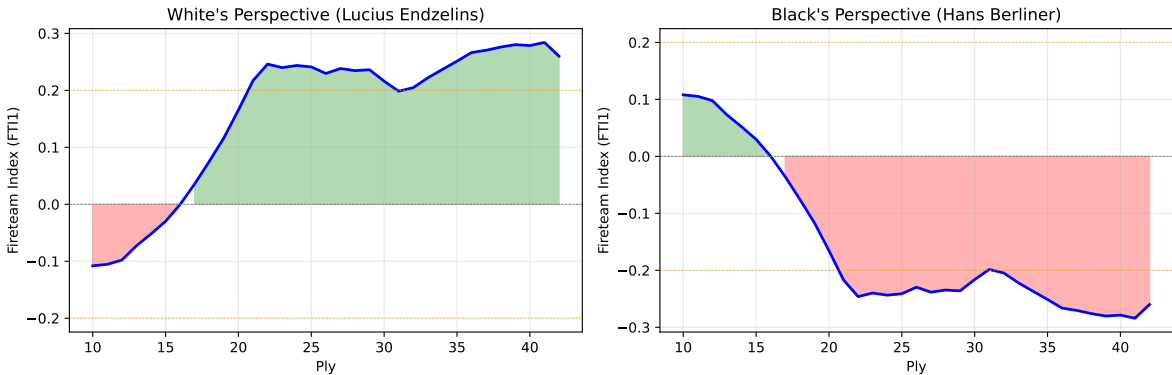
Instead of Nfe4: g4, Qd2, Bd3 Best continuation: Nxe4 Nxe4 Bf5

5.6 Fireteam Index Prediction

This game ended in a draw. The Fireteam Index is analyzed from both players' perspectives to determine whether either side achieved sustained positional dominance that might have predicted a win.

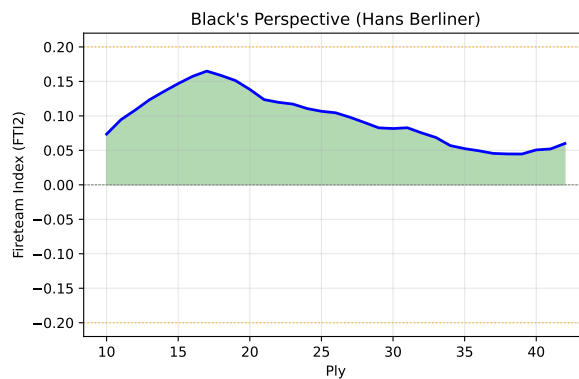
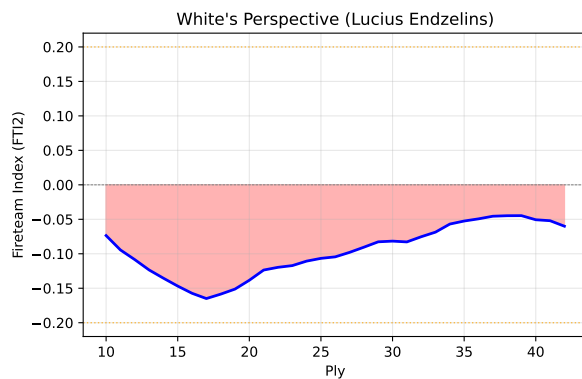
FTI1: Harmonious $(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4})$

Player	Algorithm	Prediction	Max Streak
Lucius Endzelins	Per-ply	DRAW	9 plies
Lucius Endzelins	Windowed	WIN	11 plies
Hans Berliner	Per-ply	DRAW	0 plies
Hans Berliner	Windowed	DRAW	0 plies

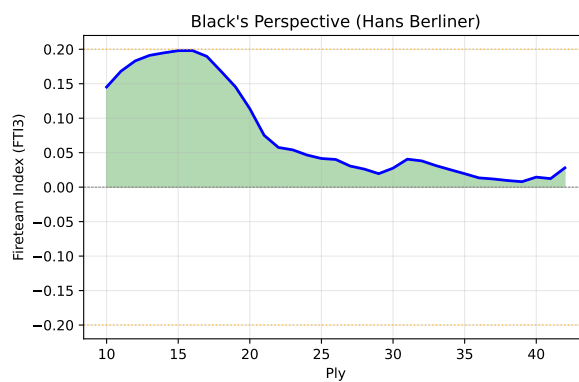
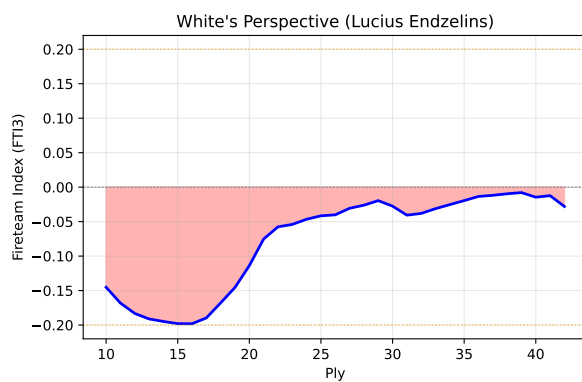


FTI2: Tactical (0.6, 0.1, 0, 0.3)

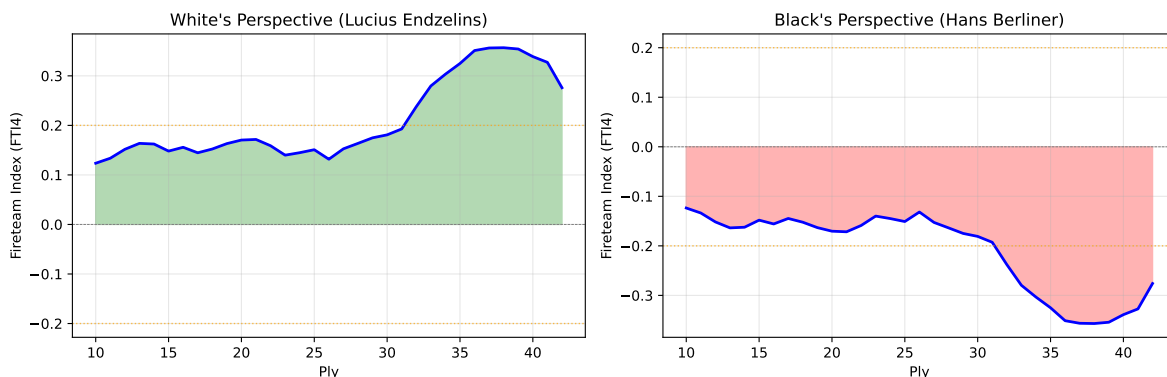
Player	Algorithm	Prediction	Max Streak
Lucius Endzelins	Per-ply	DRAW	0 plies
Lucius Endzelins	Windowed	DRAW	0 plies
Hans Berliner	Per-ply	DRAW	0 plies
Hans Berliner	Windowed	DRAW	0 plies

**FTI3: Strategic** (0.7, 0.1, 0.1, 0.1)

Player	Algorithm	Prediction	Max Streak
Lucius Endzelins	Per-ply	DRAW	0 plies
Lucius Endzelins	Windowed	DRAW	0 plies
Hans Berliner	Per-ply	DRAW	0 plies
Hans Berliner	Windowed	DRAW	0 plies

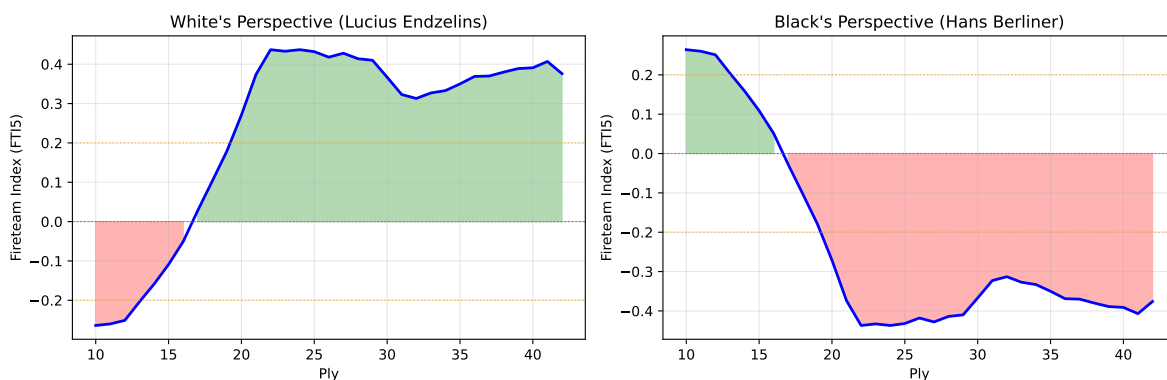
**FTI4: Dynamic** (0.2, 0.6, 0, 0.2)

Player	Algorithm	Prediction	Max Streak
Lucius Endzelins	Per-ply	WIN	13 plies
Lucius Endzelins	Windowed	WIN	11 plies
Hans Berliner	Per-ply	DRAW	0 plies
Hans Berliner	Windowed	DRAW	0 plies



FTI5: Solid (0.3, 0.2, 0.5, 0)

Player	Algorithm	Prediction	Max Streak
Lucius Endzelins	Per-ply	WIN	18 plies
Lucius Endzelins	Windowed	WIN	23 plies
Hans Berliner	Per-ply	DRAW	0 plies
Hans Berliner	Windowed	DRAW	0 plies



Interpretation. White (Lucius Endzelins) achieved sustained positional dominance, with FTI1 predicting a win. However, the game ended in a draw, suggesting Black successfully defended or White failed to convert the advantage.

5.7 Analysis Information

- Engine: Stockfish 17.1

- Depth: 22
- Analysis time: 241.6 seconds

Chapter 6

Arne Ericson vs Hans Berliner

6.1 Game Information

Event	5th Correspondence World Championship
Site	ICCF
Date	1965.07.01
Round	14
White	Arne Ericson
Black	Hans Berliner
Result	0-1
Opening	B04 – Unknown

6.2 Player Statistics

6.2.1 Arne Ericson (White)

- Total moves: 52
- Accuracy: 90.3%
- Average centipawn loss: 20.4
- Best/Excellent moves: 16 / 10
- Good moves: 11
- Inaccuracies: 11
- Mistakes: 2
- Blunders: 2

6.2.2 Hans Berliner (Black)

- Total moves: 52
- Accuracy: 95.6%
- Average centipawn loss: 8.9

- Best/Excellent moves: 32 / 8
- Good moves: 5
- Inaccuracies: 6
- Mistakes: 1
- Blunders: 0

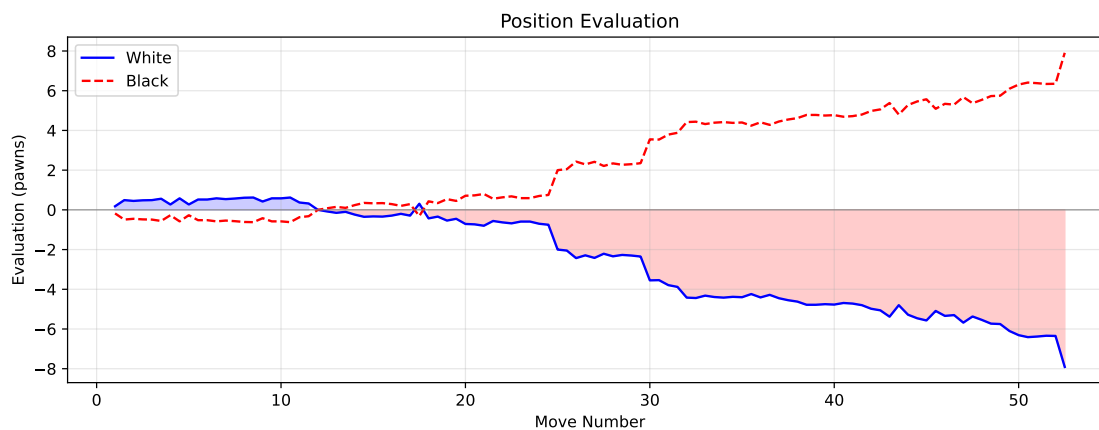
6.3 Positional Analysis

This section shows how key positional factors evolved throughout the game.

6.3.1 Position Evaluation Over Time

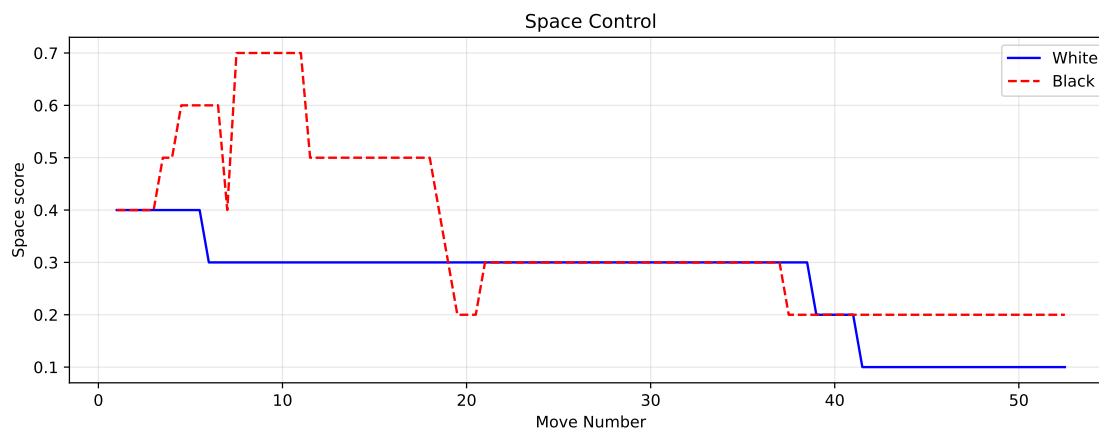
Game Character Classification. Evaluation range: minimum $m_1 = -7.91$, maximum $m_2 = +0.62$, spread $d = m_2 - m_1 = 8.53$.

This game is classified as **chaotic** (normal). A chaotic game with moderate swings.



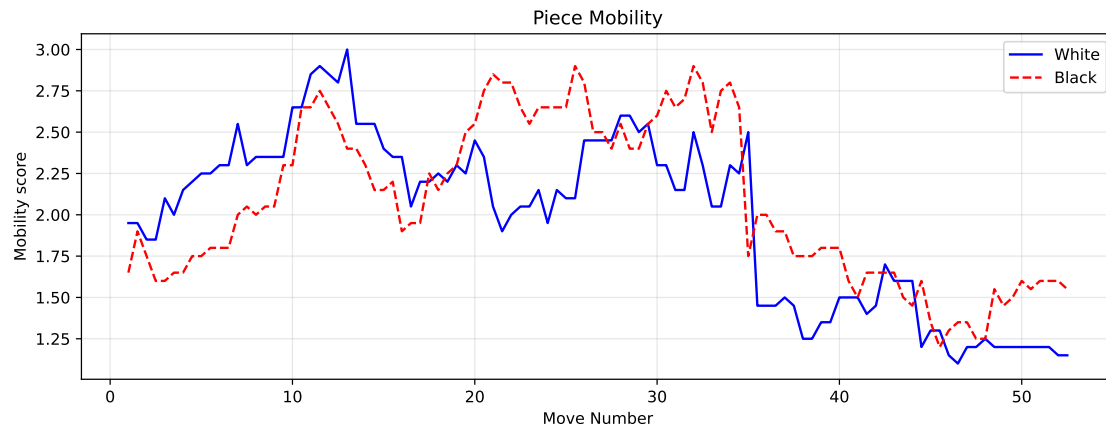
6.3.2 Space Control

- White average: 0.26 (range: 0.10 to 0.40)
- Black average: 0.35 (range: 0.20 to 0.70)



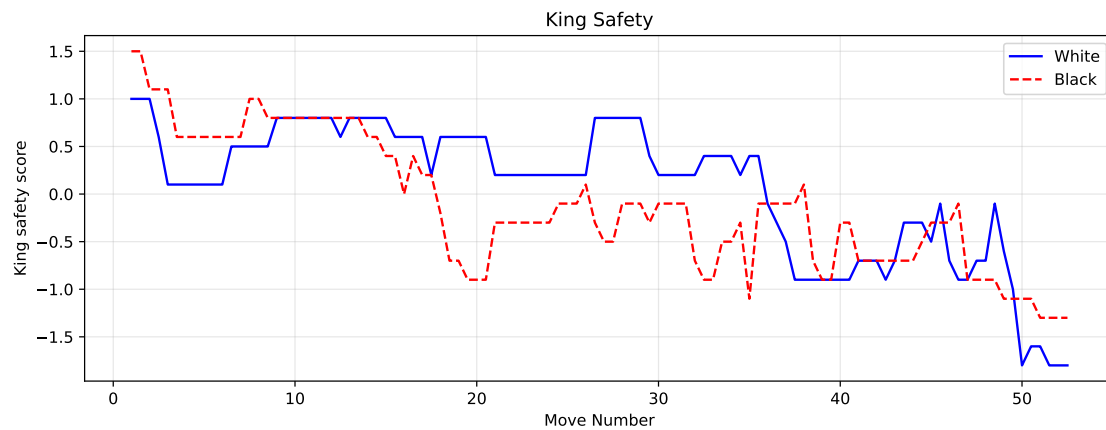
6.3.3 Mobility (Piece Activity)

- White average: 1.98 (range: 1.10 to 3.00)
- Black average: 2.08 (range: 1.20 to 2.90)



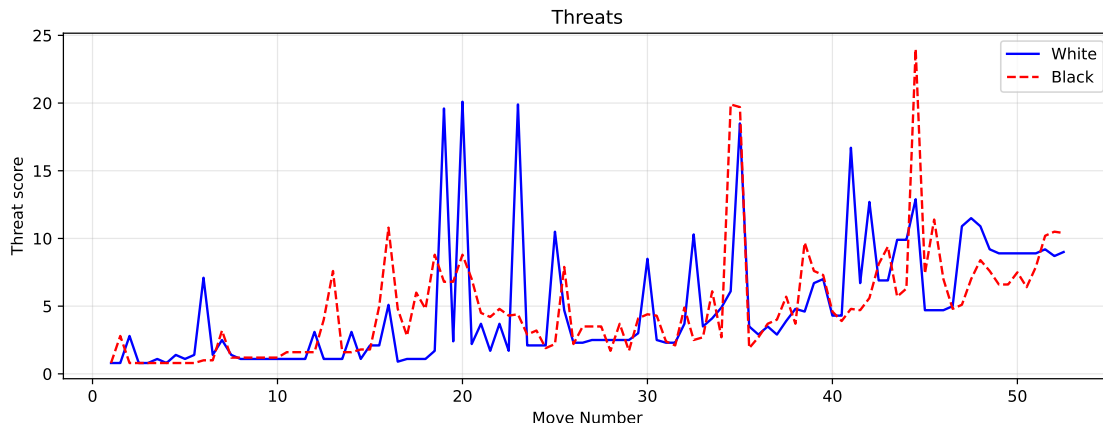
6.3.4 King Safety

- White average: 0.06
- Black average: -0.12



6.3.5 Threats

- White average: 4.87 (range: 0.80 to 20.10)
- Black average: 4.84 (range: 0.80 to 24.00)
- Average advantage: 0.02 (threats were balanced)



6.4 Annotated Game

1. e4 Nf6?! 2. e5 Nd5 3. d4 d6 4. Nf3 g6?! 5. Be2?! Bg7?! 6. c4 Nb6 7. exd6! cxd6!
8. h3! O-O 9. O-O Nc6 10. Nc3! Bf5! 11. Be3?! d5! 12. c5?! Nc4! 13. Bxc4 dxc4! 14.
d5 Nb4! 15. Bd4! Bd3 16. Bxg7! Kxg7! 17. Re1! Qc7? *Black loses 74cp. Consider
instead: Nc2, Nxd5, Rc8.*

18. Re5? *White loses 73cp. Consider instead: Ne5, Qd2, Rc1.*

18...f6! 19. d6 exd6! 20. cxd6?! Qxd6 21. Rb5! b6 22. a3! Na6! 23. Rd5! Qe7! 24.
Ne1 Bf5! 25. Qa4?? *White loses 131cp. Consider instead: Nc2, Nf3, Rd4.*

25...Nc5! 26. Qxc4?! Be6 27. Qe2 Rfe8 28. Rdd1! Nb3 29. Rab1! Bf5! 30. Qb5??
White loses 132cp. Consider instead: Qxe7+.

30...Qf7! 31. Nd3?! Rad8! 32. Nf4?! Bxb1! 33. Rxb1! Re5! 34. Qc6 Qd7! 35. Qxd7+!
Rxd7! 36. g3 Rd2 37. Kg2 g5! 38. Nh5+?! Kg6 39. g4 Nc5! 40. Ng3 Nd3! 41. Nge4
Rxb2! 42. Rd1?! Nf4+! 43. Kf1?! Ne2?! 44. a4? *White loses 63cp. Consider instead:
Rd6, Kg2, a4.*

44...Nxc3! 45. Nxc3 Rb3?! 46. Ne2 Rxh3 47. Rd7?! Ra3?! 48. Ng3 Rxa4! 49. f3
Ra1+! 50. Kg2! Ra2+! 51. Kh3 Re3! 52. Rc7 Rb3!

0-1

6.5 Critical Positions

This section highlights critical moments where the evaluation shifted substantially—either due to errors or missed opportunities. Each diagram shows the position *after* the move was played.

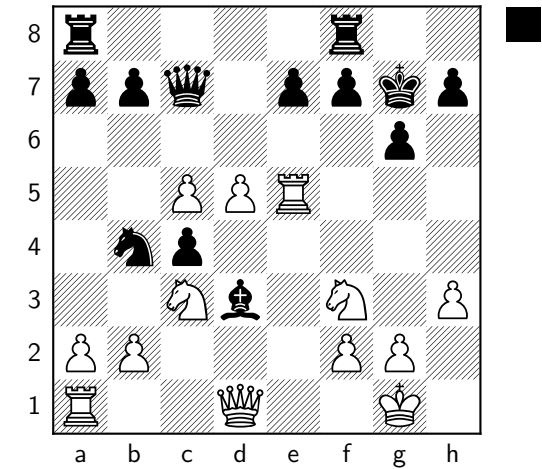
- **Instead of [move]:** The move(s) that should have been played instead.
- **Best continuation:** The optimal sequence of moves from the diagrammed position.

Positions marked with ★ represent the biggest evaluation swings in the game. The positional metrics table summarizes the spatial control, piece activity, king safety, and tactical threats for each side.

Position 1: After 18. Re5 ★

Biggest evaluation swing(s) (74cp)

Evaluation: -0.43



Metric	White	Black
Space	0.30	0.50
Mobility	2.25	2.15
King Safety	0.60	-0.20
Threats	1.10	4.80

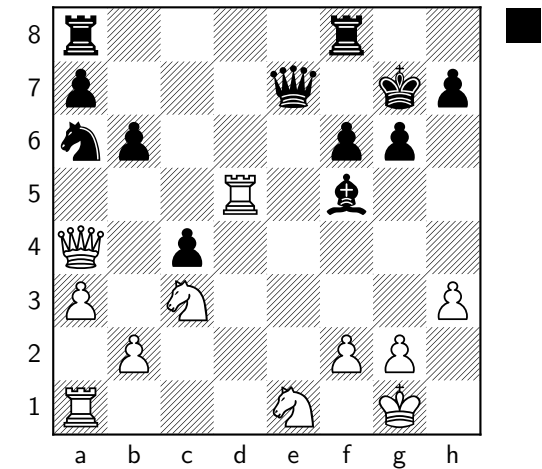
Instead of Re5: Ne5, Qd2, Rc1

Best continuation: f6 a3 fxe5

Position 2: After 25. Qa4

White blunders (-125cp)

Evaluation: -2.00



Metric	White	Black
Space	0.30	0.30
Mobility	2.10	2.65
King Safety	0.20	-0.10
Threats	10.50	2.20

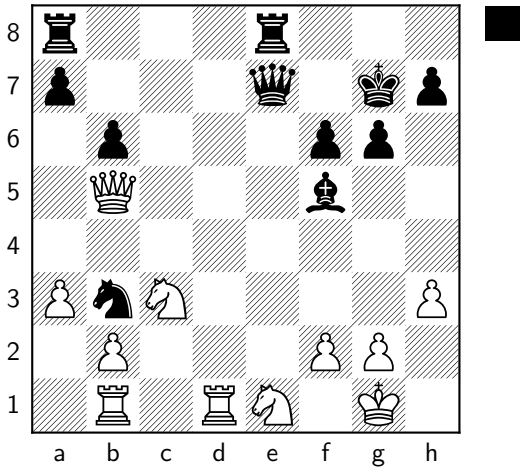
Instead of Qa4: Nc2, Nf3, Rd4

Best continuation: Nc5 Qb4 Be6

Position 3: After 30. Qb5

White blunders (-120cp)

Evaluation: -3.55



Metric	White	Black
Space	0.30	0.30
Mobility	2.30	2.60
King Safety	0.20	-0.10
Threats	8.50	4.40

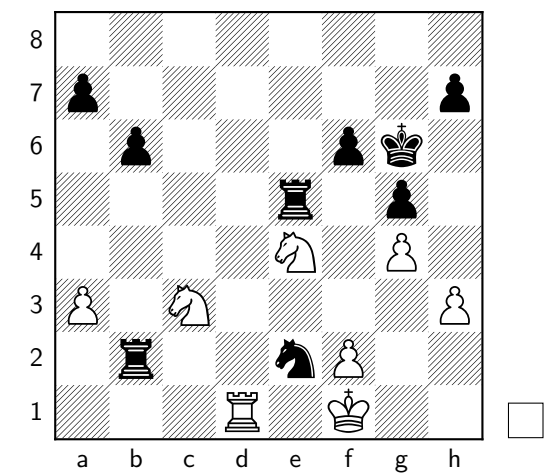
Instead of Qb5: Qxe7+

Best continuation: Qf7 Qa4 Bxb1

Position 4: After 43...Ne2 ★

Biggest evaluation swing(s) (58cp)

Evaluation: -4.80



Metric	White	Black
Space	0.10	0.20
Mobility	1.60	1.50
King Safety	-0.30	-0.70
Threats	9.90	5.70

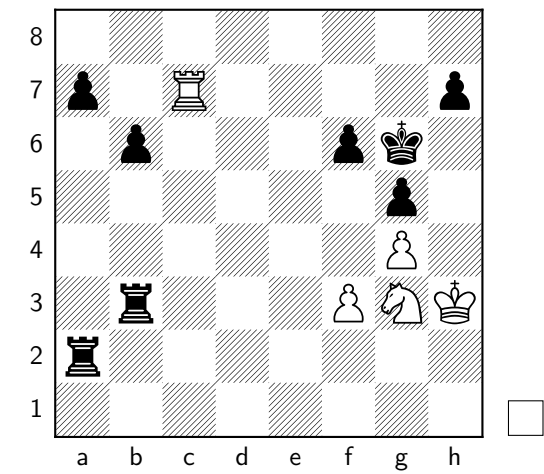
Instead of Ne2: Rb3, Re6, Rc2

Best continuation: Rd6 Nxc3 Nxc3

Position 5: After 52...Rb3

Critical moment

Evaluation: -7.91



Metric	White	Black
Space	0.10	0.20
Mobility	1.15	1.55
King Safety	-1.80	-1.30
Threats	9.00	10.40

Instead of Rb3: b5

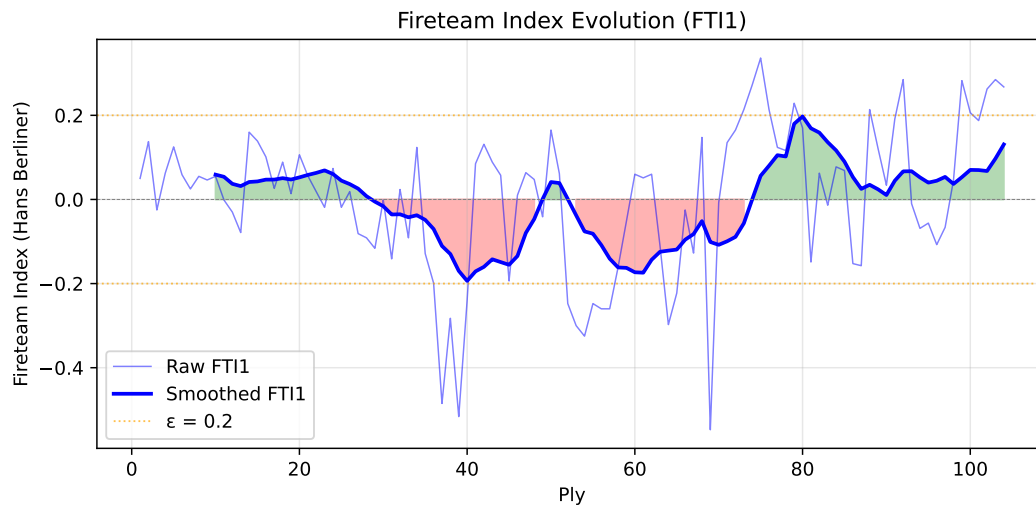
Best continuation: Re7 b5 Re2

6.6 Fireteam Index Prediction

Tracking: **Hans Berliner** (Black)

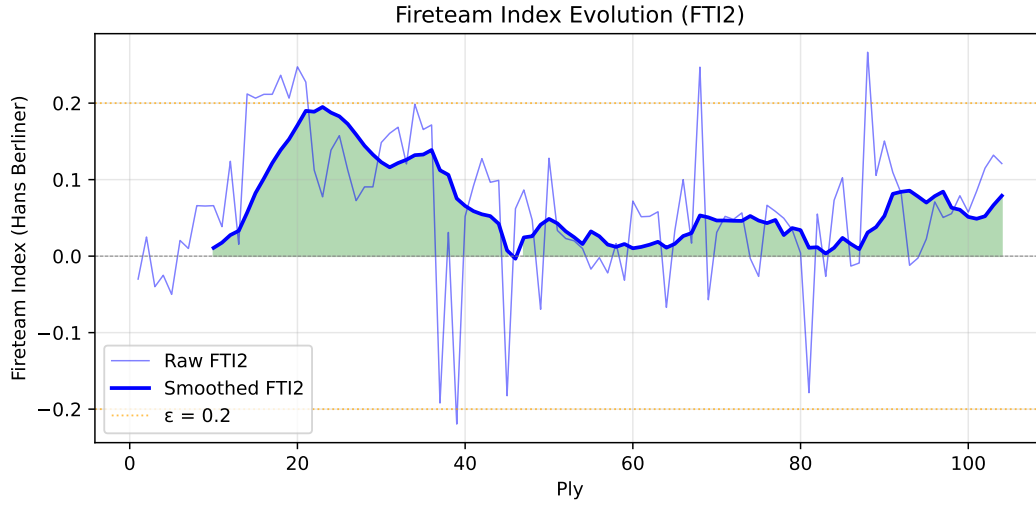
FTI1: Harmonious ($\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}$)

Algorithm	Prediction	Max Streak / Confidence
Per-ply	DRAW	3 plies
Windowed	DRAW	0 plies
Balance	DRAW	$B_n=-0.01$, $C=0.01$

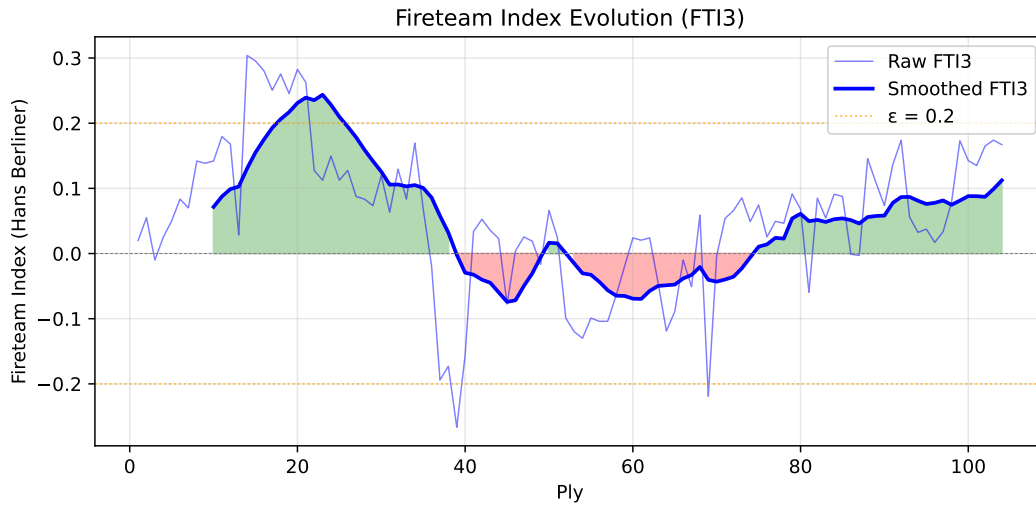


FTI2: Tactical (0.6, 0.1, 0, 0.3)

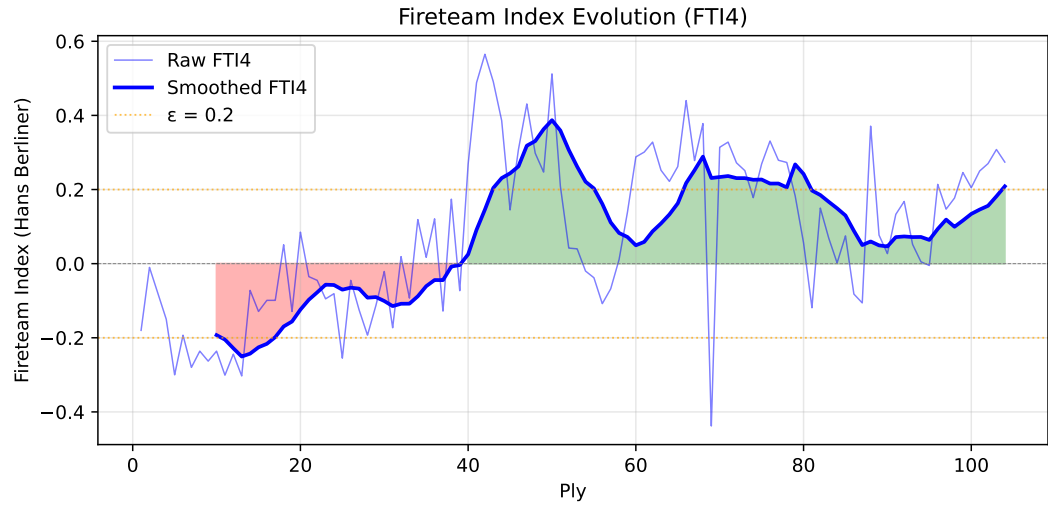
Algorithm	Prediction	Max Streak / Confidence
Per-ply	DRAW	1 plies
Windowed	DRAW	0 plies
Balance	DRAW	$B_n=+0.07$, $C=0.06$

**FTI3: Strategic (0.7, 0.1, 0.1, 0.1)**

Algorithm	Prediction	Max Streak / Confidence
Per-ply	DRAW	2 plies
Windowed	DRAW	0 plies
Balance	DRAW	$B_n = +0.03$, $C = 0.03$

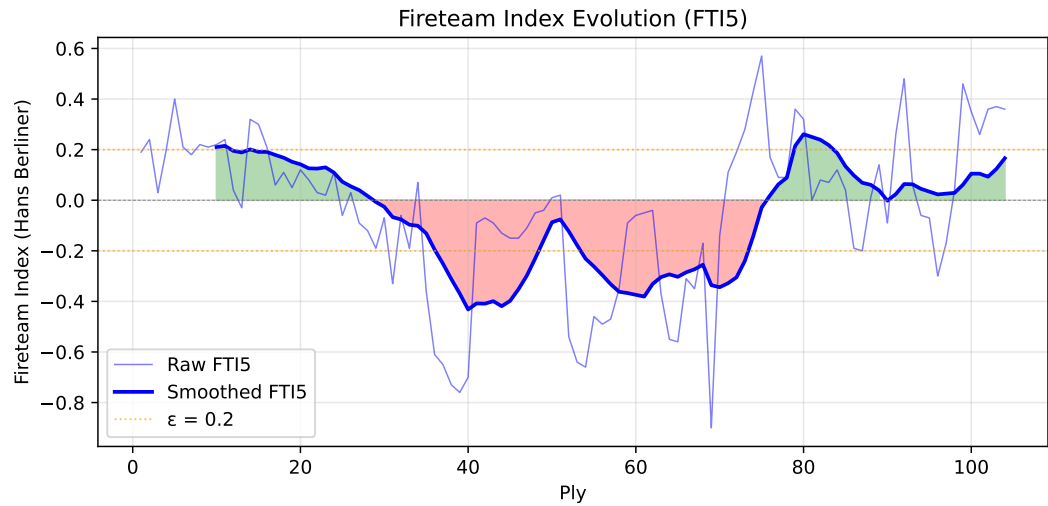
**FTI4: Dynamic (0.2, 0.6, 0, 0.2)**

Algorithm	Prediction	Max Streak / Confidence
Per-ply	DRAW	5 plies
Windowed	DRAW	8 plies
Balance	WIN	$B_n = +0.39$, $C = 0.32$



FTI5: Solid (0.3,0.2,0.5,0)

Algorithm	Prediction	Max Streak / Confidence
Per-ply	DRAW	6 plies
Windowed	DRAW	2 plies
Balance	DRAW	$B_n=-0.09$, $C=0.08$



Actual result: 0-1 (Hans Berliner’s outcome: WIN)

FTI1:	Per-ply ✗	Windowed ✗	Balance ✗
FTI2:	Per-ply ✗	Windowed ✗	Balance ✗
FTI3:	Per-ply ✗	Windowed ✗	Balance ✗
FTI4:	Per-ply ✗	Windowed ✗	Balance ✓
FTI5:	Per-ply ✗	Windowed ✗	Balance ✗

6.7 Analysis Information

- Engine: Stockfish 17.1
- Depth: 22
- Analysis time: 637.1 seconds

Chapter 7

Yakov Estrin vs Hans Berliner

7.1 Game Information

Event	5th Correspondence World Championship
Site	ICCF
Date	1965.07.01
Round	12
White	Yakov Estrin
Black	Hans Berliner
Result	0-1
Opening	C57 – Unknown

7.2 Player Statistics

7.2.1 Yakov Estrin (White)

- Total moves: 42
- Accuracy: 87.5%
- Average centipawn loss: 26.6
- Best/Excellent moves: 18 / 6
- Good moves: 9
- Inaccuracies: 5
- Mistakes: 2
- Blunders: 2

7.2.2 Hans Berliner (Black)

- Total moves: 42
- Accuracy: 92.1%
- Average centipawn loss: 16.5

- Best/Excellent moves: 21 / 7
- Good moves: 5
- Inaccuracies: 4
- Mistakes: 5
- Blunders: 0

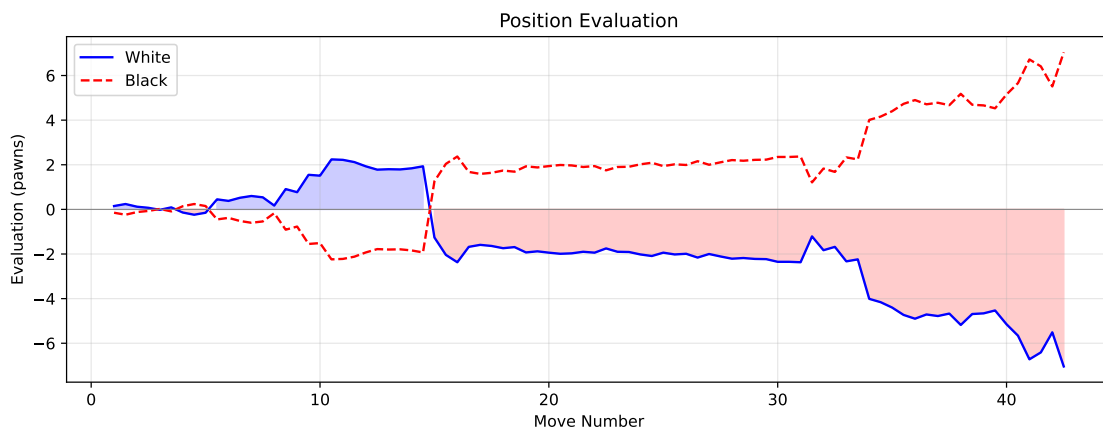
7.3 Positional Analysis

This section shows how key positional factors evolved throughout the game.

7.3.1 Position Evaluation Over Time

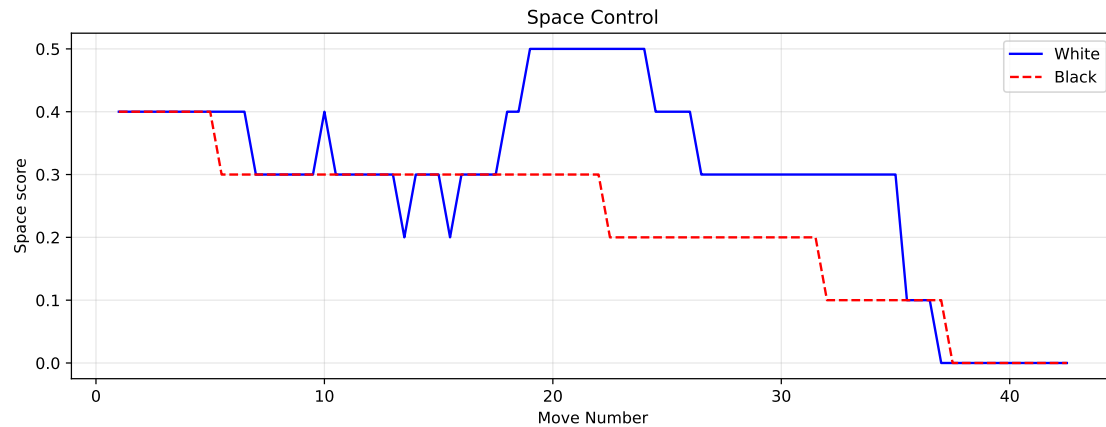
Game Character Classification. Evaluation range: minimum $m_1 = -7.04$, maximum $m_2 = +2.24$, spread $d = m_2 - m_1 = 9.28$.

This game is classified as **chaotic** (seesaw). A chaotic seesaw battle with the advantage changing hands.



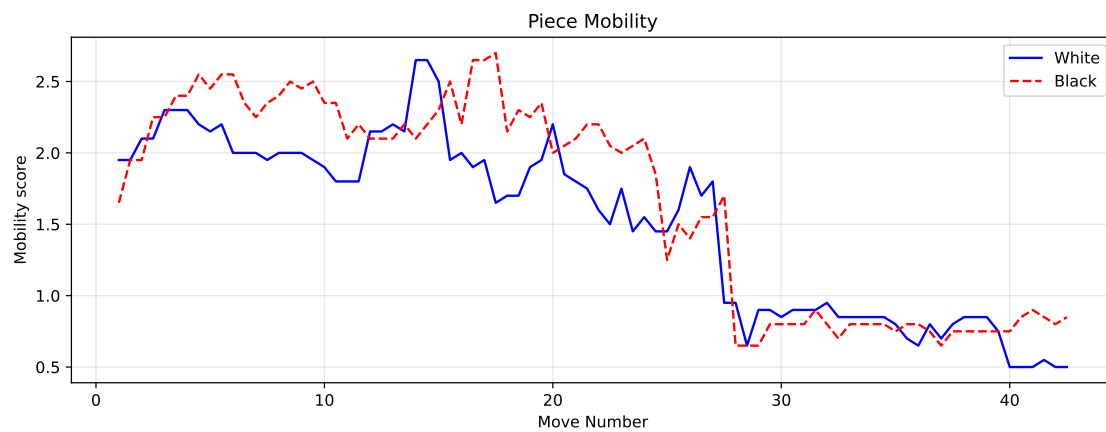
7.3.2 Space Control

- White average: 0.30 (range: 0.00 to 0.50)
- Black average: 0.22 (range: 0.00 to 0.40)



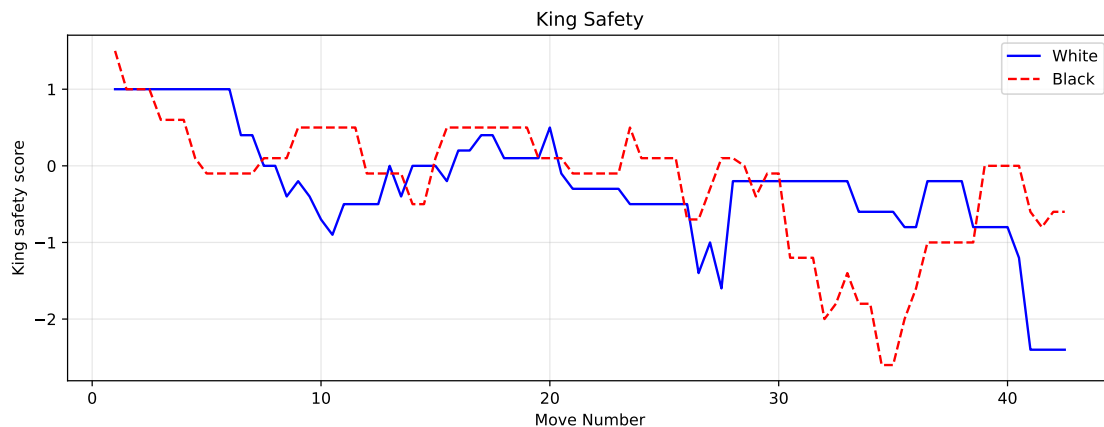
7.3.3 Mobility (Piece Activity)

- White average: 1.51 (range: 0.50 to 2.65)
- Black average: 1.67 (range: 0.65 to 2.70)



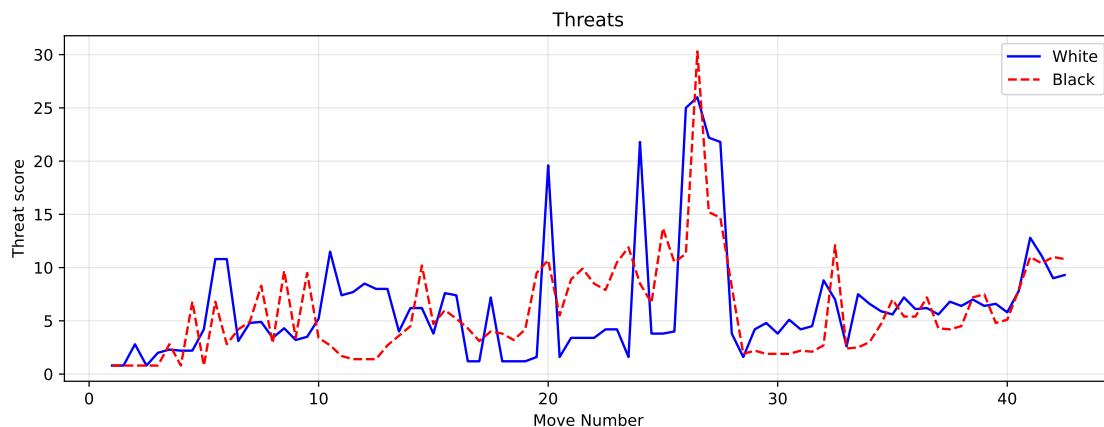
7.3.4 King Safety

- White average: -0.25
- Black average: -0.22



7.3.5 Threats

- White average: 6.35 (range: 0.80 to 26.00)
- Black average: 5.91 (range: 0.80 to 30.30)
- Average advantage: 0.43 (White maintained more threats)



7.4 Annotated Game

1. e4 e5 2. Nf3 Nc6 3. Bc4 Nf6 4. Ng5 d5 5. exd5 b5?! 6. Bf1 Nd4 7. c3! Nxd5! 8. Ne4?! Qh4? *Black loses 77cp. Consider instead: Ne6, Nc6.*
9. Ng3! Bg4? *Black loses 76cp. Consider instead: Nc6.*
10. f3! e4? *Black loses 67cp. Consider instead: Bd7, Be6, Bd6.*
11. cxd4 Bd6! 12. Bxb5+ Kd8 13. O-O exf3 14. Rxf3! Rb8 15. Be2?? *White loses 324cp. Consider instead: Bf1, a4.*
- 15...Bxf3! 16. Bxf3! Qxd4+? *Black loses 76cp. Consider instead: Re8.*
17. Kh1! Bxg3! 18. hxg3! Rb6 19. d3! Ne3! 20. Bxe3! Qxe3! 21. Bg4 h5 22. Bh3! g5 23. Nd2! g4! 24. Nc4! Qxg3! 25. Nxb6! gxh3! 26. Qf3 hxg2+ 27. Qxg2! Qxg2+ 28.

Kxg2 cxb6! 29. Rf1 Ke7! 30. Re1+! Kd6 31. Rf1! Rc8? *Black loses 116cp. Consider instead: Ke6, Ke7, Rh7.*

32. Rxf7?! Rc7! 33. Rf2? *White loses 109cp. Consider instead: Rf6+.*

33...Ke5 34. a4?? *White loses 160cp. Consider instead: Rf8.*

34...Kd4! 35. a5?! Kxd3 36. Rf3+ Kc2?! 37. b4 b5! 38. a6?! Rc4?! 39. Rf7 Rxb4 40. Rb7?! Rg4+! 41. Kf3? *White loses 113cp. Consider instead: Kh3, Kf3.*

41...b4?! 42. Rxa7! b3!

0-1

7.5 Critical Positions

This section highlights critical moments where the evaluation shifted substantially—either due to errors or missed opportunities. Each diagram shows the position *after* the move was played.

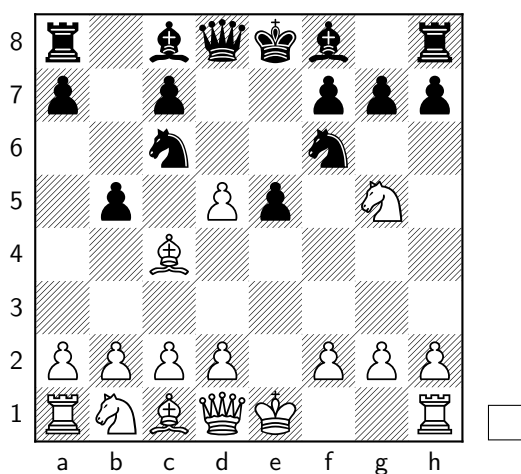
- **Instead of [move]:** The move(s) that should have been played instead.
- **Best continuation:** The optimal sequence of moves from the diagrammed position.

Positions marked with ★ represent the biggest evaluation swings in the game. The positional metrics table summarizes the spatial control, piece activity, king safety, and tactical threats for each side.

Position 1: After 5...b5 ★

Biggest evaluation swing(s) (60cp)

Evaluation: +0.45



Metric	White	Black
Space	0.40	0.30
Mobility	2.20	2.55
King Safety	1.00	-0.10
Threats	10.80	6.80

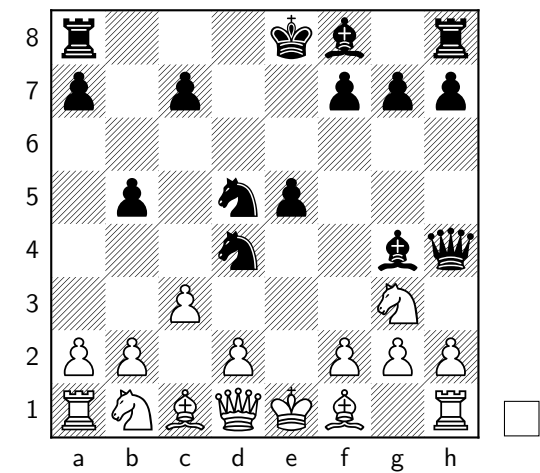
Instead of b5: Na5

Best continuation: Bf1 Qxd5 Nc3

Position 2: After 9...Bg4 ★

Biggest evaluation swing(s) (78cp)

Evaluation: +1.55



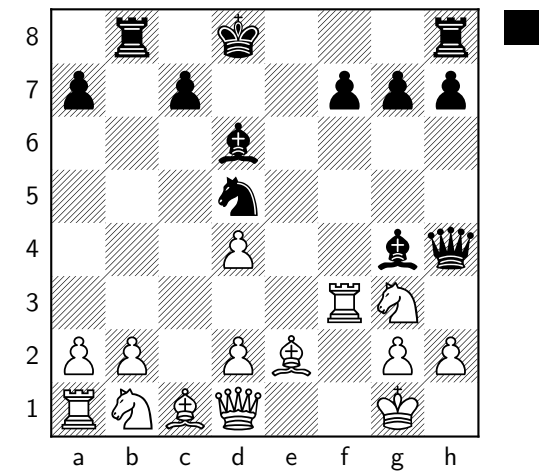
Metric	White	Black
Space	0.30	0.30
Mobility	1.95	2.50
King Safety	-0.40	0.50
Threats	3.50	9.50

Instead of Bg4: Nc6 Best continuation: f3 Bd7 cxd4

Position 3: After 15. Be2

White blunders (-319cp)

Evaluation: -1.26



Metric	White	Black
Space	0.30	0.30
Mobility	2.50	2.30
King Safety	0.00	0.10
Threats	3.80	4.70

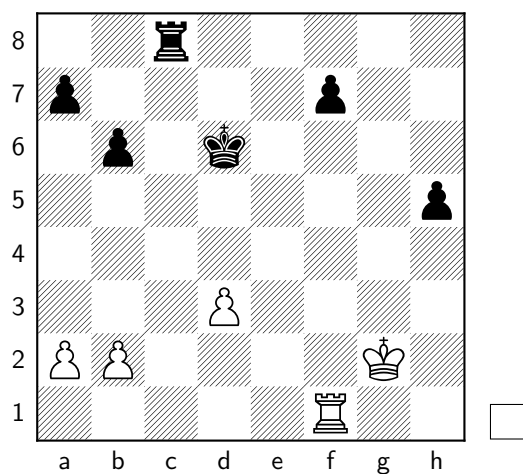
Instead of Be2: Bf1, a4

Best continuation: Bxf3 Bxf3 Qxd4+

Position 4: After 31...Rc8

Critical moment

Evaluation: -1.21



Metric	White	Black
Space	0.30	0.20
Mobility	0.90	0.90
King Safety	-0.20	-1.20
Threats	4.50	2.10

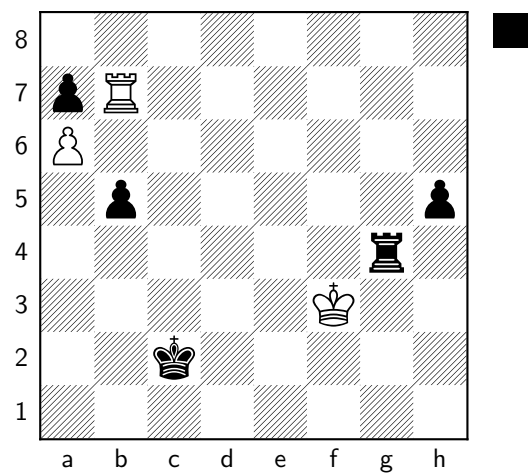
Instead of Rc8: Ke6, Ke7, Rh7

Best continuation: Rxf7 Rc7 Rf5

Position 5: After 41. Kf3

Critical moment

Evaluation: -6.72



Metric	White	Black
Space	0.00	0.00
Mobility	0.50	0.90
King Safety	-2.40	-0.60
Threats	12.80	11.00

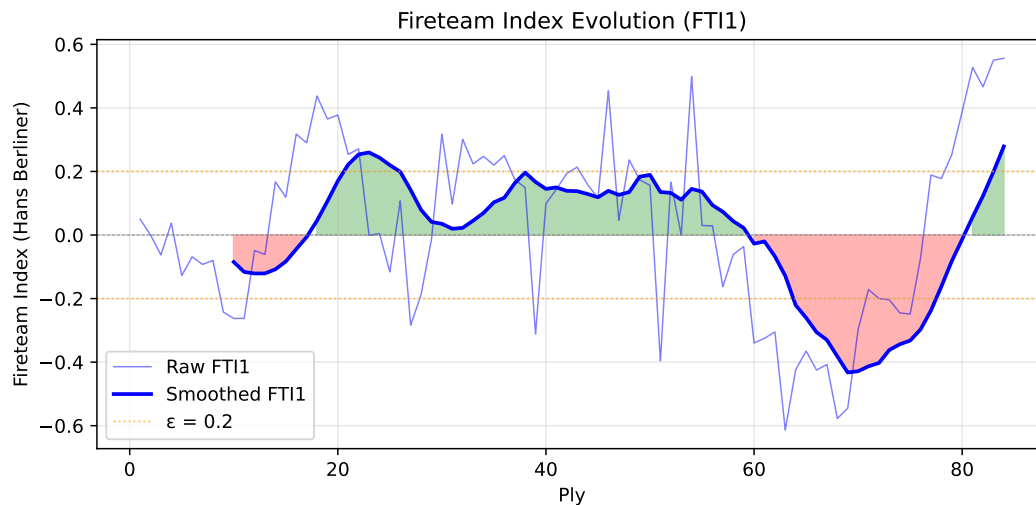
Instead of Kf3: Kh3 Best continuation: b4 Rxa7 b3

7.6 Fireteam Index Prediction

Tracking: **Hans Berliner** (Black)

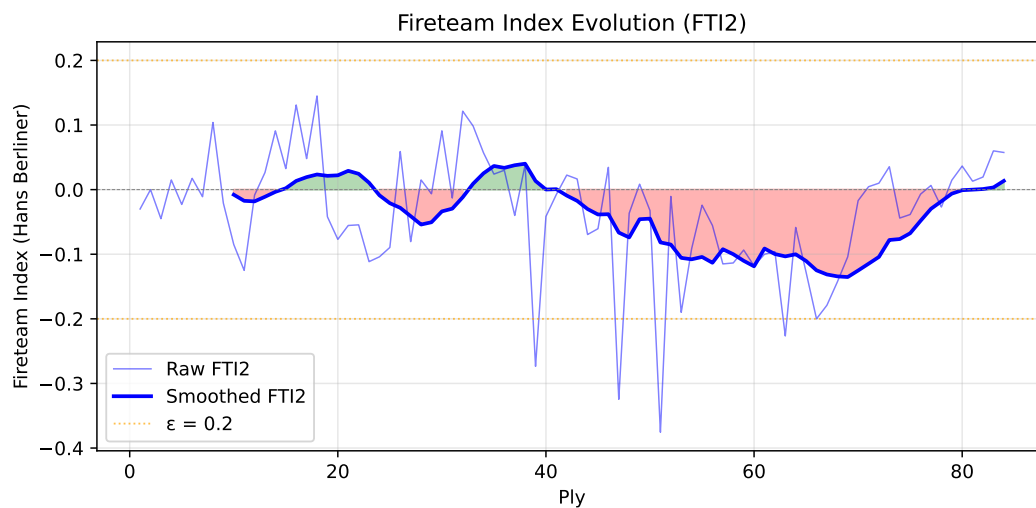
FTI1: Harmonious ($\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}$)

Algorithm	Prediction	Max Streak / Confidence
Per-ply	DRAW	6 plies
Windowed	DRAW	2 plies
Balance	DRAW	$B_n=+0.07$, $C=0.03$



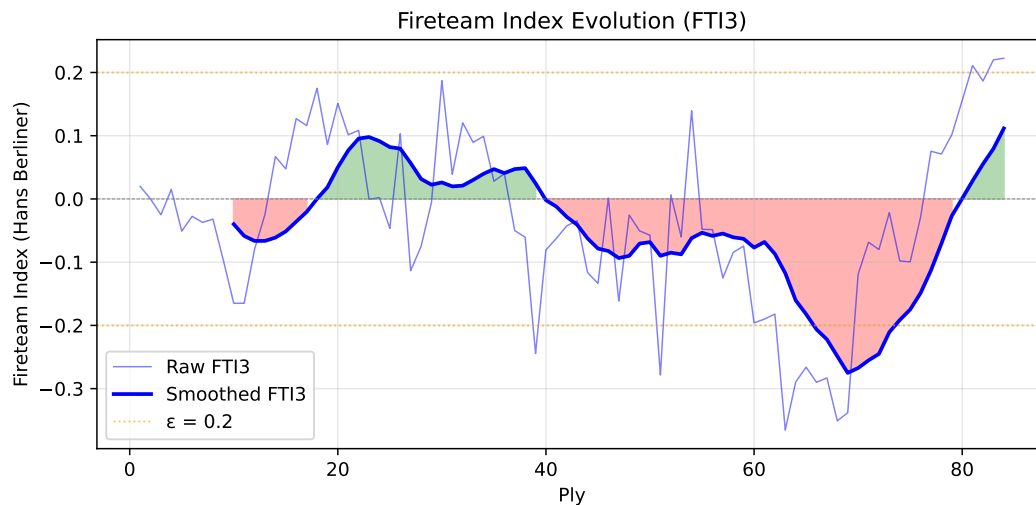
FTI2: Tactical (0.6, 0.1, 0, 0.3)

Algorithm	Prediction	Max Streak / Confidence
Per-ply	DRAW	0 plies
Windowed	DRAW	0 plies
Balance	DRAW	$B_n = -0.06$, $C = 0.06$



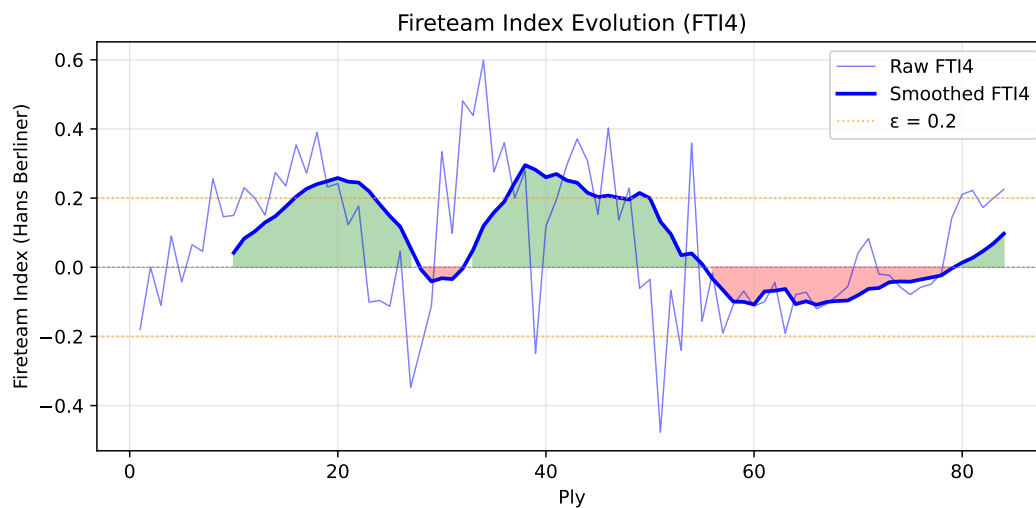
FTI3: Strategic (0.7, 0.1, 0.1, 0.1)

Algorithm	Prediction	Max Streak / Confidence
Per-ply	DRAW	0 plies
Windowed	DRAW	0 plies
Balance	DRAW	$B_n = -0.09$, $C = 0.08$



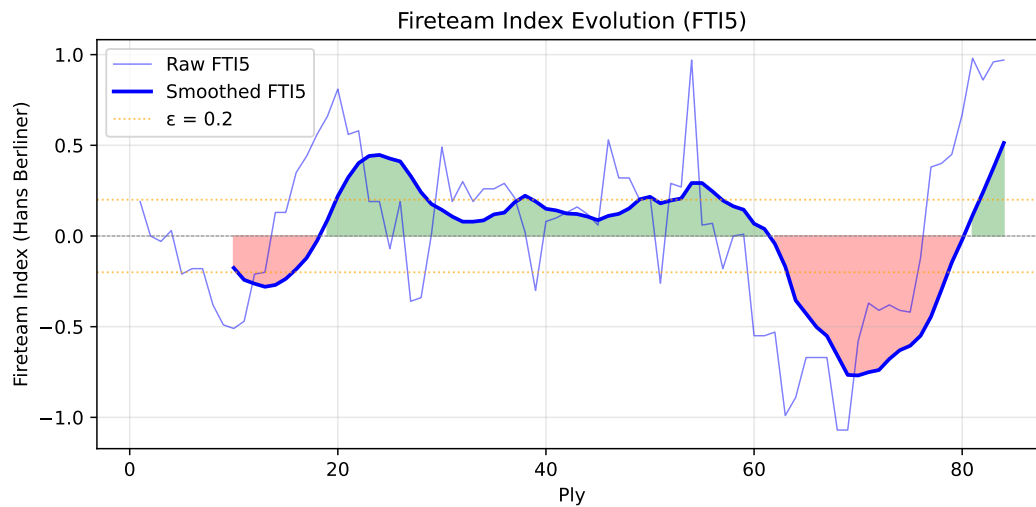
FTI4: Dynamic (0.2, 0.6, 0, 0.2)

Algorithm	Prediction	Max Streak / Confidence
Per-ply	DRAW	5 plies
Windowed	DRAW	5 plies
Balance	DRAW	$B_n=+0.24$, $C=0.13$



FTI5: Solid (0.3, 0.2, 0.5, 0)

Algorithm	Prediction	Max Streak / Confidence
Per-ply	DRAW	8 plies
Windowed	DRAW	7 plies
Balance	DRAW	$B_n=+0.10$, $C=0.04$



Actual result: 0-1 (Hans Berliner's outcome: WIN)

FTI1: Per-ply	×	Windowed	×	Balance	×
FTI2: Per-ply	×	Windowed	×	Balance	×
FTI3: Per-ply	×	Windowed	×	Balance	×
FTI4: Per-ply	×	Windowed	×	Balance	×
FTI5: Per-ply	×	Windowed	×	Balance	×

7.7 Analysis Information

- Engine: Stockfish 17.1
- Depth: 22
- Analysis time: 512.2 seconds

Chapter 8

Berliner Hans J (USA) vs Husak Karel (CZE)

8.1 Game Information

Event	Ch World (corr.)
Site	corr.
Date	1965.??.??
Round	2
White	Berliner Hans J (USA)
Black	Husak Karel (CZE)
Result	1/2-1/2
Opening	D87 – Unknown

8.2 Player Statistics

8.2.1 Berliner Hans J (USA) (White)

- Total moves: 33
- Accuracy: 96.3%
- Average centipawn loss: 7.5
- Best/Excellent moves: 15 / 8
- Good moves: 7
- Inaccuracies: 3
- Mistakes: 0
- Blunders: 0

8.2.2 Husak Karel (CZE) (Black)

- Total moves: 33
- Accuracy: 97.1%

- Average centipawn loss: 5.9
- Best/Excellent moves: 21 / 3
- Good moves: 6
- Inaccuracies: 2
- Mistakes: 1
- Blunders: 0

8.3 Positional Analysis

This section shows how key positional factors evolved throughout the game.

8.3.1 Position Evaluation Over Time

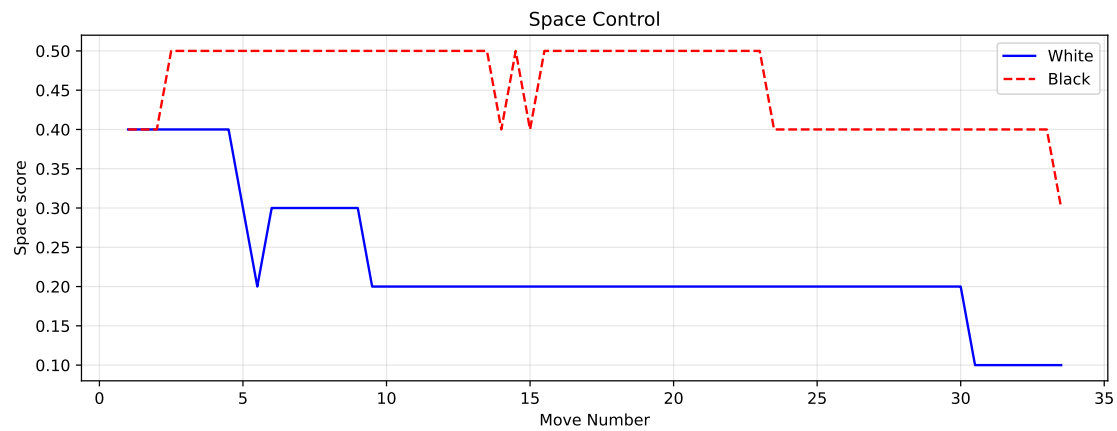
Game Character Classification. Evaluation range: minimum $m_1 = -0.35$, maximum $m_2 = +0.76$, spread $d = m_2 - m_1 = 1.11$.

This game is classified as **tense** (one-sided). A tense, one-sided game. White maintained the advantage throughout.



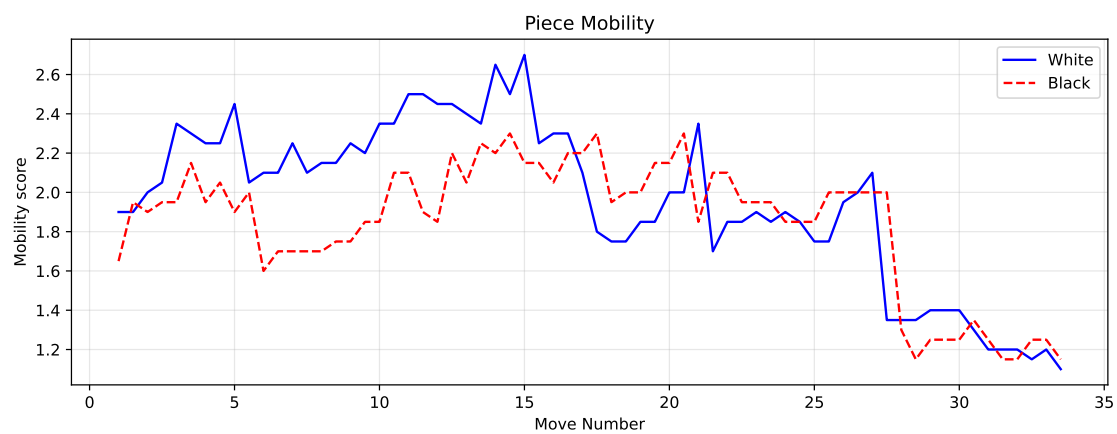
8.3.2 Space Control

- White average: 0.23 (range: 0.10 to 0.40)
- Black average: 0.46 (range: 0.30 to 0.50)



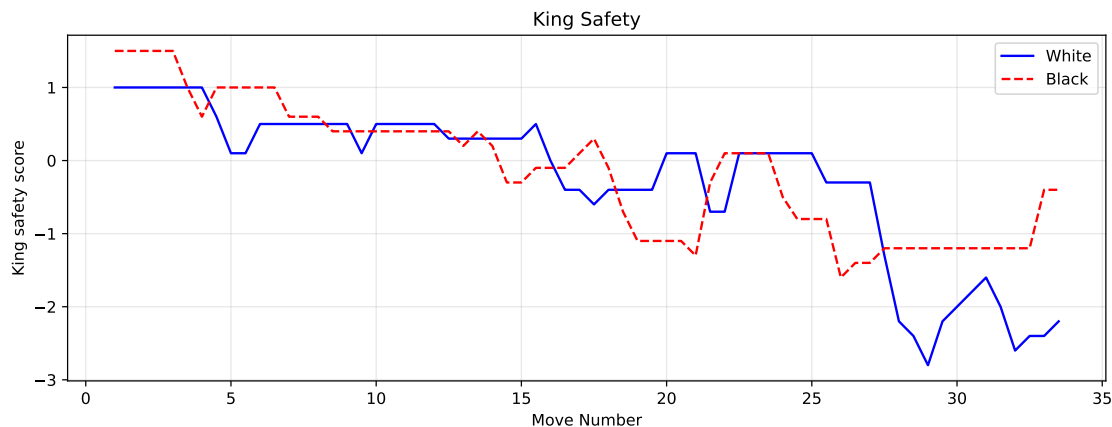
8.3.3 Mobility (Piece Activity)

- White average: 1.96 (range: 1.10 to 2.70)
- Black average: 1.85 (range: 1.15 to 2.30)



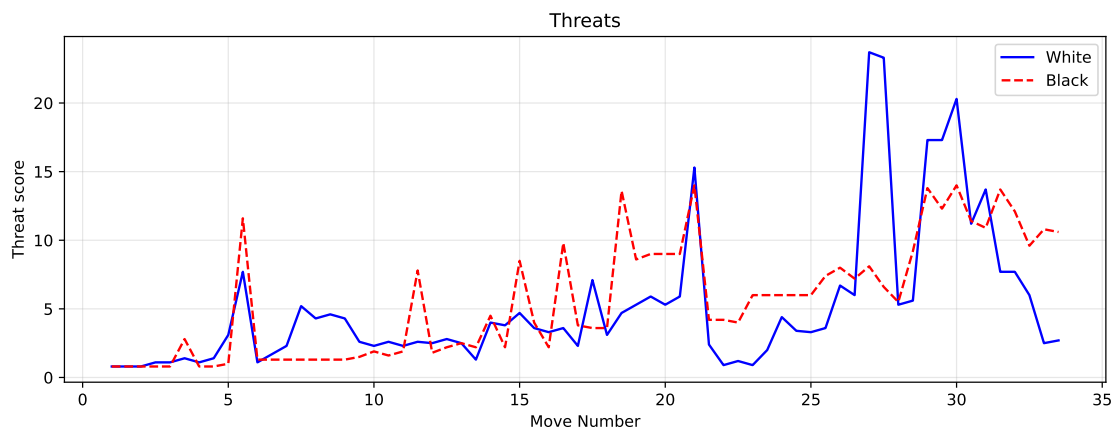
8.3.4 King Safety

- White average: -0.25
- Black average: -0.12



8.3.5 Threats

- White average: 5.17 (range: 0.80 to 23.70)
- Black average: 5.64 (range: 0.80 to 14.00)
- Average advantage: -0.47 (Black maintained more threats)



8.4 Annotated Game

1. d4 Nf6 2. c4 g6 3. Nc3 d5 4. cxd5 Nxd5 5. e4 Nxc3 6. bxc3 Bg7 7. Bc4! c5 8. Ne2 O-O 9. h4?! cxd4 10. cxd4! Nc6! 11. Be3! Na5! 12. Bd5 Qd6? *Black loses 63cp. Consider instead: e6, h5, h6.*

13. h5 Be6?! 14. hxg6?! hxg6! 15. Bxe6! Qxe6! 16. f3?! Nc4?! 17. Bh6 Bxh6 18. Rxh6 Kg7! 19. Qc1 Qa6 20. Kf2 Rh8! 21. Rxh8! Rxh8! 22. Qc3 Rc8! 23. Rc1 b5 24. d5+ f6! 25. Qd4 Qa3! 26. Rh1 Ne5 27. Qe3! Qxe3+! 28. Kxe3! Rc2! 29. Nd4! Nc4+! 30. Kd3! Rxg2! 31. Nxb5! Ne5+! 32. Ke3! Rxa2! 33. Rc1 f5

1/2-1/2

8.5 Critical Positions

This section highlights critical moments where the evaluation shifted substantially—either due to errors or missed opportunities. Each diagram shows the position *after* the move was played.

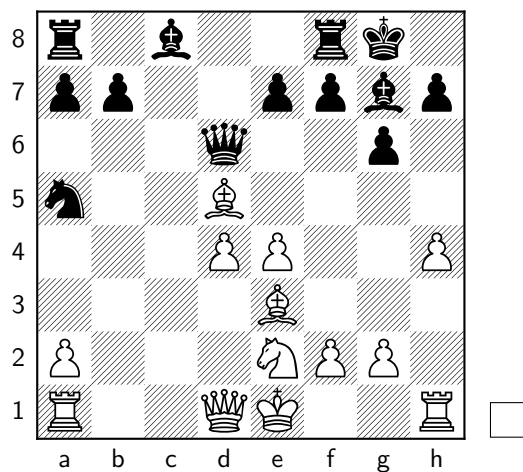
- **Instead of [move]:** The move(s) that should have been played instead.
- **Best continuation:** The optimal sequence of moves from the diagrammed position.

Positions marked with ★ represent the biggest evaluation swings in the game. The positional metrics table summarizes the spatial control, piece activity, king safety, and tactical threats for each side.

Position 1: After 12...Qd6 ★

Biggest evaluation swing(s) (61cp)

Evaluation: +0.45



Metric	White	Black
Space	0.20	0.50
Mobility	2.45	2.20
King Safety	0.30	0.40
Threats	2.80	2.20

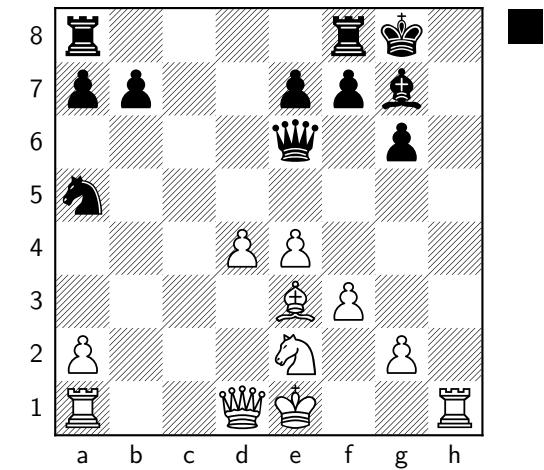
Instead of Qd6: e6, h5, h6

Best continuation: h5 e6 Bb3

Position 2: After 16. f3 ★

Biggest evaluation swing(s) (55cp)

Evaluation: -0.35



Metric	White	Black
Space	0.20	0.50
Mobility	2.30	2.05
King Safety	0.00	-0.10
Threats	3.30	2.20

Instead of f3: e5, Qa4, Qb1

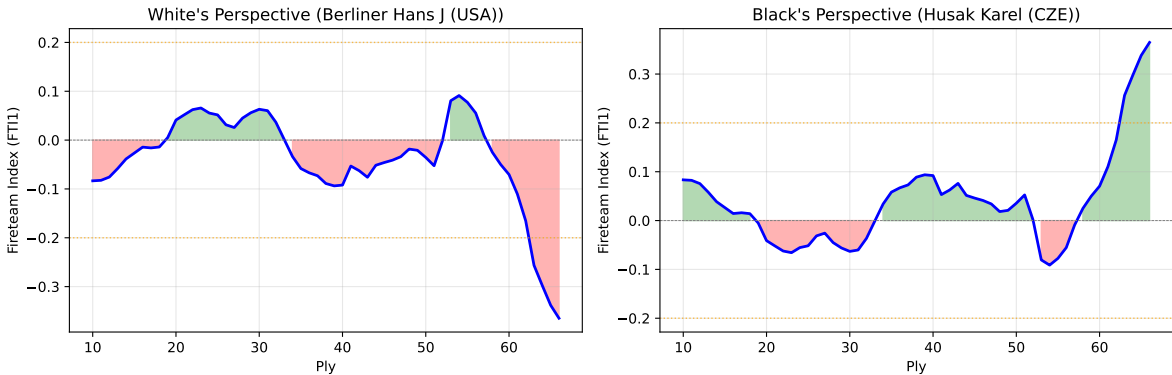
Best continuation: Rfd8 Kf2 b5

8.6 Fireteam Index Prediction

This game ended in a draw. The Fireteam Index is analyzed from both players' perspectives to determine whether either side achieved sustained positional dominance that might have predicted a win.

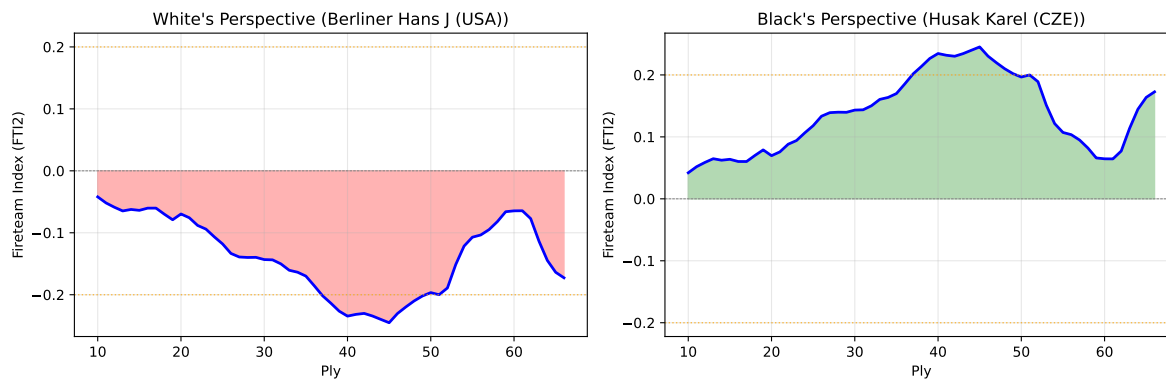
FTI1: Harmonious ($\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}$)

Player	Algorithm	Prediction	Max Streak
Berliner Hans J (USA)	Per-ply	DRAW	3 plies
Berliner Hans J (USA)	Windowed	DRAW	0 plies
Husak Karel (CZE)	Per-ply	DRAW	5 plies
Husak Karel (CZE)	Windowed	DRAW	4 plies

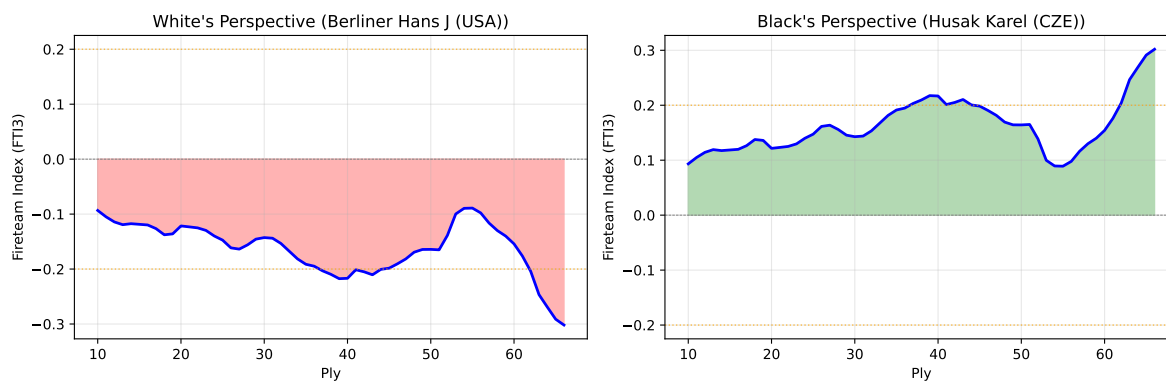


FTI2: Tactical (0.6, 0.1, 0, 0.3)

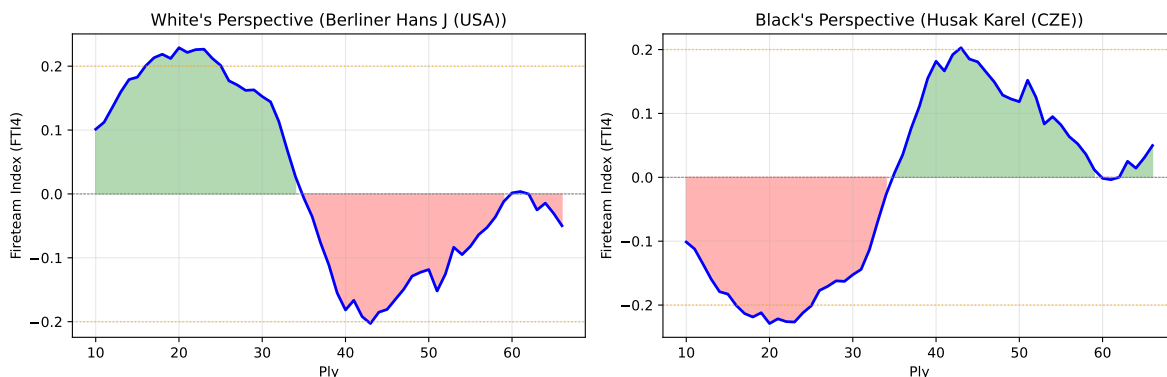
Player	Algorithm	Prediction	Max Streak
Berliner Hans J (USA)	Per-ply	DRAW	0 plies
Berliner Hans J (USA)	Windowed	DRAW	0 plies
Husak Karel (CZE)	Per-ply	DRAW	6 plies
Husak Karel (CZE)	Windowed	WIN	13 plies

**FTI3: Strategic** (0.7, 0.1, 0.1, 0.1)

Player	Algorithm	Prediction	Max Streak
Berliner Hans J (USA)	Per-ply	DRAW	0 plies
Berliner Hans J (USA)	Windowed	DRAW	0 plies
Husak Karel (CZE)	Per-ply	DRAW	7 plies
Husak Karel (CZE)	Windowed	DRAW	8 plies

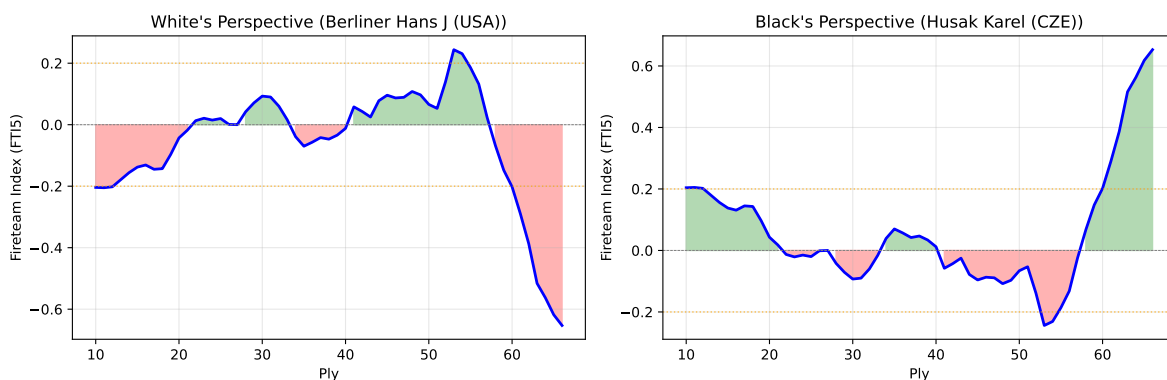
**FTI4: Dynamic** (0.2, 0.6, 0, 0.2)

Player	Algorithm	Prediction	Max Streak
Berliner Hans J (USA)	Per-ply	DRAW	2 plies
Berliner Hans J (USA)	Windowed	DRAW	9 plies
Husak Karel (CZE)	Per-ply	DRAW	2 plies
Husak Karel (CZE)	Windowed	DRAW	1 plies



FTI5: Solid (0.3, 0.2, 0.5, 0)

Player	Algorithm	Prediction	Max Streak
Berliner Hans J (USA)	Per-ply	DRAW	5 plies
Berliner Hans J (USA)	Windowed	DRAW	2 plies
Husak Karel (CZE)	Per-ply	WIN	13 plies
Husak Karel (CZE)	Windowed	DRAW	7 plies



Interpretation. Neither player achieved sustained positional dominance according to FTI1. Both algorithms predicted DRAW for both sides, which matches the actual result. This suggests a balanced game where neither side established lasting control.

8.7 Analysis Information

- Engine: Stockfish 17.1

- Depth: 22
- Analysis time: 270.0 seconds

Chapter 9

Berliner Hans J (USA) vs Hybl Jaroslav (CZE)

9.1 Game Information

Event	Ch World (corr.)
Site	corr.
Date	1965.??.
Round	1
White	Berliner Hans J (USA)
Black	Hybl Jaroslav (CZE) (2172)
Result	1/2-1/2
Opening	D86 – Unknown

9.2 Player Statistics

9.2.1 Berliner Hans J (USA) (White)

- Total moves: 24
- Accuracy: 97.3%
- Average centipawn loss: 5.4
- Best/Excellent moves: 12 / 4
- Good moves: 7
- Inaccuracies: 1
- Mistakes: 0
- Blunders: 0

9.2.2 Hybl Jaroslav (CZE) (Black)

- Total moves: 24
- Accuracy: 97.7%

- Average centipawn loss: 4.7
- Best/Excellent moves: 13 / 4
- Good moves: 6
- Inaccuracies: 0
- Mistakes: 1
- Blunders: 0

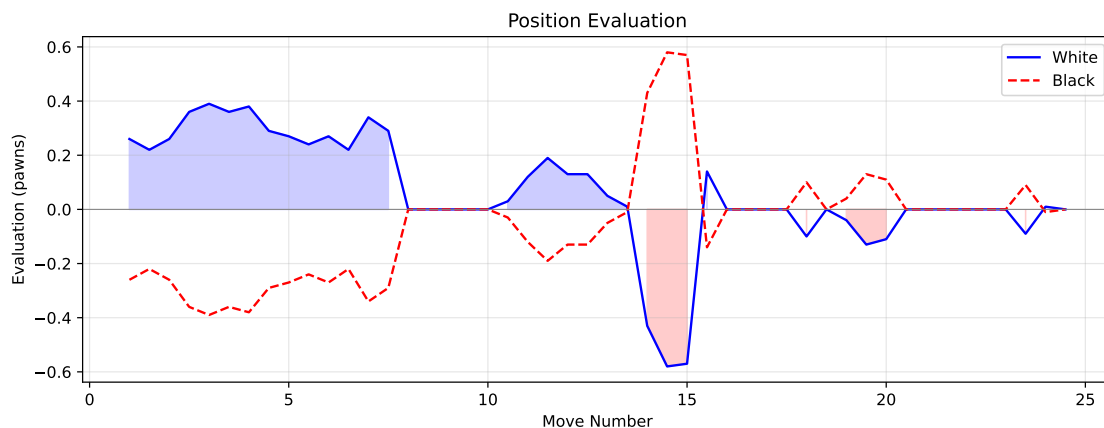
9.3 Positional Analysis

This section shows how key positional factors evolved throughout the game.

9.3.1 Position Evaluation Over Time

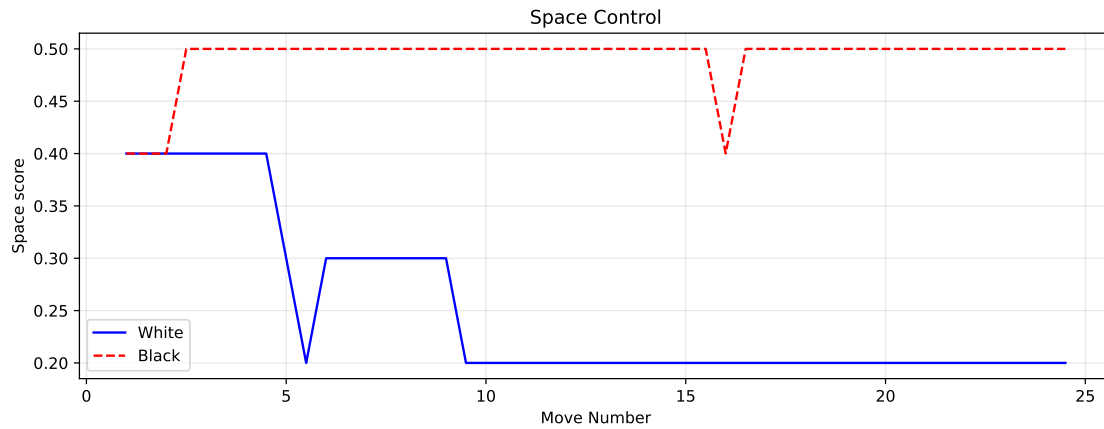
Game Character Classification. Evaluation range: minimum $m_1 = -0.58$, maximum $m_2 = +0.39$, spread $d = m_2 - m_1 = 0.97$.

This game is classified as **balanced** (one-sided). A balanced, one-sided game. Black maintained the advantage throughout.



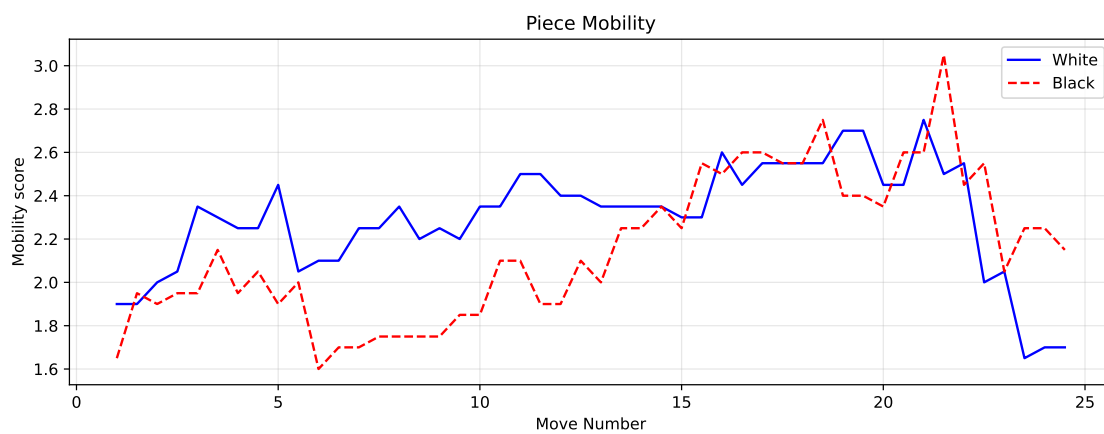
9.3.2 Space Control

- White average: 0.25 (range: 0.20 to 0.40)
- Black average: 0.49 (range: 0.40 to 0.50)



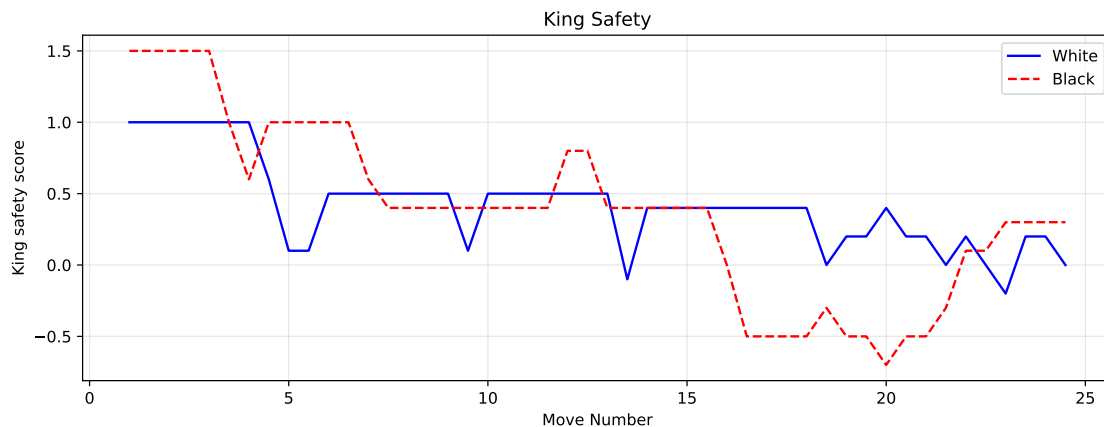
9.3.3 Mobility (Piece Activity)

- White average: 2.29 (range: 1.65 to 2.75)
- Black average: 2.16 (range: 1.60 to 3.05)



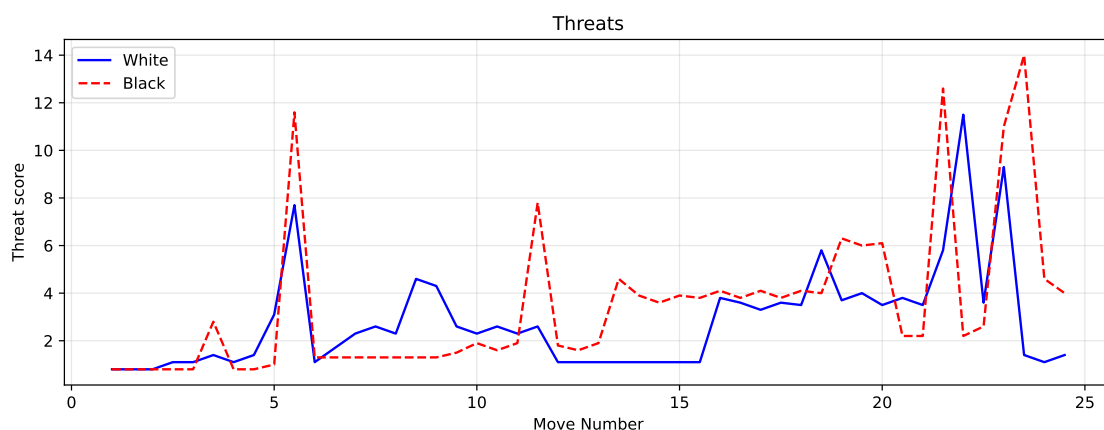
9.3.4 King Safety

- White average: 0.42
- Black average: 0.38



9.3.5 Threats

- White average: 2.82 (range: 0.80 to 11.50)
- Black average: 3.49 (range: 0.80 to 14.00)
- Average advantage: -0.67 (Black maintained more threats)



9.4 Annotated Game

1. d4 Nf6 2. c4 g6 3. Nc3 d5 4. cxd5 Nxd5 5. e4 Nxc3 6. bxc3 Bg7 7. Bc4! O-O! 8. h4 c5! 9. Ne2! cxd4 10. cxd4! Nc6! 11. Be3! Na5 12. Bd3 Nc6 13. e5 Qa5+! 14. Kf1?! Rd8! 15. h5! Be6? *Black loses 68cp. Consider instead: Bxe5, Nxe5, Qa3.*

16. hxc3 hxc3! 17. Rh4! Rd7 18. Qb1 Nxe5 19. dxe5! Rad8! 20. Nd4! Qxe5! 21. Qc2! Rxd4 22. Rxd4! Rxd4! 23. Bxd4! Qxd4! 24. Rd1! Qh4

1/2-1/2

9.5 Critical Positions

This section highlights critical moments where the evaluation shifted substantially—either due to errors or missed opportunities. Each diagram shows the position *after* the move was played.

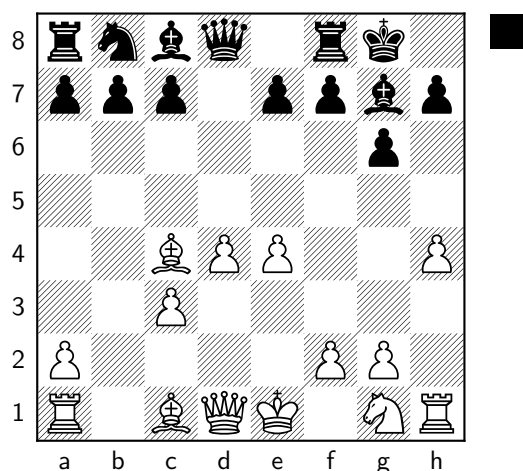
- **Instead of [move]:** The move(s) that should have been played instead.
- **Best continuation:** The optimal sequence of moves from the diagrammed position.

Positions marked with ★ represent the biggest evaluation swings in the game. The positional metrics table summarizes the spatial control, piece activity, king safety, and tactical threats for each side.

Position 1: After 8. h4 ★

Biggest evaluation swing(s) (29cp)

Evaluation: +0.00



Metric	White	Black
Space	0.30	0.50
Mobility	2.35	1.75
King Safety	0.50	0.40
Threats	2.30	1.30

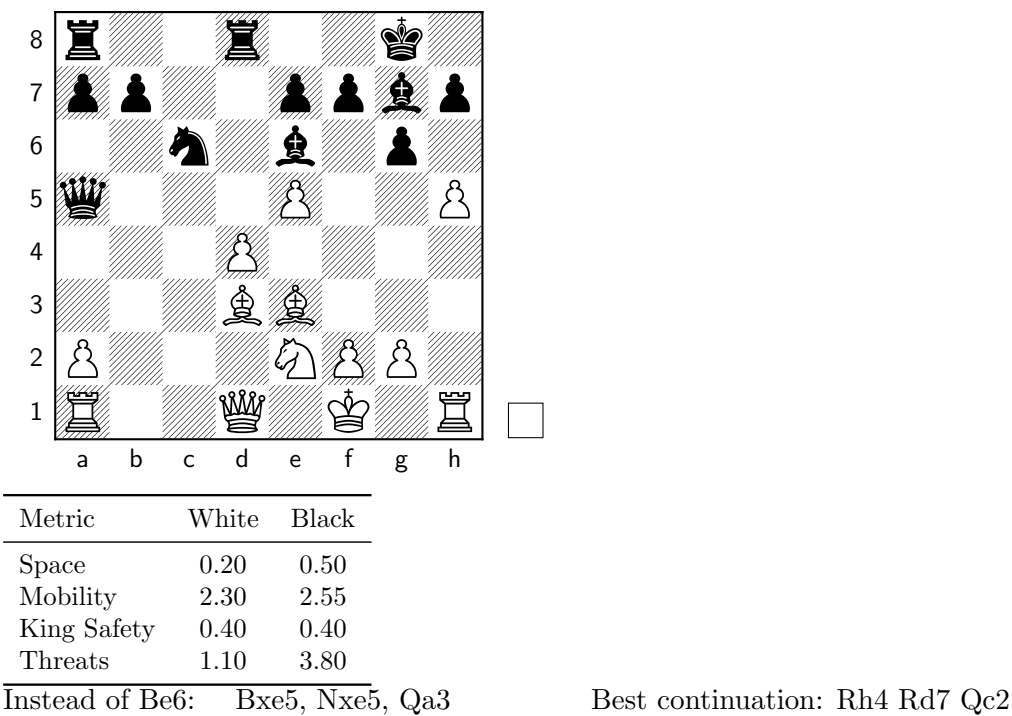
Instead of h4: Ne2, Be3, Nf3

Best continuation: c5 Ne2 cxd4

Position 2: After 15...Be6 ★

Biggest evaluation swing(s) (71cp)

Evaluation: +0.14

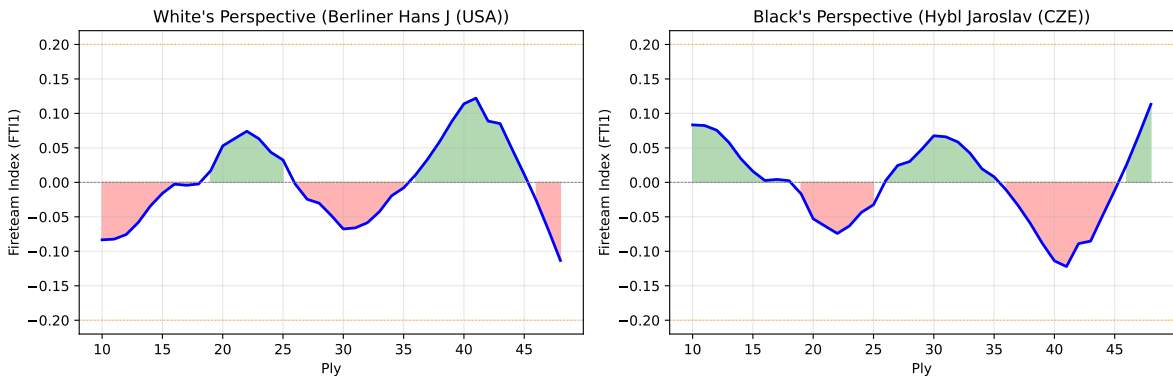


9.6 Fireteam Index Prediction

This game ended in a draw. The Fireteam Index is analyzed from both players' perspectives to determine whether either side achieved sustained positional dominance that might have predicted a win.

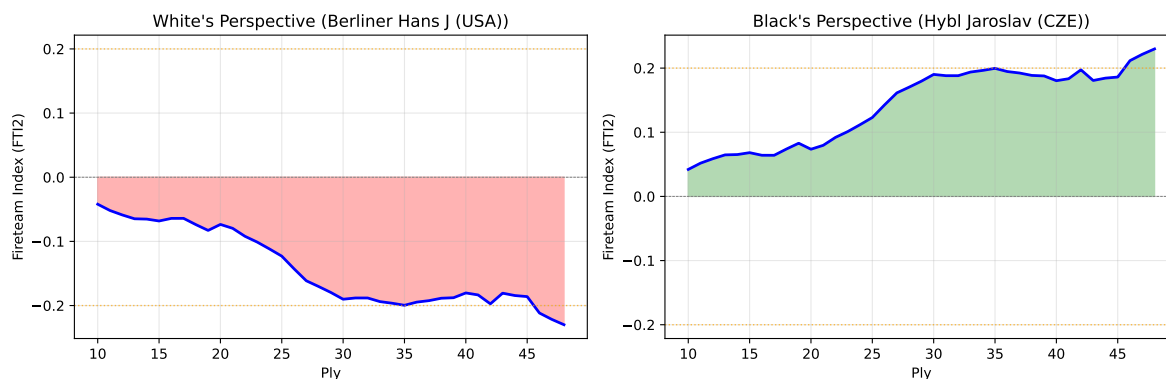
FTI1: Harmonious ($\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}$)

Player	Algorithm	Prediction	Max Streak
Berliner Hans J (USA)	Per-ply	DRAW	0 plies
Berliner Hans J (USA)	Windowed	DRAW	0 plies
Hybl Jaroslav (CZE)	Per-ply	DRAW	5 plies
Hybl Jaroslav (CZE)	Windowed	DRAW	0 plies

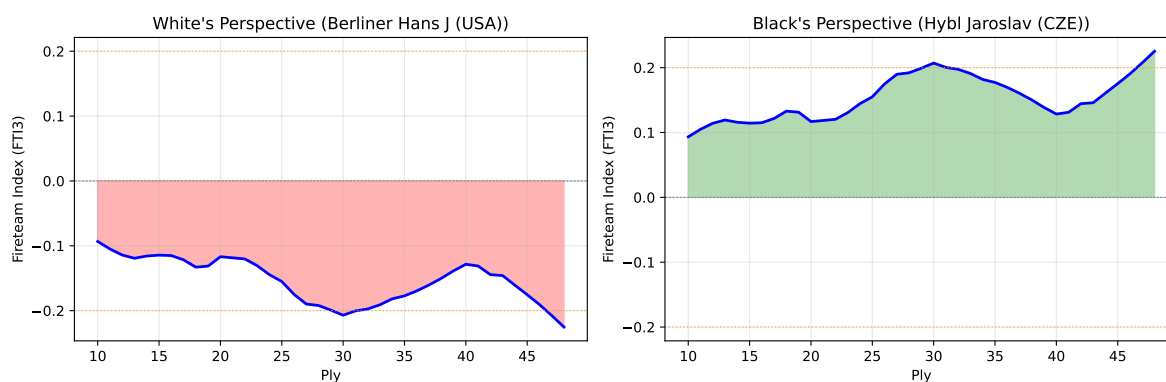


FTI2: Tactical (0.6, 0.1, 0, 0.3)

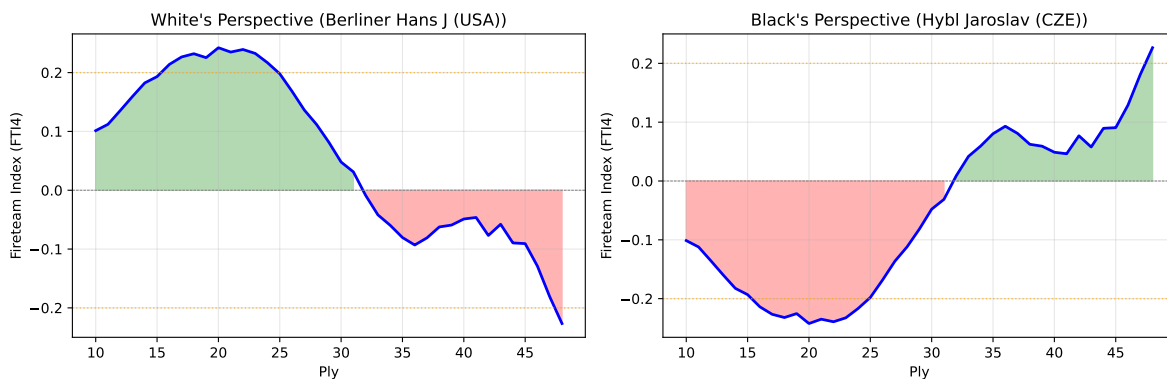
Player	Algorithm	Prediction	Max Streak
Berliner Hans J (USA)	Per-ply	DRAW	0 plies
Berliner Hans J (USA)	Windowed	DRAW	0 plies
Hybl Jaroslav (CZE)	Per-ply	DRAW	5 plies
Hybl Jaroslav (CZE)	Windowed	DRAW	3 plies

**FTI3: Strategic** (0.7, 0.1, 0.1, 0.1)

Player	Algorithm	Prediction	Max Streak
Berliner Hans J (USA)	Per-ply	DRAW	0 plies
Berliner Hans J (USA)	Windowed	DRAW	0 plies
Hybl Jaroslav (CZE)	Per-ply	DRAW	5 plies
Hybl Jaroslav (CZE)	Windowed	DRAW	2 plies

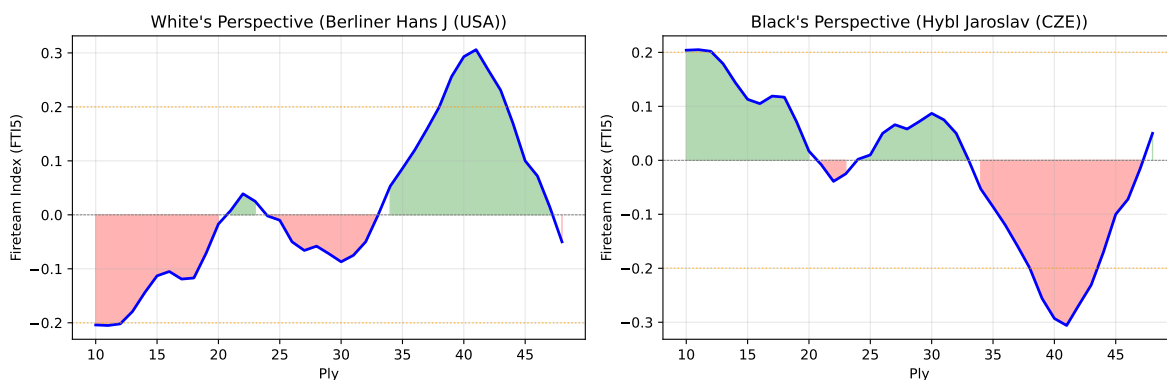
**FTI4: Dynamic** (0.2, 0.6, 0, 0.2)

Player	Algorithm	Prediction	Max Streak
Berliner Hans J (USA)	Per-ply	DRAW	2 plies
Berliner Hans J (USA)	Windowed	DRAW	8 plies
Hybl Jaroslav (CZE)	Per-ply	DRAW	3 plies
Hybl Jaroslav (CZE)	Windowed	DRAW	1 plies



FTI5: Solid (0.3, 0.2, 0.5, 0)

Player	Algorithm	Prediction	Max Streak
Berliner Hans J (USA)	Per-ply	DRAW	5 plies
Berliner Hans J (USA)	Windowed	DRAW	6 plies
Hybl Jaroslav (CZE)	Per-ply	DRAW	5 plies
Hybl Jaroslav (CZE)	Windowed	DRAW	0 plies



Interpretation. Neither player achieved sustained positional dominance according to FTI1. Both algorithms predicted DRAW for both sides, which matches the actual result. This suggests a balanced game where neither side established lasting control.

9.7 Analysis Information

- Engine: Stockfish 17.1

- Depth: 22
- Analysis time: 199.7 seconds

Chapter 10

Hans Berliner vs Kenneth Charles Messere

10.1 Game Information

Event	5th Correspondence World Championship
Site	ICCF
Date	1965.07.01
Round	13
White	Hans Berliner
Black	Kenneth Charles Messere
Result	1-0
Opening	D86 – Unknown

10.2 Player Statistics

10.2.1 Hans Berliner (White)

- Total moves: 22
- Accuracy: 89.6%
- Average centipawn loss: 22.0
- Best/Excellent moves: 12 / 1
- Good moves: 6
- Inaccuracies: 1
- Mistakes: 0
- Blunders: 2

10.2.2 Kenneth Charles Messere (Black)

- Total moves: 21
- Accuracy: 81.4%

- Average centipawn loss: 41.2
- Best/Excellent moves: 6 / 2
- Good moves: 7
- Inaccuracies: 0
- Mistakes: 3
- Blunders: 3

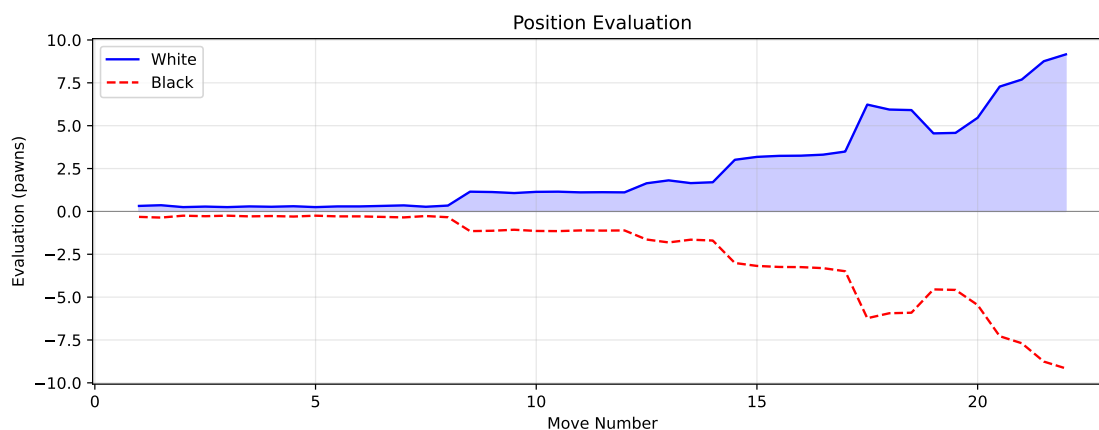
10.3 Positional Analysis

This section shows how key positional factors evolved throughout the game.

10.3.1 Position Evaluation Over Time

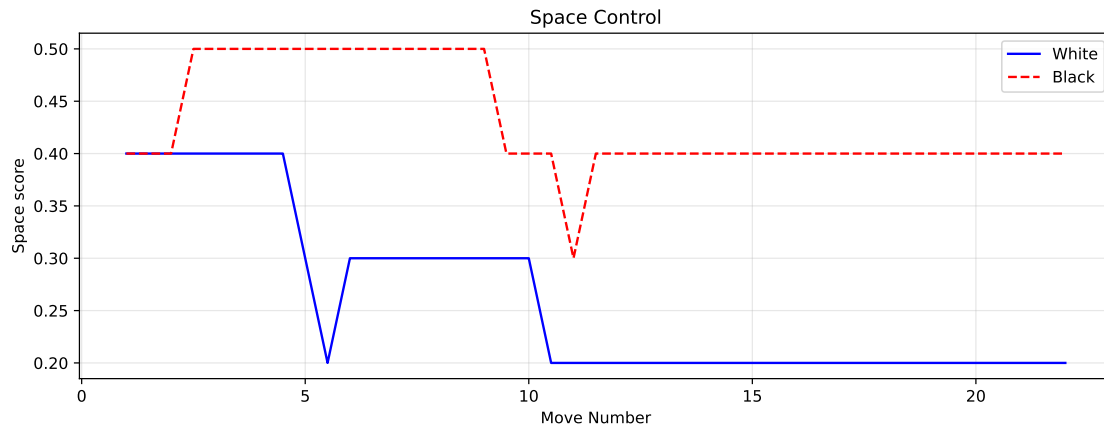
Game Character Classification. Evaluation range: minimum $m_1 = +0.25$, maximum $m_2 = +9.16$, spread $d = m_2 - m_1 = 8.91$.

This game is classified as **chaotic** (one-sided). A chaotic, one-sided game. White maintained the advantage throughout.



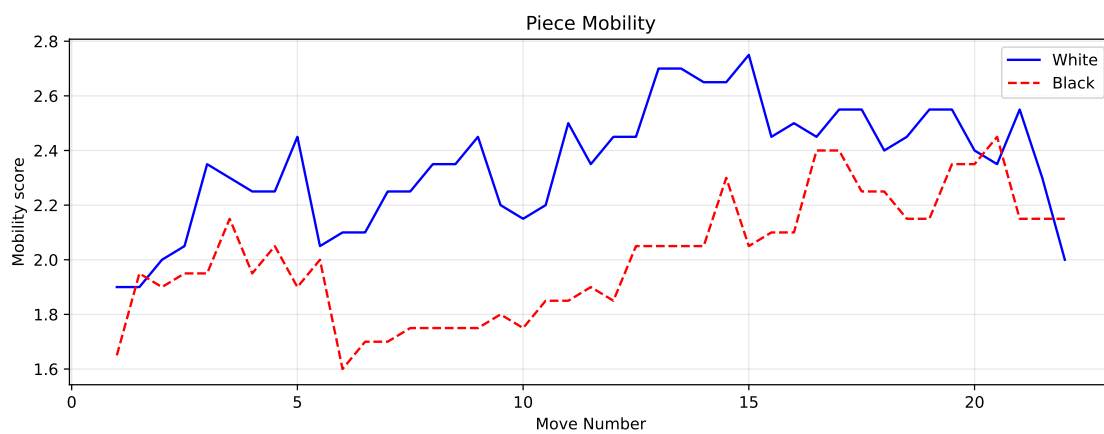
10.3.2 Space Control

- White average: 0.26 (range: 0.20 to 0.40)
- Black average: 0.43 (range: 0.30 to 0.50)



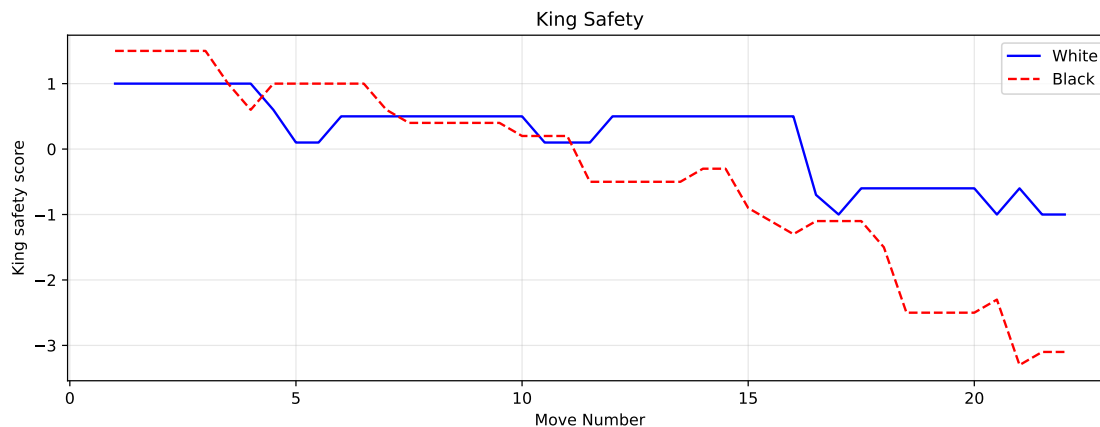
10.3.3 Mobility (Piece Activity)

- White average: 2.35 (range: 1.90 to 2.75)
- Black average: 2.01 (range: 1.60 to 2.45)



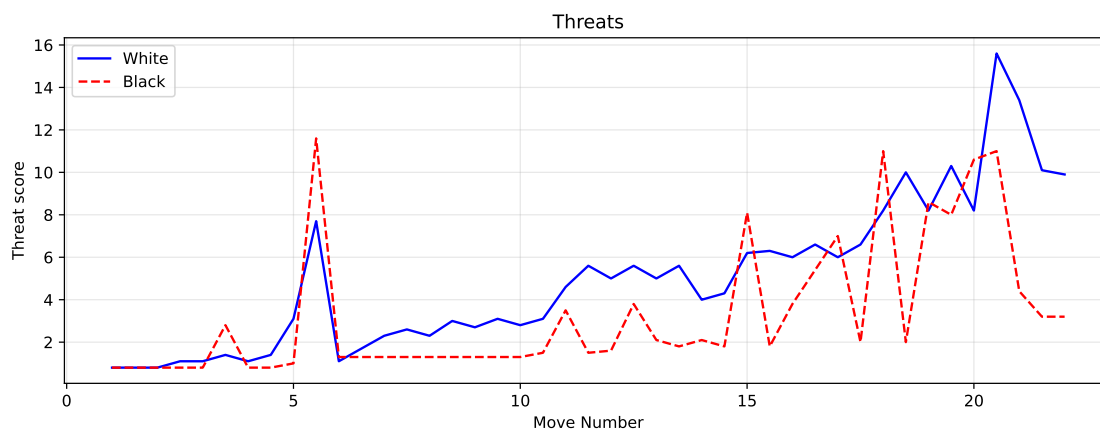
10.3.4 King Safety

- White average: 0.19
- Black average: -0.37



10.3.5 Threats

- White average: 5.01 (range: 0.80 to 15.60)
- Black average: 3.31 (range: 0.80 to 11.60)
- Average advantage: 1.69 (White maintained more threats)



10.4 Annotated Game

1. d4 Nf6 2. c4 g6 3. Nc3 d5 4. cxd5 Nxd5 5. e4 Nxc3 6. bxc3 Bg7 7. Bc4 O-O! 8. Ne2! b6? *Black loses 85cp. Consider instead: c5, Nc6, Re8.*
9. h4! e5! 10. h5! exd4! 11. hxg6! hxg6! 12. cxd4 Qe7? *Black loses 63cp. Consider instead: b5.*
13. Qd3! Re8 14. Bh6! Nc6?? *Black loses 133cp. Consider instead: Ba6, Bxh6, Be6.*
15. Bxg7! Kxg7 16. Qd2! Qxe4 17. O-O-O! Qe7?? *Black loses 247cp. Consider instead: Nb4, Qf5, Bf5.*
18. Qh6+?! Kf6 19. Rd3?? *White loses 162cp. Consider instead: Qh4+.*
- 19...Bg4! 20. Rhh3 Nxd4?? *Black loses 167cp. Consider instead: Qe4, Qd7.*
21. Rxd4! Bxe2? *Black loses 111cp. Consider instead: Qe5, Bf5, Qe6.*

22. Rf4+?? White loses 238cp. Consider instead: Qf4+.

1-0

10.5 Critical Positions

This section highlights critical moments where the evaluation shifted substantially—either due to errors or missed opportunities. Each diagram shows the position *after* the move was played.

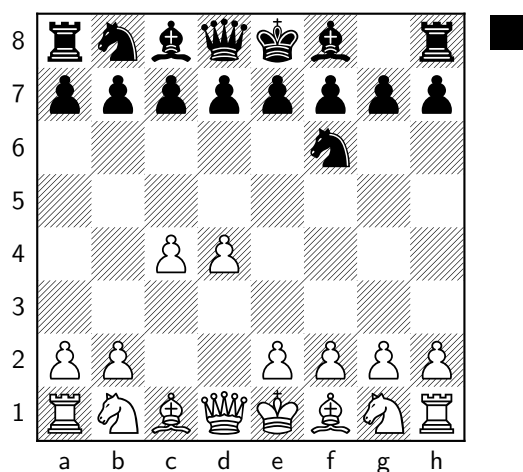
- **Instead of [move]:** The move(s) that should have been played instead.
- **Best continuation:** The optimal sequence of moves from the diagrammed position.

Positions marked with ★ represent the biggest evaluation swings in the game. The positional metrics table summarizes the spatial control, piece activity, king safety, and tactical threats for each side.

Position 1: After 2. c4 ★

Biggest evaluation swing(s) (11cp)

Evaluation: +0.25



Metric	White	Black
Space	0.40	0.40
Mobility	2.00	1.90
King Safety	1.00	1.50
Threats	0.80	0.80

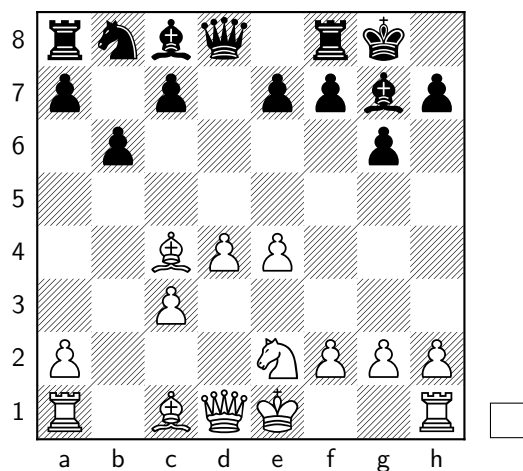
Instead of c4: Nf3, e3

Best continuation: g6 Nf3 Bg7

Position 2: After 8...b6 ★

Biggest evaluation swing(s) (81cp)

Evaluation: +1.15



Metric	White	Black
Space	0.30	0.50
Mobility	2.35	1.75
King Safety	0.50	0.40
Threats	3.00	1.30

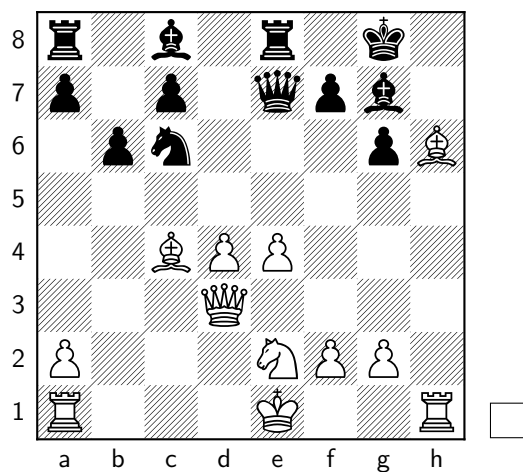
Instead of b6: c5, Nc6, Re8

Best continuation: h4 Ba6 Bxa6

Position 3: After 14...Nc6

Black blunders (+131cp)

Evaluation: +3.01



Metric	White	Black
Space	0.20	0.40
Mobility	2.65	2.30
King Safety	0.50	-0.30
Threats	4.30	1.80

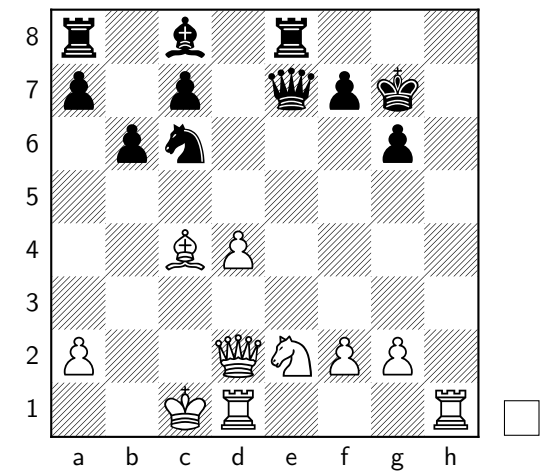
Instead of Nc6: Ba6, Bxh6, Be6

Best continuation: Bxg7 Kxg7 Qd2

Position 4: After 17...Qe7

Black blunders (+274cp)

Evaluation: +6.23



Metric	White	Black
Space	0.20	0.40
Mobility	2.55	2.25
King Safety	-0.60	-1.10
Threats	6.60	2.00

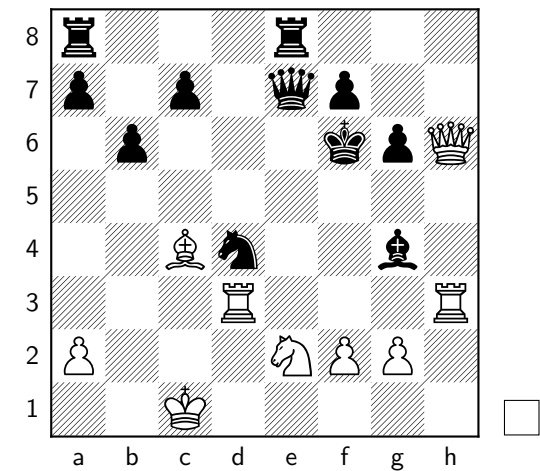
Instead of Qe7: Nb4, Qf5, Bf5

Best continuation: Qh6+ Kf6 Qh4+

Position 5: After 20...Nxd4

Black blunders (+182cp)

Evaluation: +7.28



Metric	White	Black
Space	0.20	0.40
Mobility	2.35	2.45
King Safety	-1.00	-2.30
Threats	15.60	11.00

Instead of Nxd4: Qe4, Qd7

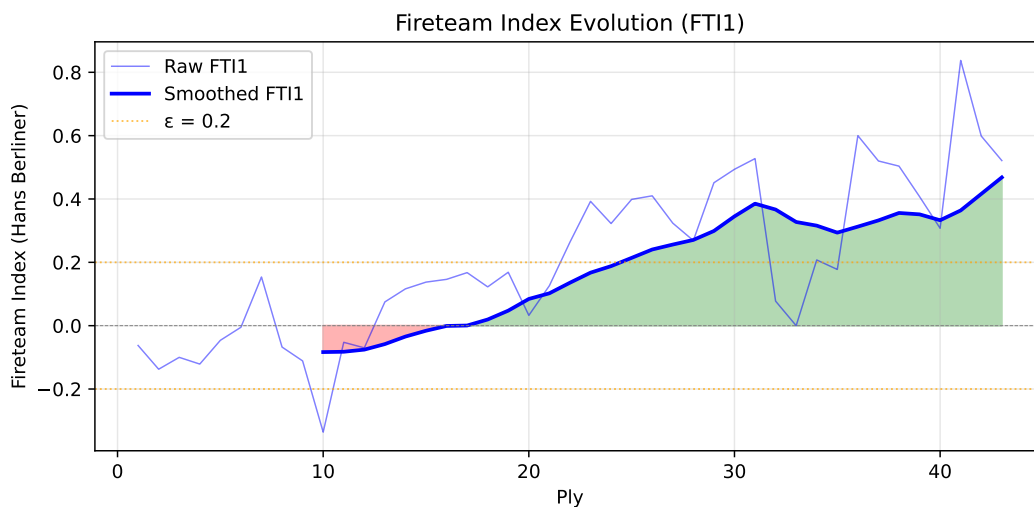
Best continuation: Qh4+ Kg7 Qh6+

10.6 Fireteam Index Prediction

Tracking: **Hans Berliner** (White)

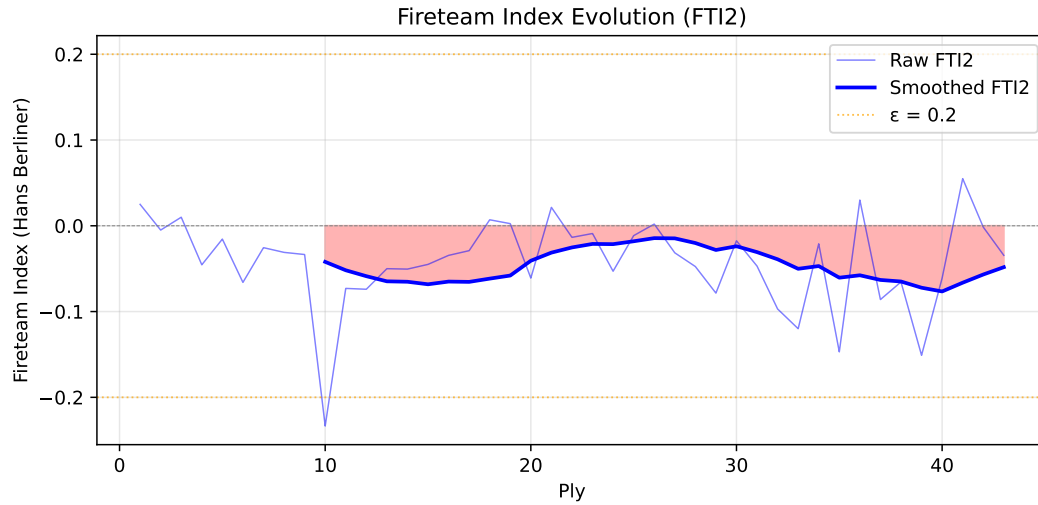
FTI1: Harmonious ($\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}$)

Algorithm	Prediction	Max Streak / Confidence
Per-ply	WIN	10 plies
Windowed	WIN	17 plies
Balance	WIN	$B_n=+0.70$, $C=0.70$



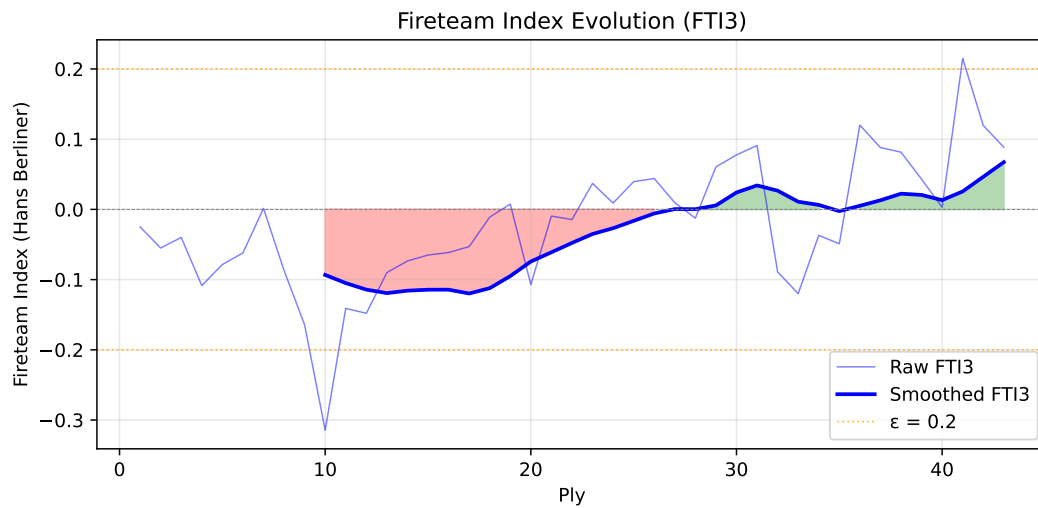
FTI2: Tactical (0.6, 0.1, 0, 0.3)

Algorithm	Prediction	Max Streak / Confidence
Per-ply	DRAW	0 plies
Windowed	DRAW	0 plies
Balance	DRAW	$B_n=+0.00$, $C=0.00$



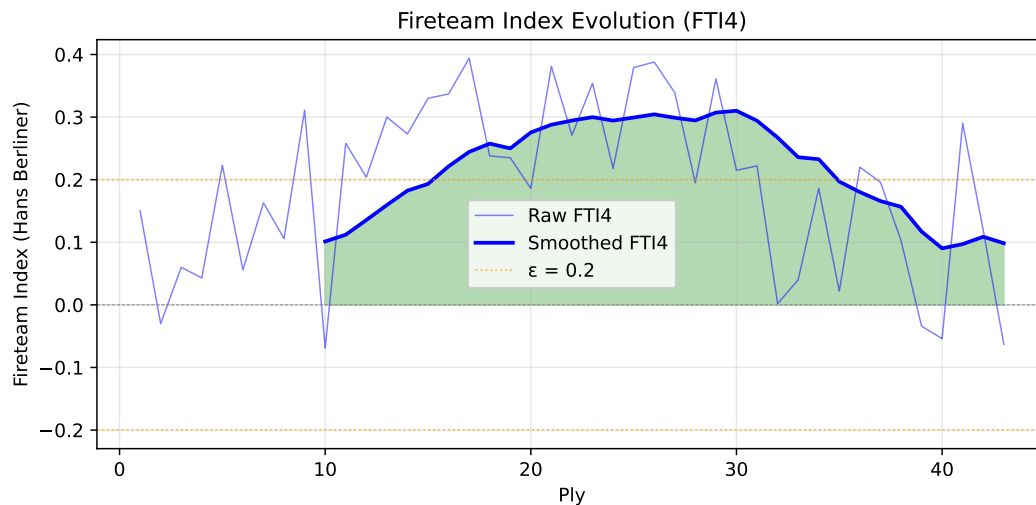
FTI3: Strategic (0.7, 0.1, 0.1)

Algorithm	Prediction	Max Streak / Confidence
Per-ply	DRAW	0 plies
Windowed	DRAW	0 plies
Balance	DRAW	$B_n = +0.04$, $C = 0.04$



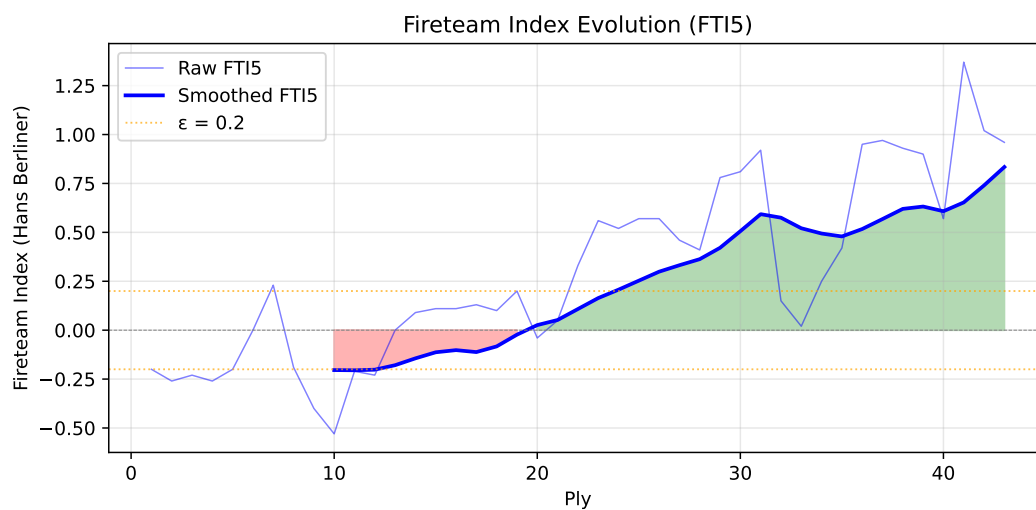
FTI4: Dynamic (0.2, 0.6, 0, 0.2)

Algorithm	Prediction	Max Streak / Confidence
Per-ply	DRAW	3 plies
Windowed	WIN	13 plies
Balance	WIN	$B_n = +0.56$, $C = 0.56$



FTI5: Solid (0.3, 0.2, 0.5, 0)

Algorithm	Prediction	Max Streak / Confidence
Per-ply	WIN	10 plies
Windowed	WIN	19 plies
Balance	WIN	$B_n=+0.74$, $C=0.74$



Actual result: 1-0 (Hans Berliner's outcome: WIN)

FTI1: Per-ply	✓	Windowed	✓	Balance	✓
FTI2: Per-ply	✗	Windowed	✗	Balance	✗
FTI3: Per-ply	✗	Windowed	✗	Balance	✗
FTI4: Per-ply	✗	Windowed	✓	Balance	✓
FTI5: Per-ply	✓	Windowed	✓	Balance	✓

10.7 Analysis Information

- Engine: Stockfish 17.1
- Depth: 22
- Analysis time: 509.0 seconds

Chapter 11

Hans Berliner vs Julius Nielsen

11.1 Game Information

Event	5th Correspondence World Championship
Site	ICCF
Date	1965.07.01
Round	7
White	Hans Berliner
Black	Julius Nielsen
Result	1-0
Opening	D35 – Unknown

11.2 Player Statistics

11.2.1 Hans Berliner (White)

- Total moves: 48
- Accuracy: 94.0%
- Average centipawn loss: 12.4
- Best/Excellent moves: 22 / 10
- Good moves: 9
- Inaccuracies: 3
- Mistakes: 4
- Blunders: 0

11.2.2 Julius Nielsen (Black)

- Total moves: 47
- Accuracy: 90.9%
- Average centipawn loss: 19.2

- Best/Excellent moves: 12 / 13
- Good moves: 11
- Inaccuracies: 9
- Mistakes: 1
- Blunders: 1

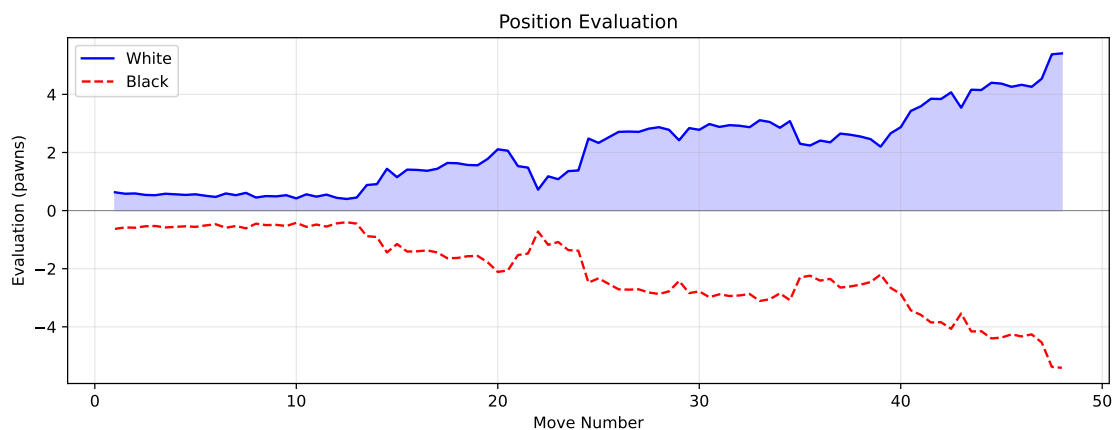
11.3 Positional Analysis

This section shows how key positional factors evolved throughout the game.

11.3.1 Position Evaluation Over Time

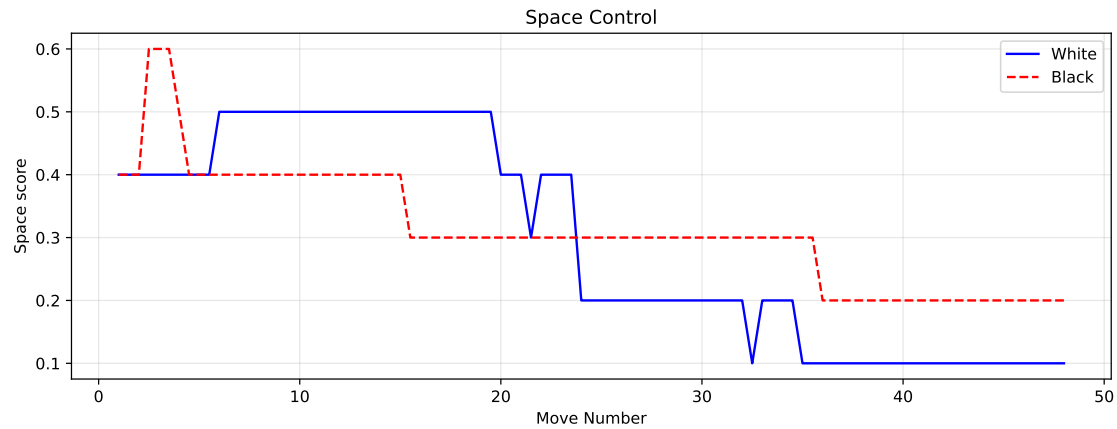
Game Character Classification. Evaluation range: minimum $m_1 = +0.40$, maximum $m_2 = +5.41$, spread $d = m_2 - m_1 = 5.01$.

This game is classified as **tactical** (one-sided). A tactical, one-sided game. White maintained the advantage throughout.



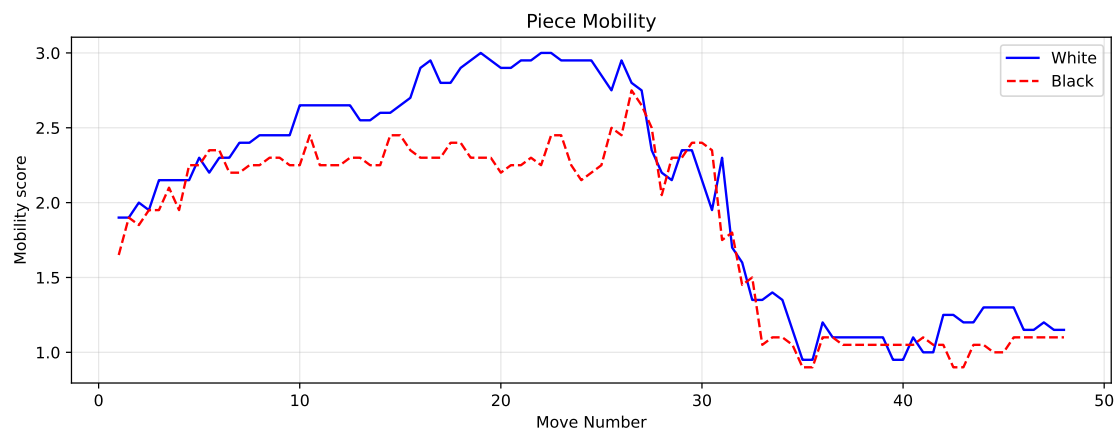
11.3.2 Space Control

- White average: 0.30 (range: 0.10 to 0.50)
- Black average: 0.31 (range: 0.20 to 0.60)



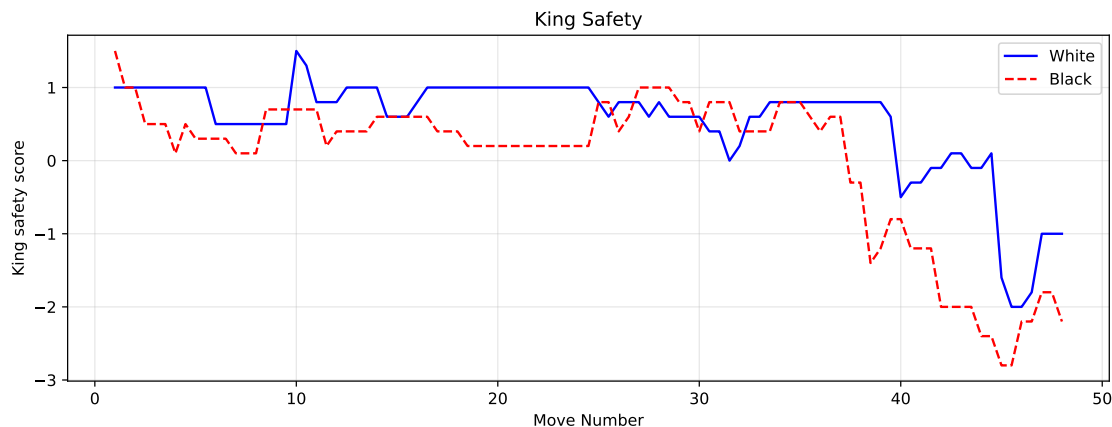
11.3.3 Mobility (Piece Activity)

- White average: 2.07 (range: 0.95 to 3.00)
- Black average: 1.84 (range: 0.90 to 2.75)



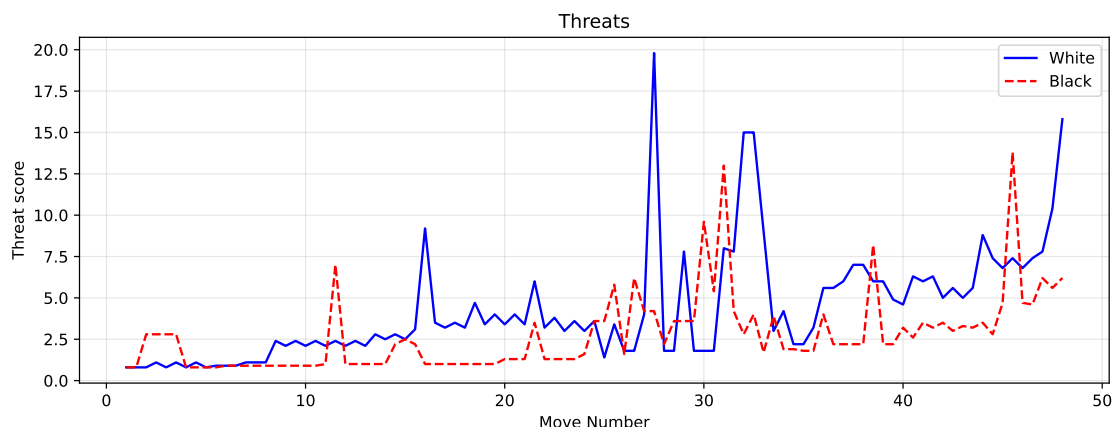
11.3.4 King Safety

- White average: 0.54
- Black average: 0.01



11.3.5 Threats

- White average: 4.33 (range: 0.80 to 19.80)
- Black average: 2.78 (range: 0.80 to 13.80)
- Average advantage: 1.55 (White maintained more threats)



11.4 Annotated Game

1. d4 d5 2. c4 e6 3. Nc3 Nf6 4. cxd5 exd5 5. Bg5 c6 6. e3 Nbd7 7. Bd3! Be7 8. Nge2 O-O 9. Qc2 Re8 10. O-O Nf8 11. f3 h6 12. Bh4 Ne6! 13. Rad1 b6?! 14. Bf2! Qc7?! 15. Kh1?! c5 16. Bg3! Qb7 17. Be5! Rd8?! 18. Ng3! Ne8! 19. Nf5 Bf6 20. f4! Nf8 21. Be2? *White loses 72cp. Consider instead: g4.*

21...cxd4 22. Bxd4? *White loses 72cp. Consider instead: exd4, Nxd4.*

22...Be6?! 23. Bf3 Qc8 24. e4 Ng6?? *Black loses 128cp. Consider instead: Kh8.*

25. Ne3?! dxe4?! 26. Qxe4! Rxd4 27. Qxa8 Qxa8! 28. Bxa8! Nxf4! 29. Bc6?! Nc7?! 30. Ng4! Bxg4! 31. Rxd4! Bxd4 32. Rxf4 Bxc3! 33. bxc3! Be6 34. Ra4 a5 35. c4? *White loses 76cp. Consider instead: Kg1, a3, Rd4.*

35...Kf8! 36. c5! bxc5! 37. Rxa5 Ke7 38. Kg1! Kd6 39. Bf3 Nd5?! 40. Kf2 Nb4?! 41. a4! Bb3 42. Ra8 c4 43. a5? *White loses 68cp.*

43...c3 44. Rc8 c2 45. Ke3! Be6?! 46. Rc3! Bf5 47. Kd2! h5? *Black loses 104cp.*

Consider instead: Bd3, Be6.

48. Rc4!

1-0

11.5 Critical Positions

This section highlights critical moments where the evaluation shifted substantially—either due to errors or missed opportunities. Each diagram shows the position *after* the move was played.

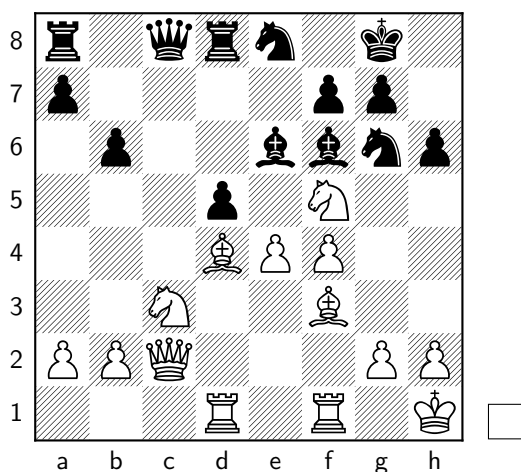
- **Instead of [move]:** The move(s) that should have been played instead.
- **Best continuation:** The optimal sequence of moves from the diagrammed position.

Positions marked with ★ represent the biggest evaluation swings in the game. The positional metrics table summarizes the spatial control, piece activity, king safety, and tactical threats for each side.

Position 1: After 24...Ng6

Black blunders (+110cp)

Evaluation: +2.48



Metric	White	Black
Space	0.20	0.30
Mobility	2.95	2.20
King Safety	1.00	0.20
Threats	3.60	3.60

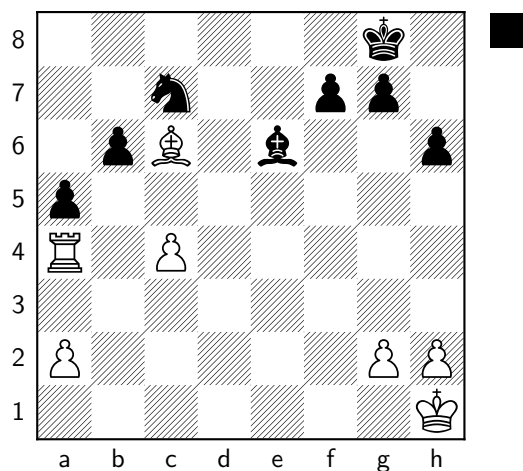
Instead of Ng6: Kh8

Best continuation: Nxf6+ gxf6 f5

Position 2: After 35. c4 ★

Biggest evaluation swing(s) (78cp)

Evaluation: +2.30



Metric	White	Black
Space	0.10	0.30
Mobility	0.95	0.90
King Safety	0.80	0.80
Threats	2.20	1.80

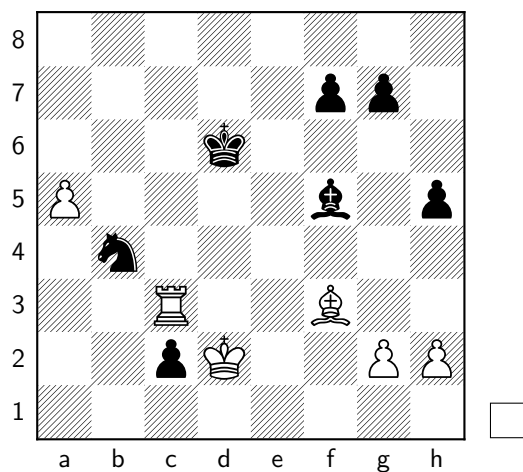
Instead of c4: Kg1, a3, Rd4

Best continuation: Kf8 c5 bxc5

Position 3: After 47...h5 ★

Biggest evaluation swing(s) (84cp)

Evaluation: +5.38



Metric	White	Black
Space	0.10	0.20
Mobility	1.15	1.10
King Safety	-1.00	-1.80
Threats	10.40	5.60

Instead of h5: Bd3, Be6

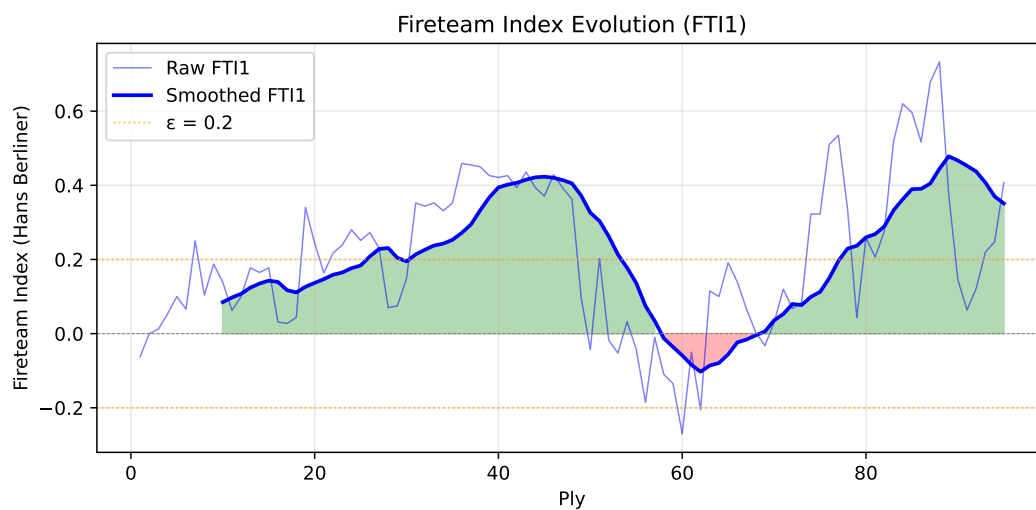
Best continuation: Rc4 c1=Q+ Rxc1

11.6 Fireteam Index Prediction

Tracking: **Hans Berliner** (White)

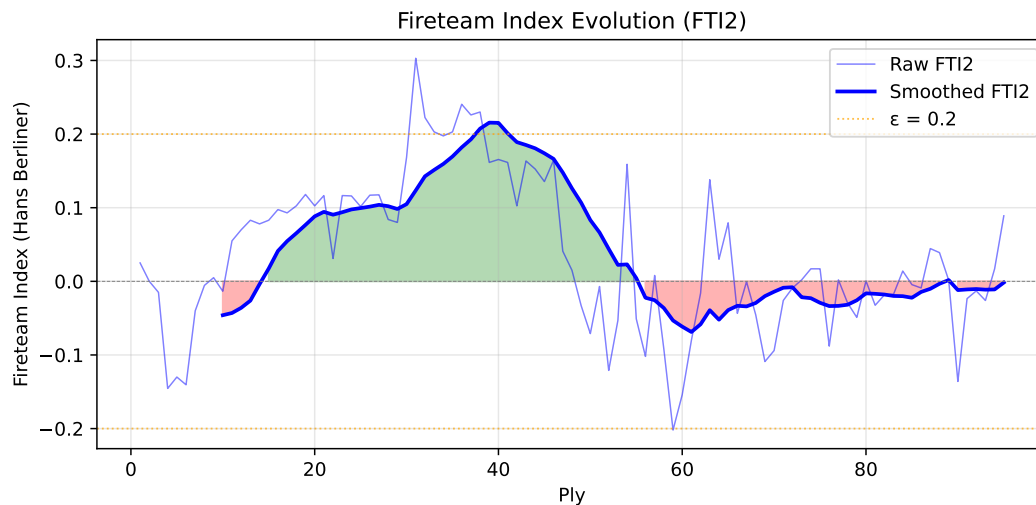
FTI1: Harmonious ($\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}$)

Algorithm	Prediction	Max Streak / Confidence
Per-ply	WIN	18 plies
Windowed	WIN	18 plies
Balance	WIN	$B_n=+0.54$, $C=0.48$



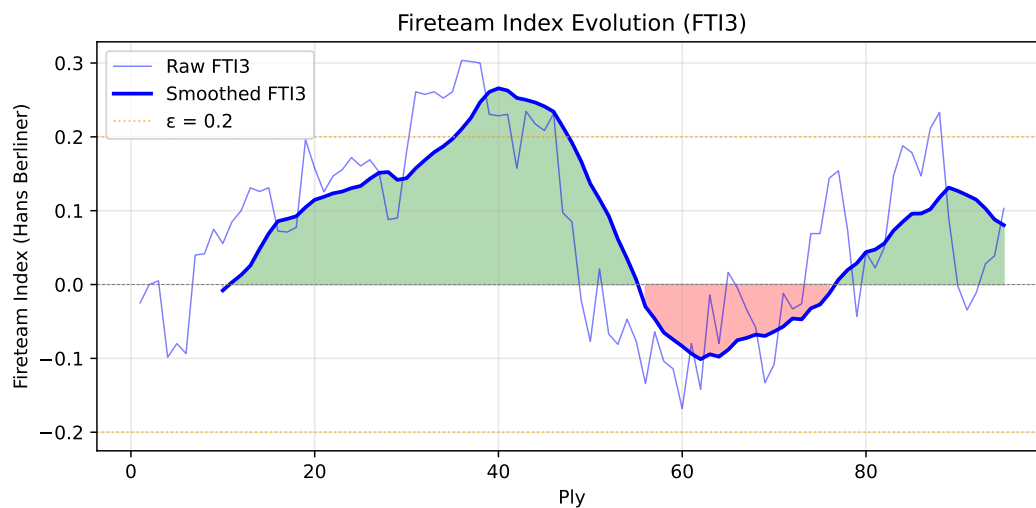
FTI2: Tactical (0.6, 0.1, 0, 0.3)

Algorithm	Prediction	Max Streak / Confidence
Per-ply	DRAW	1 plies
Windowed	DRAW	0 plies
Balance	DRAW	$B_n=+0.08$, $C=0.07$



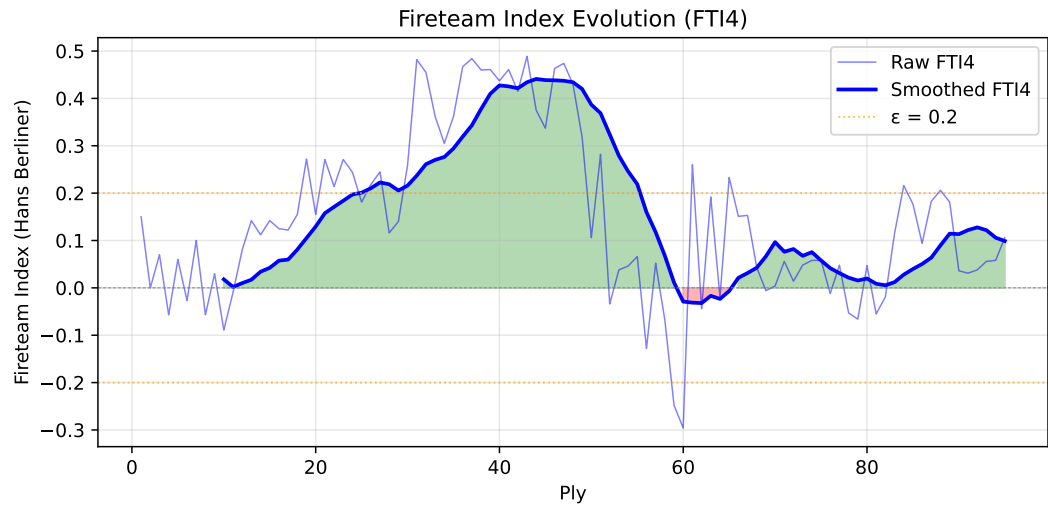
FTI3: Strategic (0.7, 0.1, 0.1, 0.1)

Algorithm	Prediction	Max Streak / Confidence
Per-ply	DRAW	8 plies
Windowed	DRAW	5 plies
Balance	WIN	$B_n=+0.22$, $C=0.22$



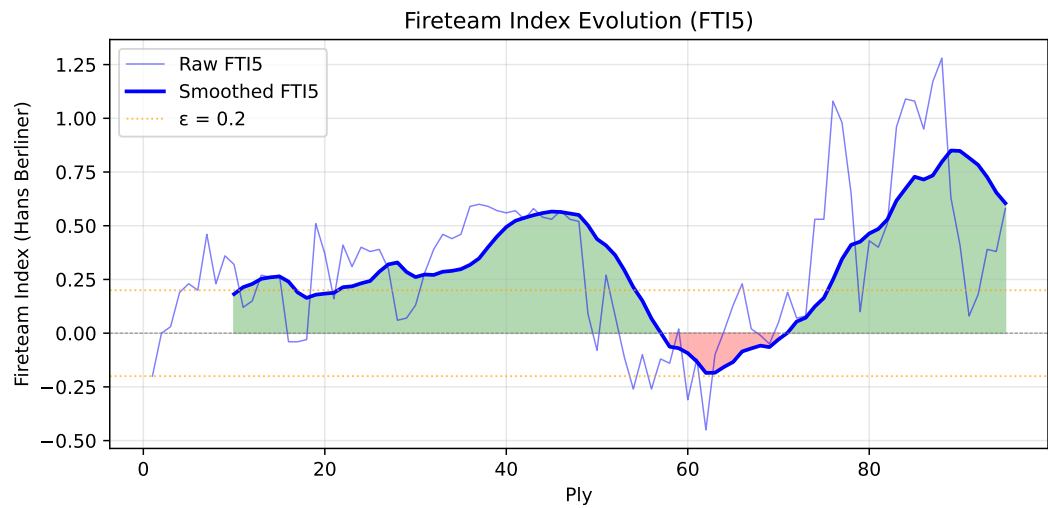
FTI4: Dynamic (0.2, 0.6, 0, 0.2)

Algorithm	Prediction	Max Streak / Confidence
Per-ply	WIN	20 plies
Windowed	WIN	22 plies
Balance	WIN	$B_n=+0.38$, $C=0.33$



FTI5: Solid (0.3,0.2,0.5,0)

Algorithm	Prediction	Max Streak / Confidence
Per-ply	WIN	18 plies
Windowed	WIN	28 plies
Balance	WIN	$B_n=+0.54$, $C=0.48$



Actual result: 1-0 (Hans Berliner’s outcome: WIN)

FTI1:	Per-ply ✓	Windowed ✓	Balance ✓
FTI2:	Per-ply ✗	Windowed ✗	Balance ✗
FTI3:	Per-ply ✗	Windowed ✗	Balance ✓
FTI4:	Per-ply ✓	Windowed ✓	Balance ✓
FTI5:	Per-ply ✓	Windowed ✓	Balance ✓

11.7 Analysis Information

- Engine: Stockfish 17.1
- Depth: 22
- Analysis time: 863.0 seconds

Chapter 12

Hans Berliner vs Sture Valentin Nyman

12.1 Game Information

Event	5th Correspondence World Championship
Site	ICCF
Date	1965.07.01
Round	15
White	Hans Berliner
Black	Sture Valentin Nyman
Result	1-0
Opening	A56 – Unknown

12.2 Player Statistics

12.2.1 Hans Berliner (White)

- Total moves: 46
- Accuracy: 94.3%
- Average centipawn loss: 11.8
- Best/Excellent moves: 18 / 12
- Good moves: 9
- Inaccuracies: 6
- Mistakes: 1
- Blunders: 0

12.2.2 Sture Valentin Nyman (Black)

- Total moves: 45
- Accuracy: 91.6%

- Average centipawn loss: 17.6
- Best/Excellent moves: 15 / 6
- Good moves: 14
- Inaccuracies: 8
- Mistakes: 2
- Blunders: 0

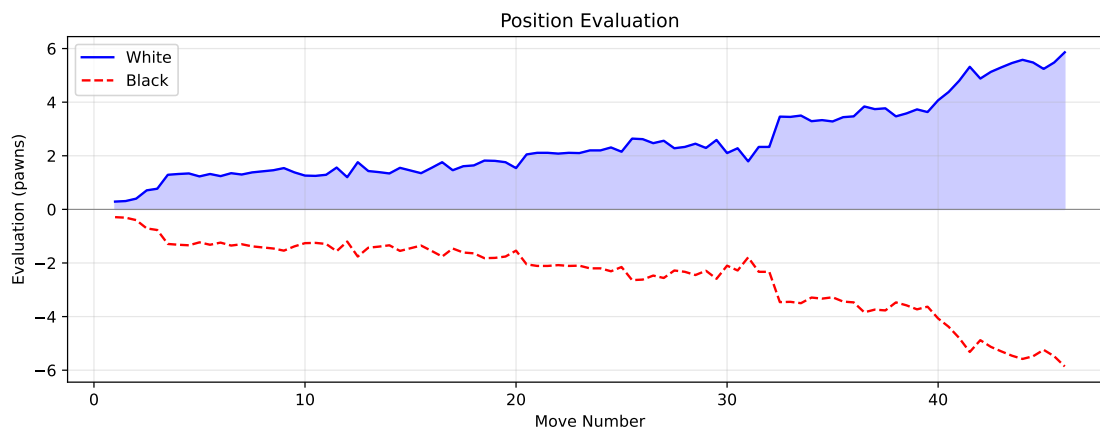
12.3 Positional Analysis

This section shows how key positional factors evolved throughout the game.

12.3.1 Position Evaluation Over Time

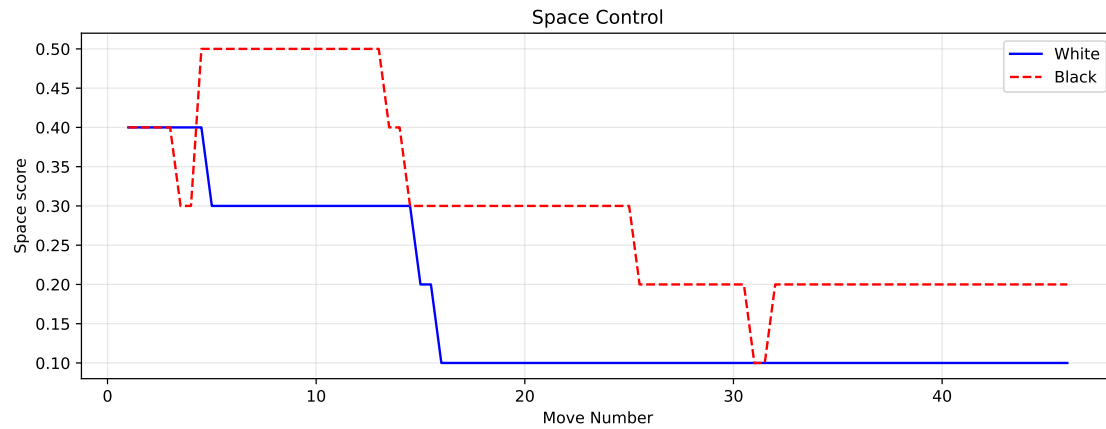
Game Character Classification. Evaluation range: minimum $m_1 = +0.29$, maximum $m_2 = +5.86$, spread $d = m_2 - m_1 = 5.57$.

This game is classified as **tactical** (one-sided). A tactical, one-sided game. White maintained the advantage throughout.



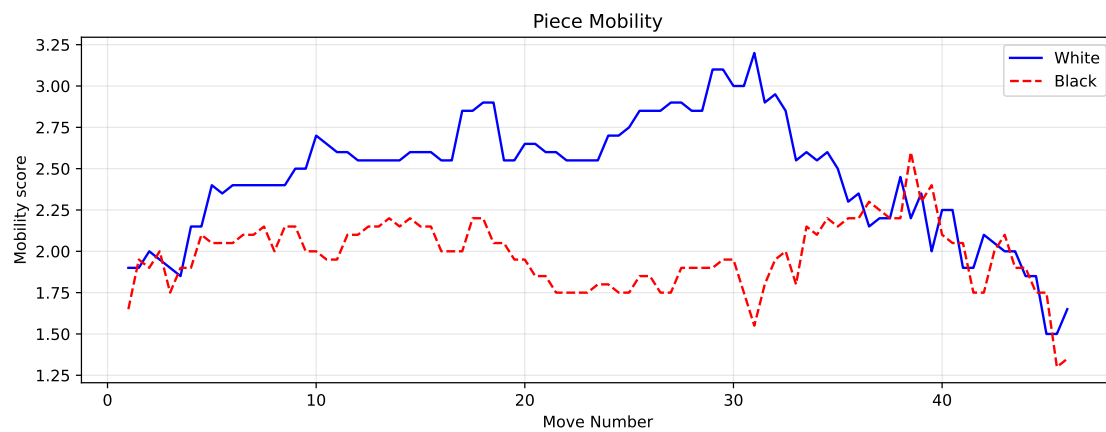
12.3.2 Space Control

- White average: 0.17 (range: 0.10 to 0.40)
- Black average: 0.30 (range: 0.10 to 0.50)



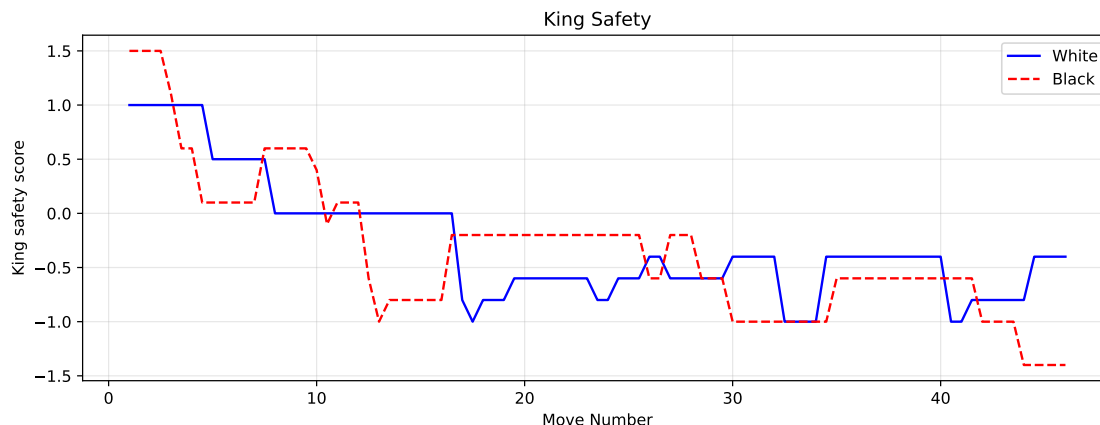
12.3.3 Mobility (Piece Activity)

- White average: 2.46 (range: 1.50 to 3.20)
- Black average: 1.98 (range: 1.30 to 2.60)



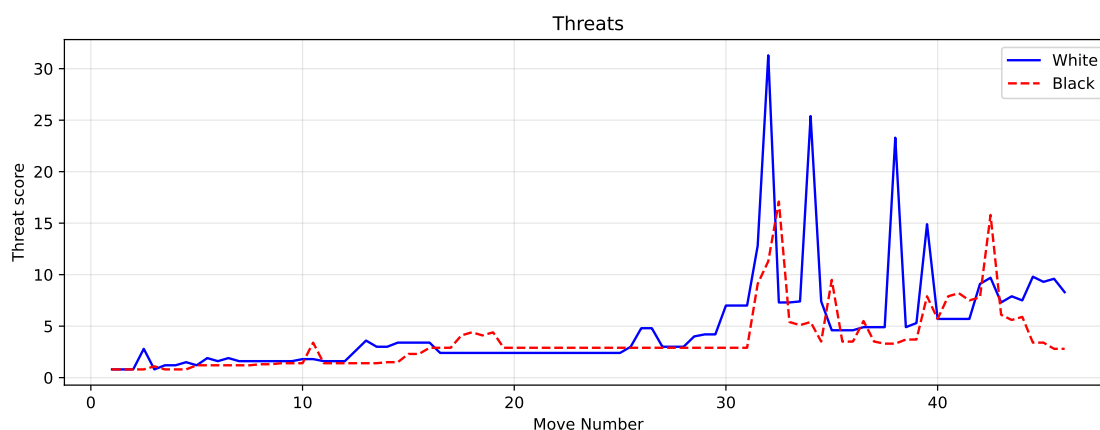
12.3.4 King Safety

- White average: -0.27
- Black average: -0.34



12.3.5 Threats

- White average: 4.79 (range: 0.80 to 31.30)
- Black average: 3.55 (range: 0.80 to 17.10)
- Average advantage: 1.24 (White maintained more threats)



12.4 Annotated Game

1. d4 Nf6 2. c4 c5?! 3. d5 e5? *Black loses 67cp. Consider instead: b5, d6, g6.*

4. Nc3 d6 5. e4 g6 6. Bd3 Bg7 7. Nge2 O-O 8. f3 Na6 9. h4 Nc7 10. Bg5 h6! 11. Be3! Bd7 12. Qd2?! Kh7?! 13. h5?! g5 14. a3 b6 15. g4 Ng8 16. b4! Kh8 17. Kf2?! Qf6 18. Kg2! Rfb8 19. Rhb1! Qd8! 20. Ra2 Qc8 21. Bc2 Bf8 22. Rab2 Ne8! 23. Qd3 Ngf6 24. Ng3! Ng8 25. Bf2 f6?! 26. Nf5 Ne7! 27. Ne3 Qc7! 28. Qd1! Kh7 29. Ba4 Nc8?! 30. Nf5?! Ng7 31. bxc5?! Bxa4?! 32. cxb6 Qxc4? *Black loses 102cp. Consider instead: Nxb6, axb6.*

33. Nxa4! axb6 34. Rb4 Qc7! 35. Nxb6 Bxb6 36. Qb3! b5?! 37. Nc3 Na7! 38. Nxb5?! Nxb5! 39. Rxb5! Rxb5 40. Qxb5! Qc2?! 41. Rb4! Qa2?! 42. Qd7? *White loses 60cp. Consider instead: a4.*

42...Qxa3! 43. Rb7 Rg8! 44. Qe7 Qa8 45. Rd7 Qf8! 46. Qe6

1-0

12.5 Critical Positions

This section highlights critical moments where the evaluation shifted substantially—either due to errors or missed opportunities. Each diagram shows the position *after* the move was played.

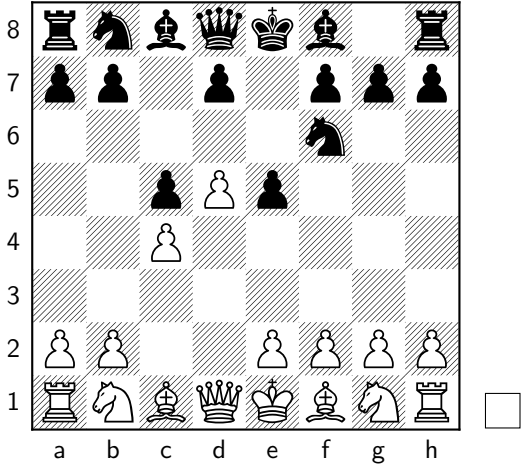
- **Instead of [move]:** The move(s) that should have been played instead.
- **Best continuation:** The optimal sequence of moves from the diagrammed position.

Positions marked with ★ represent the biggest evaluation swings in the game. The positional metrics table summarizes the spatial control, piece activity, king safety, and tactical threats for each side.

Position 1: After 3...e5 ★

Biggest evaluation swing(s) (52cp)

Evaluation: +1.29



Metric	White	Black
Space	0.40	0.30
Mobility	1.85	1.90
King Safety	1.00	0.60
Threats	1.20	0.80

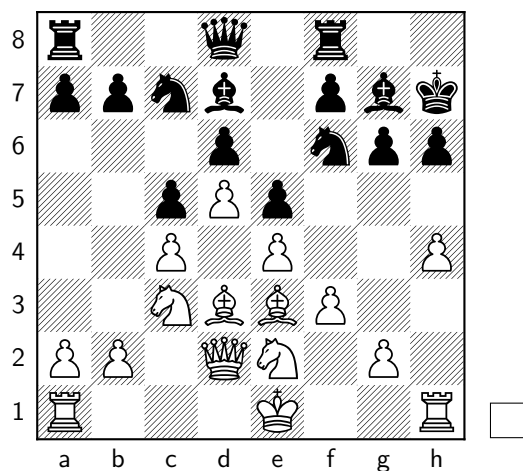
Instead of e5: b5, d6, g6

Best continuation: Nc3 d6 e4

Position 2: After 12...Kh7 ★

Biggest evaluation swing(s) (56cp)

Evaluation: +1.76



Metric	White	Black
Space	0.30	0.50
Mobility	2.55	2.15
King Safety	0.00	-0.60
Threats	2.60	1.40

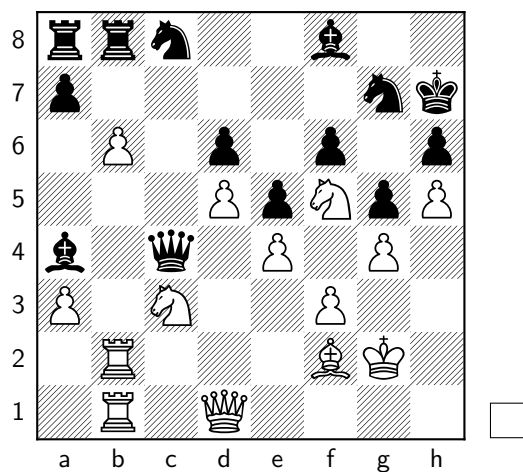
Instead of Kh7: h5

Best continuation: g4 Rb8 Ng3

Position 3: After 32...Qxc4

Critical moment

Evaluation: +3.46



Metric	White	Black
Space	0.10	0.20
Mobility	2.85	2.00
King Safety	-1.00	-1.00
Threats	7.30	17.10

Instead of Qxc4: Nxb6, axb6

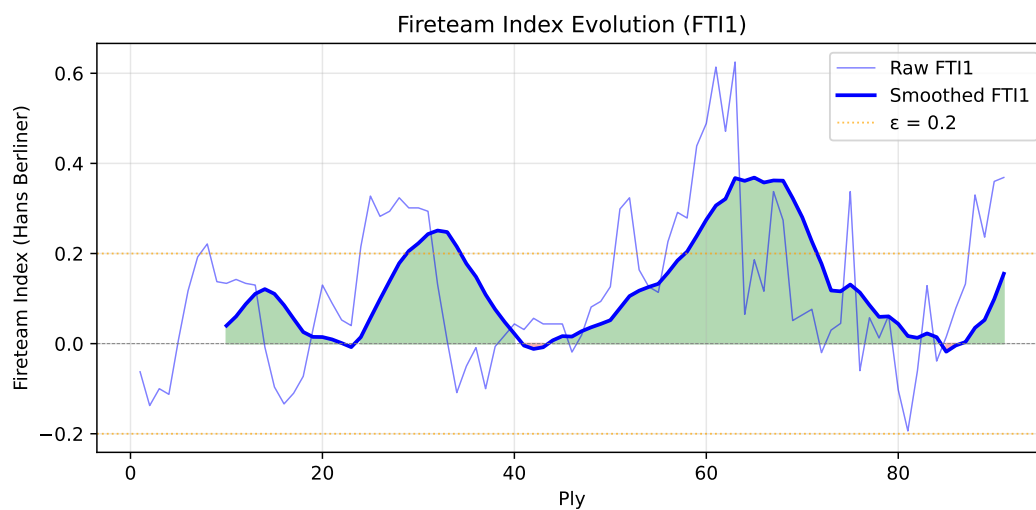
Best continuation: Nxa4 axb6 Rb4

12.6 Fireteam Index Prediction

Tracking: **Hans Berliner** (White)

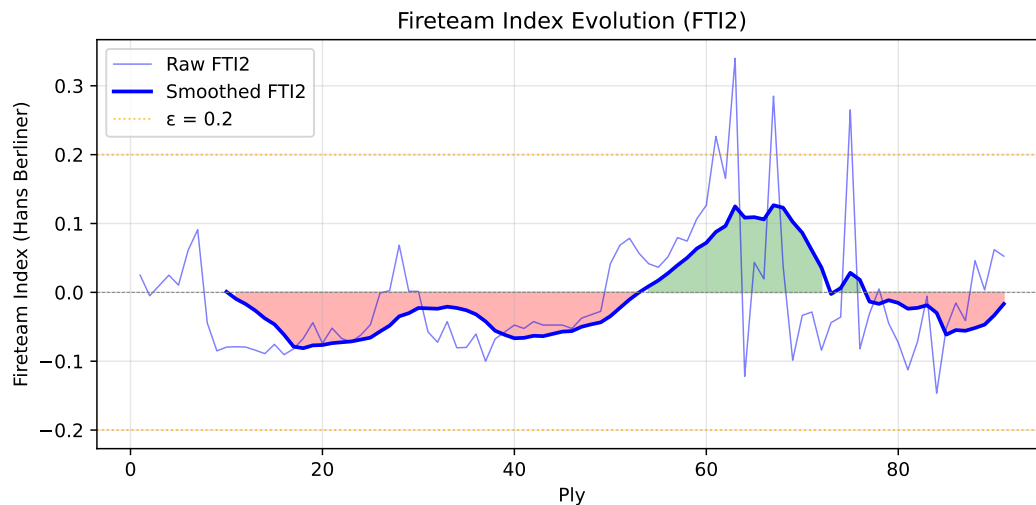
FTI1: Harmonious ($\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}$)

Algorithm	Prediction	Max Streak / Confidence
Per-ply	DRAW	7 plies
Windowed	WIN	11 plies
Balance	WIN	$B_n=+0.33$, $C=0.33$

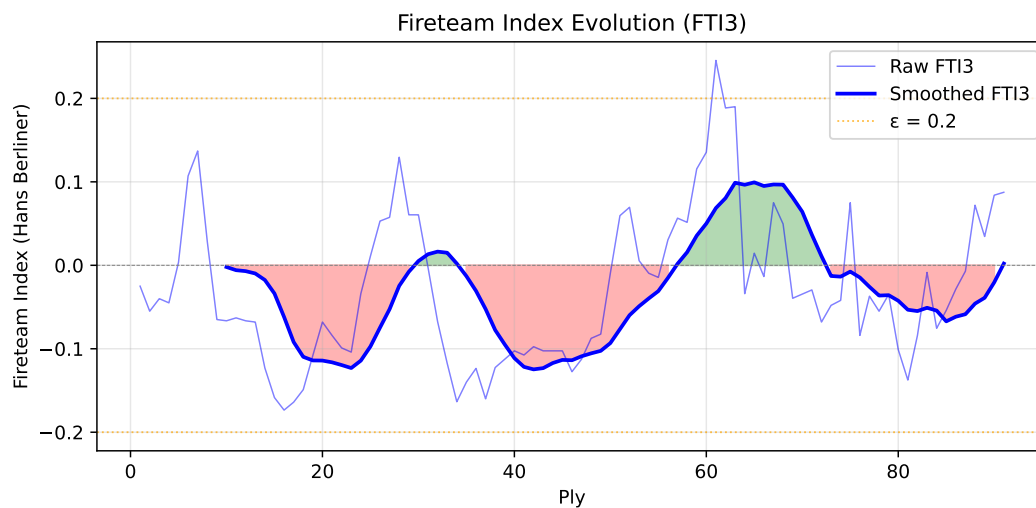


FTI2: Tactical (0.6, 0.1, 0, 0.3)

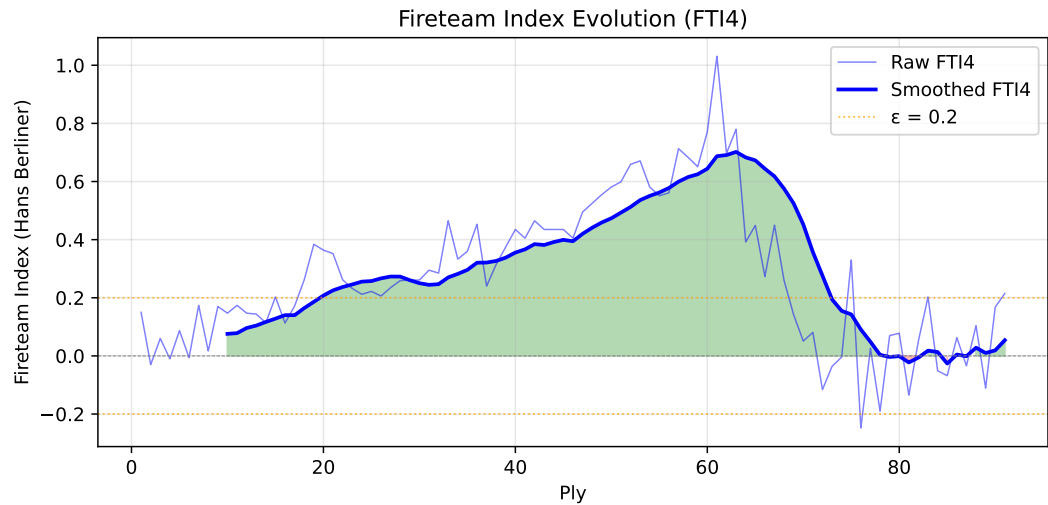
Algorithm	Prediction	Max Streak / Confidence
Per-ply	DRAW	1 plies
Windowed	DRAW	0 plies
Balance	DRAW	$B_n=+0.05$, $C=0.05$


FTI3: Strategic (0.7, 0.1, 0.1, 0.1)

Algorithm	Prediction	Max Streak / Confidence
Per-ply	DRAW	0 plies
Windowed	DRAW	0 plies
Balance	DRAW	$B_n = +0.01$, $C = 0.01$

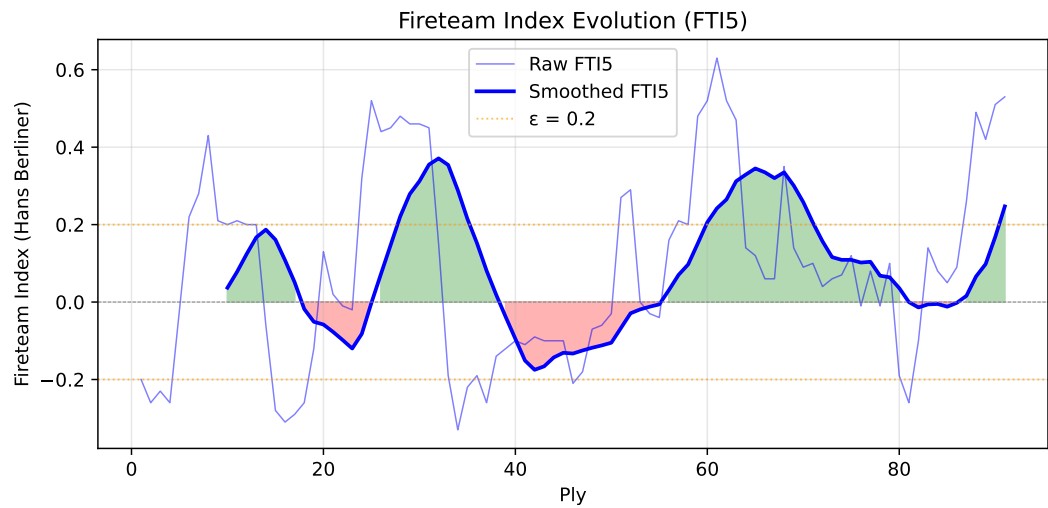

FTI4: Dynamic (0.2, 0.6, 0, 0.2)

Algorithm	Prediction	Max Streak / Confidence
Per-ply	WIN	31 plies
Windowed	WIN	40 plies
Balance	WIN	$B_n = +0.71$, $C = 0.61$



FTI5: Solid (0.3,0.2,0.5,0)

Algorithm	Prediction	Max Streak / Confidence
Per-ply	DRAW	8 plies
Windowed	DRAW	9 plies
Balance	DRAW	$B_n=+0.20$, $C=0.13$



Actual result: 1-0 (Hans Berliner's outcome: WIN)

FTI1:	Per-ply ✗	Windowed ✓	Balance ✓
FTI2:	Per-ply ✗	Windowed ✗	Balance ✗
FTI3:	Per-ply ✗	Windowed ✗	Balance ✗
FTI4:	Per-ply ✓	Windowed ✓	Balance ✓
FTI5:	Per-ply ✗	Windowed ✗	Balance ✗

12.7 Analysis Information

- Engine: Stockfish 17.1
- Depth: 22
- Analysis time: 534.8 seconds

Chapter 13

Josef Richter vs Hans Berliner

13.1 Game Information

Event	5th Correspondence World Championship
Site	ICCF
Date	1965.07.01
Round	8
White	Josef Richter
Black	Hans Berliner
Result	0-1
Opening	B03 – Unknown

13.2 Player Statistics

13.2.1 Josef Richter (White)

- Total moves: 39
- Accuracy: 90.8%
- Average centipawn loss: 19.2
- Best/Excellent moves: 13 / 8
- Good moves: 9
- Inaccuracies: 3
- Mistakes: 6
- Blunders: 0

13.2.2 Hans Berliner (Black)

- Total moves: 39
- Accuracy: 95.4%
- Average centipawn loss: 9.3

- Best/Excellent moves: 19 / 7
- Good moves: 9
- Inaccuracies: 3
- Mistakes: 1
- Blunders: 0

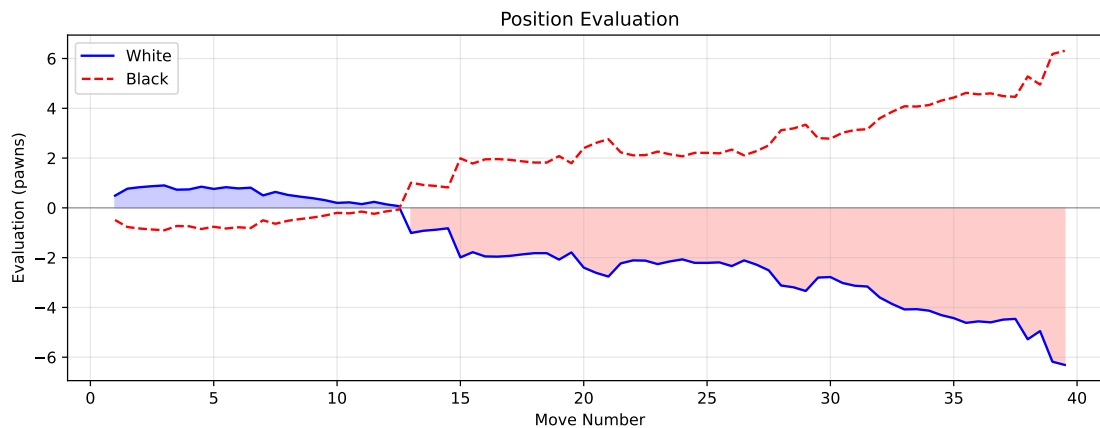
13.3 Positional Analysis

This section shows how key positional factors evolved throughout the game.

13.3.1 Position Evaluation Over Time

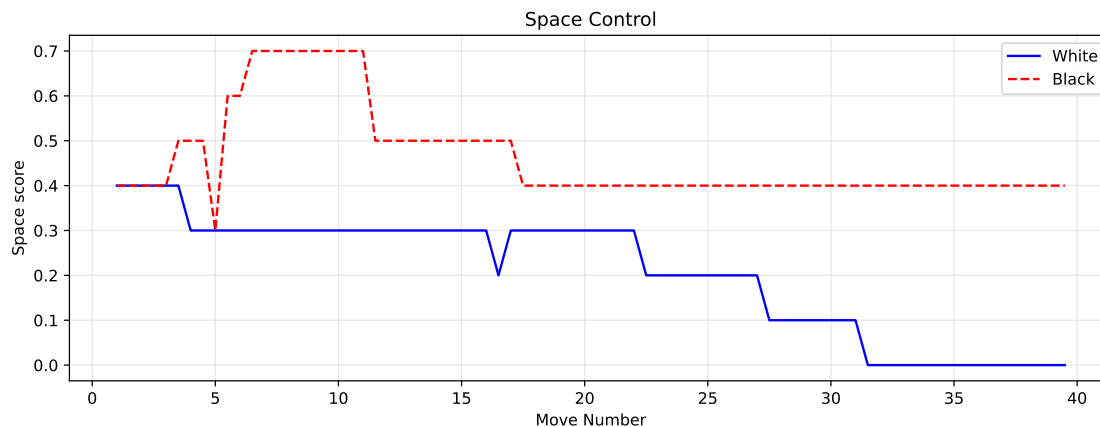
Game Character Classification. Evaluation range: minimum $m_1 = -6.31$, maximum $m_2 = +0.90$, spread $d = m_2 - m_1 = 7.21$.

This game is classified as **chaotic** (normal). A chaotic game with moderate swings.



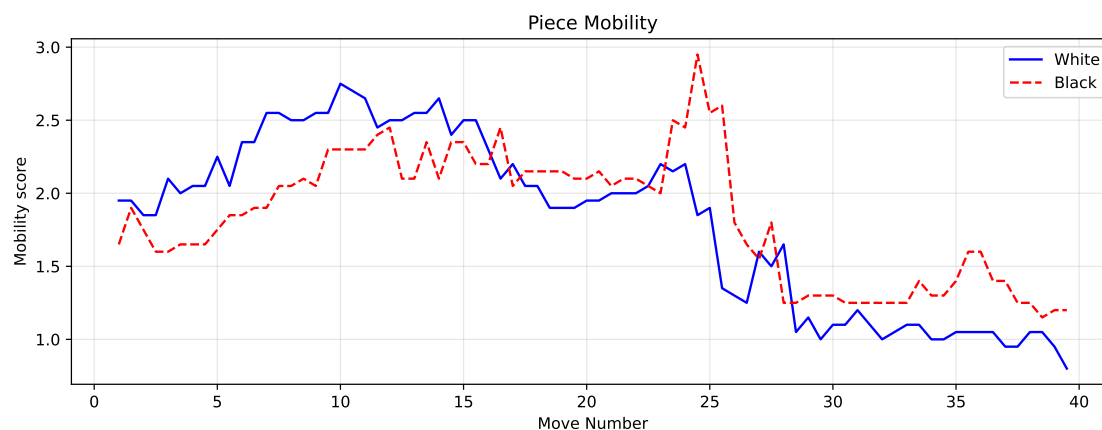
13.3.2 Space Control

- White average: 0.21 (range: 0.00 to 0.40)
- Black average: 0.46 (range: 0.30 to 0.70)



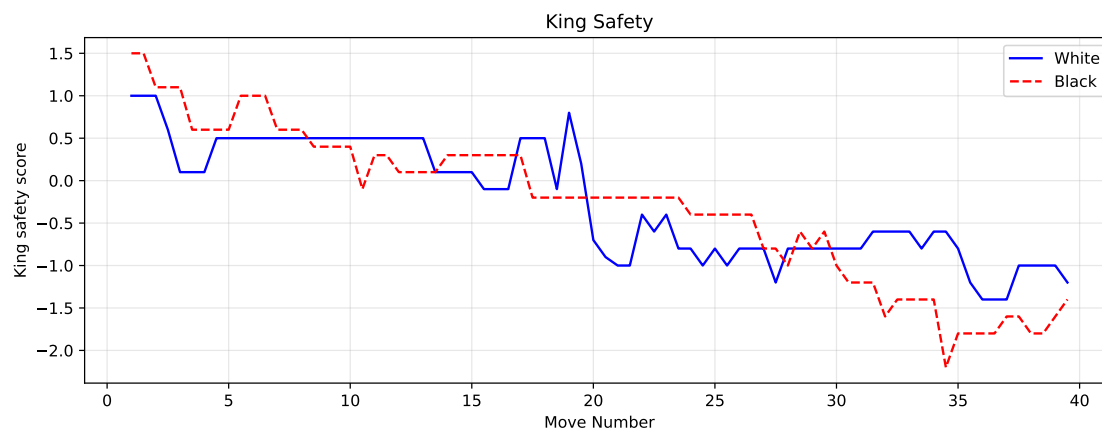
13.3.3 Mobility (Piece Activity)

- White average: 1.82 (range: 0.80 to 2.75)
- Black average: 1.84 (range: 1.15 to 2.95)



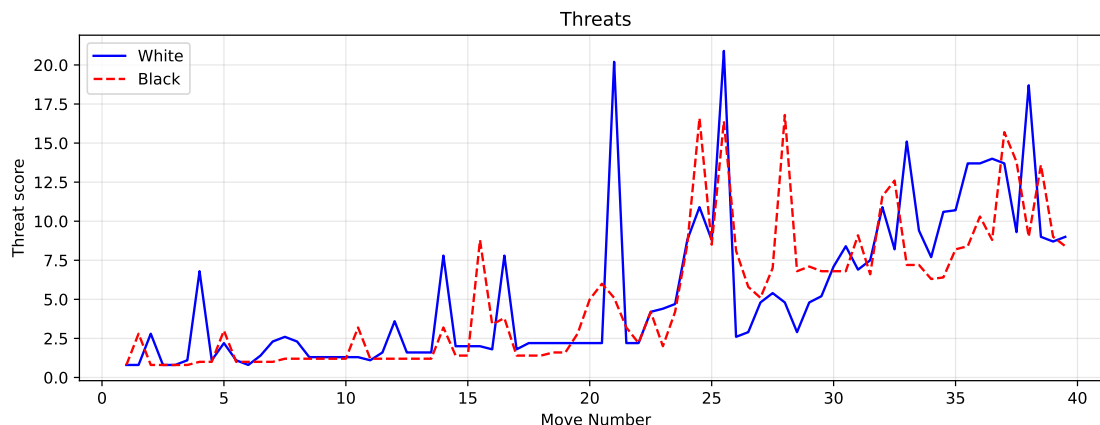
13.3.4 King Safety

- White average: -0.24
- Black average: -0.29



13.3.5 Threats

- White average: 5.37 (range: 0.80 to 20.90)
- Black average: 5.09 (range: 0.80 to 16.80)
- Average advantage: 0.28 (White maintained more threats)



13.4 Annotated Game

1. e4 Nf6 2. e5 Nd5 3. d4 d6 4. c4 Nb6 5. exd6 cxd6 6. Nc3 g6 7. Bg5?! Bg7! 8. Rc1 O-O 9. h3 Nc6! 10. Nf3 h6! 11. Be3 e5 12. d5 Ne7! 13. Qb3? *White loses 105cp. Consider instead: Bd3, g4, Be2.*

13...Nf5! 14. Bxb6! axb6 15. Be2? *White loses 90cp. Consider instead: Bd3, Ne4.*

15...Nd4?! 16. Qd1 Nxf3+ 17. Bxf3 f5 18. Be2 e4 19. O-O Qh4 20. Kh2? *White loses 61cp. Consider instead: Qe1, f4, Qd2.*

20...Be5+! 21. g3! Qg5?! 22. Kh1! f4 23. Bg4! fxg3 24. Nxe4! Bxg4 25. Qxg4 Qxg4 26. hxg4 Rxa2 27. fxg3 Rxb2! 28. Rxf8+?! Kxf8! 29. g5! hxg5? *Black loses 63cp. Consider instead: Kg7.*

30. Rf1+ Kg7! 31. Nxe5?! Bxg3! 32. Rf7+? *White loses 65cp. Consider instead: Kg1, Rf3, Ne6+.*

32...Kh6! 33. Ne4 Be5! 34. Nf2 Kg5! 35. Nd3! Rh2+! 36. Kg1! Bg3! 37. Ne1! Rb2 38. Nd3? *White loses 80cp. Consider instead: Nf3+, Rf1.*

38...Rd2?! 39. Nc1? *White loses 65cp. Consider instead: Nb4, Rf3, Nc1.*

39...Bf4!

0-1

13.5 Critical Positions

This section highlights critical moments where the evaluation shifted substantially—either due to errors or missed opportunities. Each diagram shows the position *after* the move was played.

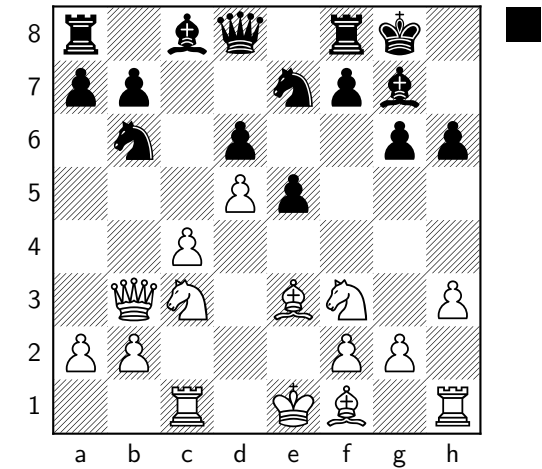
- **Instead of [move]:** The move(s) that should have been played instead.
- **Best continuation:** The optimal sequence of moves from the diagrammed position.

Positions marked with ★ represent the biggest evaluation swings in the game. The positional metrics table summarizes the spatial control, piece activity, king safety, and tactical threats for each side.

Position 1: After 13. Qb3

Critical moment

Evaluation: -1.01



Metric	White	Black
Space	0.30	0.50
Mobility	2.55	2.10
King Safety	0.50	0.10
Threats	1.60	1.20

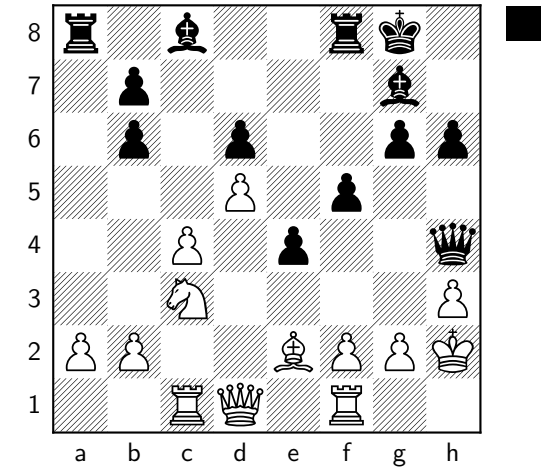
Instead of Qb3: Bd3, g4, Be2

Best continuation: Nf5 Bd2 Nd4

Position 2: After 20. Kh2 ★

Biggest evaluation swing(s) (61cp)

Evaluation: -2.40



Metric	White	Black
Space	0.30	0.40
Mobility	1.95	2.10
King Safety	-0.70	-0.20
Threats	2.20	5.00

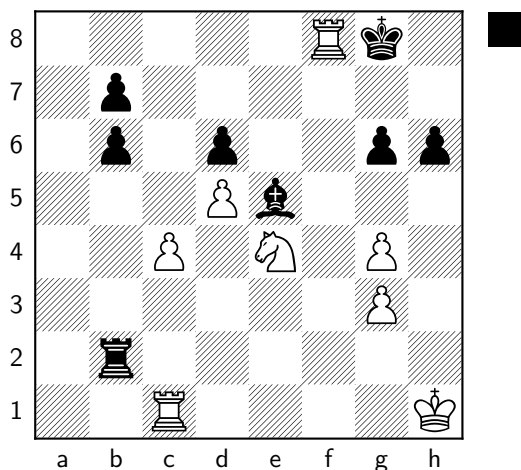
Instead of Kh2: Qe1, f4, Qd2

Best continuation: Be5+ g3 Qe7

Position 3: After 28. Rxf8+ ★

Biggest evaluation swing(s) (61cp)

Evaluation: -3.12



Metric	White	Black
Space	0.10	0.40
Mobility	1.65	1.25
King Safety	-0.80	-1.00
Threats	4.80	16.80

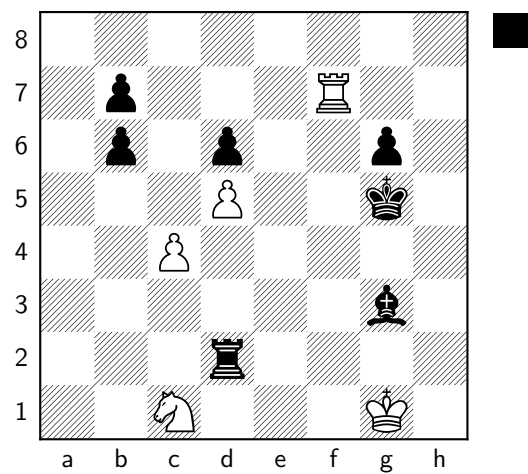
Instead of Rxf8+: Rg1, g5, Kg1

Best continuation: Kxf8 g5 Kg7

Position 4: After 39. Nc1

Critical moment

Evaluation: -6.18



Metric	White	Black
Space	0.00	0.40
Mobility	0.95	1.20
King Safety	-1.00	-1.60
Threats	8.70	9.00

Instead of Nc1: Nb4, Rf3

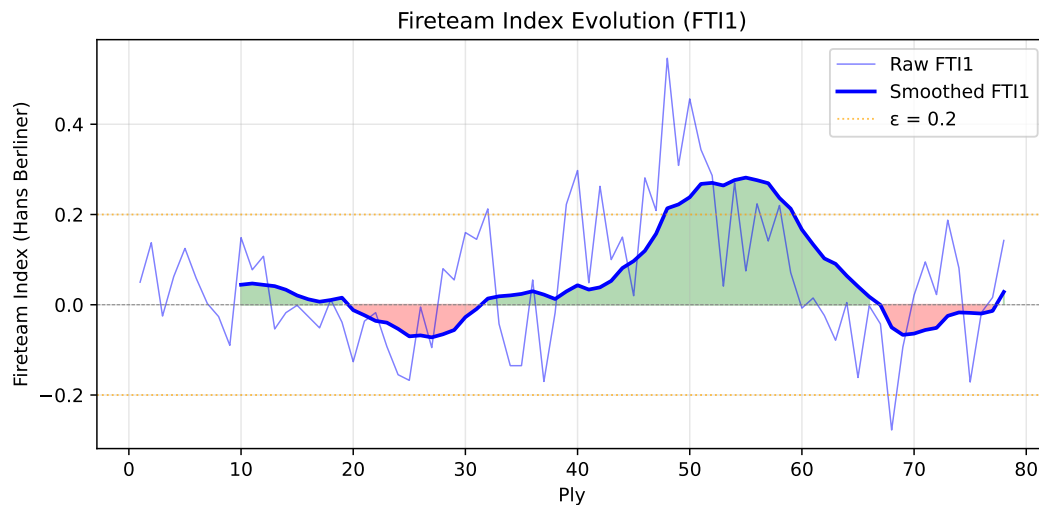
Best continuation: Bf4 Nb3 Rd3

13.6 Fireteam Index Prediction

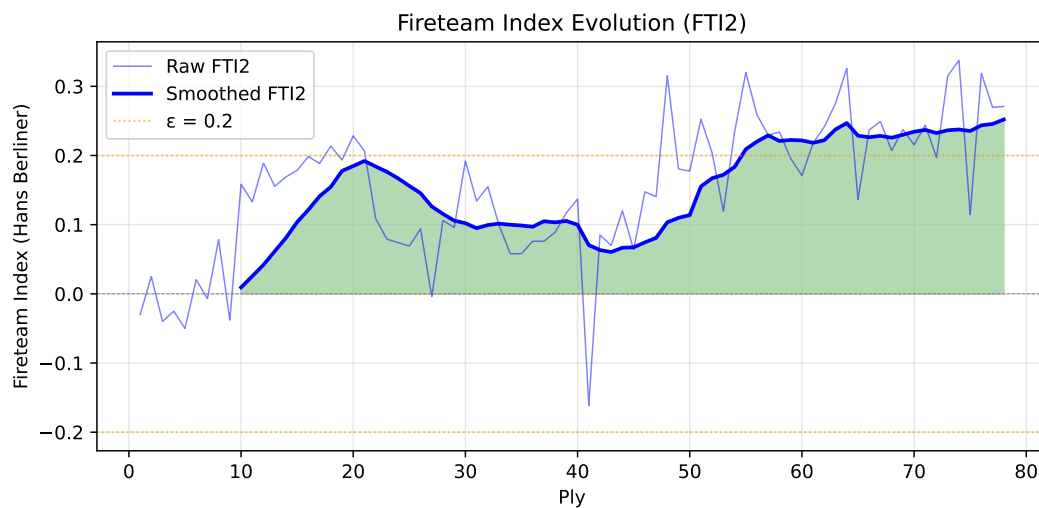
Tracking: Hans Berliner (Black)

FTI1: Harmonious ($\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}$)

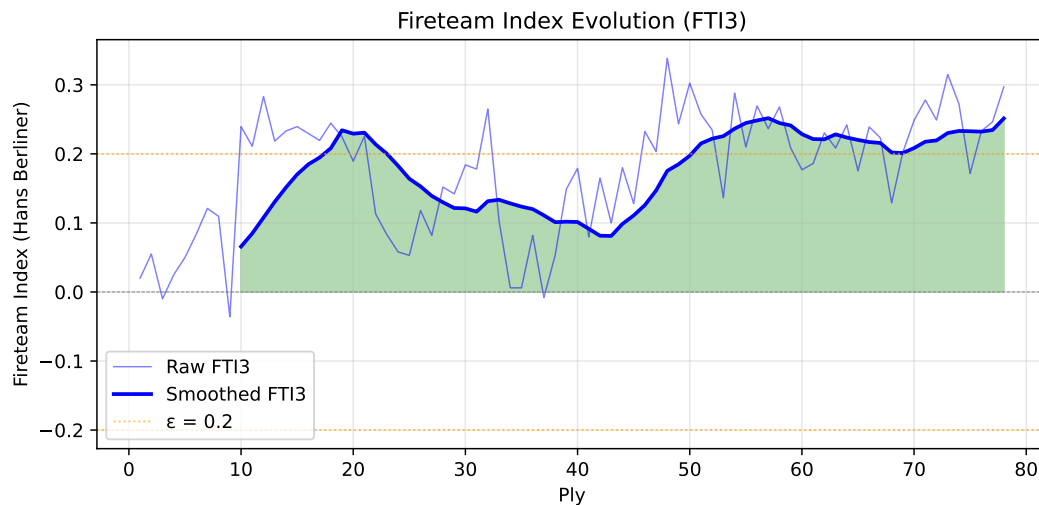
Algorithm	Prediction	Max Streak / Confidence
Per-ply	DRAW	5 plies
Windowed	DRAW	7 plies
Balance	WIN	$B_n=+0.21$, $C=0.19$


FTI2: Tactical (0.6, 0.1, 0, 0.3)

Algorithm	Prediction	Max Streak / Confidence
Per-ply	DRAW	3 plies
Windowed	DRAW	1 plies
Balance	WIN	$B_n=+0.42$, $C=0.42$

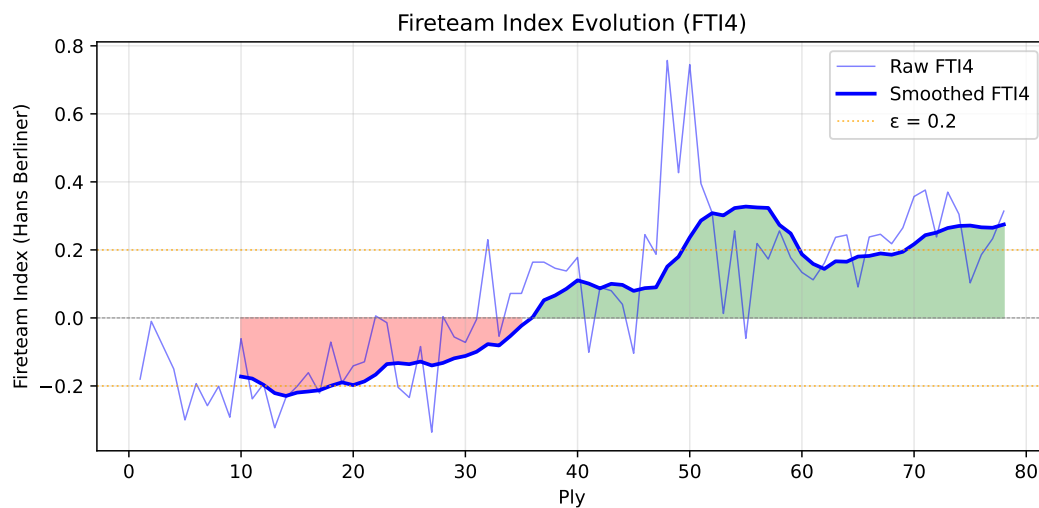

FTI3: Strategic (0.7, 0.1, 0.1, 0.1)

Algorithm	Prediction	Max Streak / Confidence
Per-ply	DRAW	2 plies
Windowed	DRAW	1 plies
Balance	WIN	$B_n=+0.52$, $C=0.52$



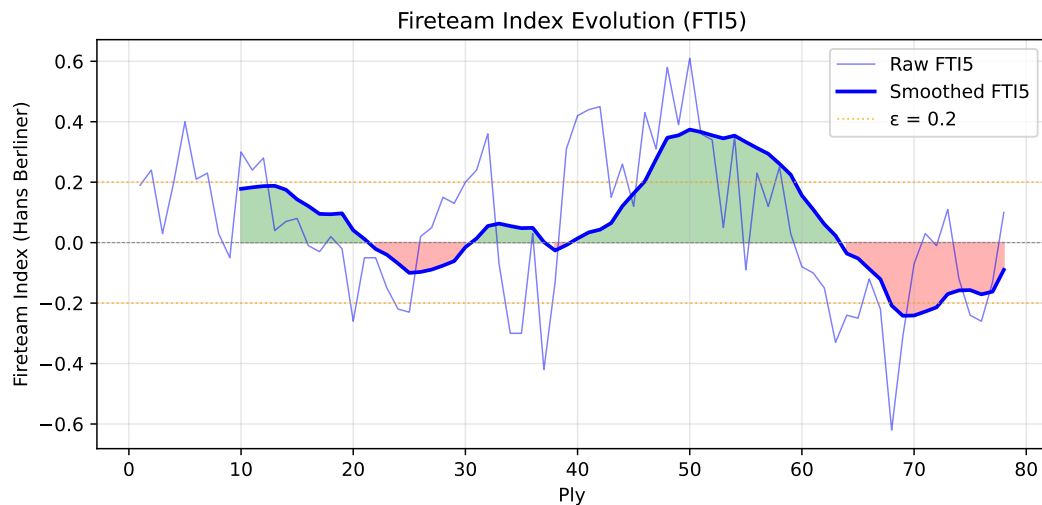
FTI4: Dynamic (0.2, 0.6, 0, 0.2)

Algorithm	Prediction	Max Streak / Confidence
Per-ply	DRAW	5 plies
Windowed	DRAW	8 plies
Balance	WIN	$B_n = +0.31$, $C = 0.28$



FTI5: Solid (0.3, 0.2, 0.5, 0)

Algorithm	Prediction	Max Streak / Confidence
Per-ply	DRAW	7 plies
Windowed	WIN	12 plies
Balance	DRAW	$B_n = +0.06$, $C = 0.04$



Actual result: 0-1 (Hans Berliner's outcome: WIN)

FTI1:	Per-ply	×	Windowed	×	Balance	✓
FTI2:	Per-ply	×	Windowed	×	Balance	✓
FTI3:	Per-ply	×	Windowed	×	Balance	✓
FTI4:	Per-ply	×	Windowed	×	Balance	✓
FTI5:	Per-ply	×	Windowed	✓	Balance	×

13.7 Analysis Information

- Engine: Stockfish 17.1
- Depth: 22
- Analysis time: 549.3 seconds

Chapter 14

Berliner Hans J (USA) vs Rokhlin Yakov (RUS)

14.1 Game Information

Event	Ch World (corr.)
Site	corr.
Date	1965.??.??
Round	9
White	Berliner Hans J (USA)
Black	Rokhlin Yakov (RUS)
Result	1-0
Opening	E31 – Unknown

14.2 Player Statistics

14.2.1 Berliner Hans J (USA) (White)

- Total moves: 46
- Accuracy: 90.0%
- Average centipawn loss: 21.0
- Best/Excellent moves: 21 / 6
- Good moves: 7
- Inaccuracies: 6
- Mistakes: 5
- Blunders: 1

14.2.2 Rokhlin Yakov (RUS) (Black)

- Total moves: 46
- Accuracy: 88.2%

- Average centipawn loss: 25.1
- Best/Excellent moves: 16 / 4
- Good moves: 14
- Inaccuracies: 6
- Mistakes: 4
- Blunders: 2

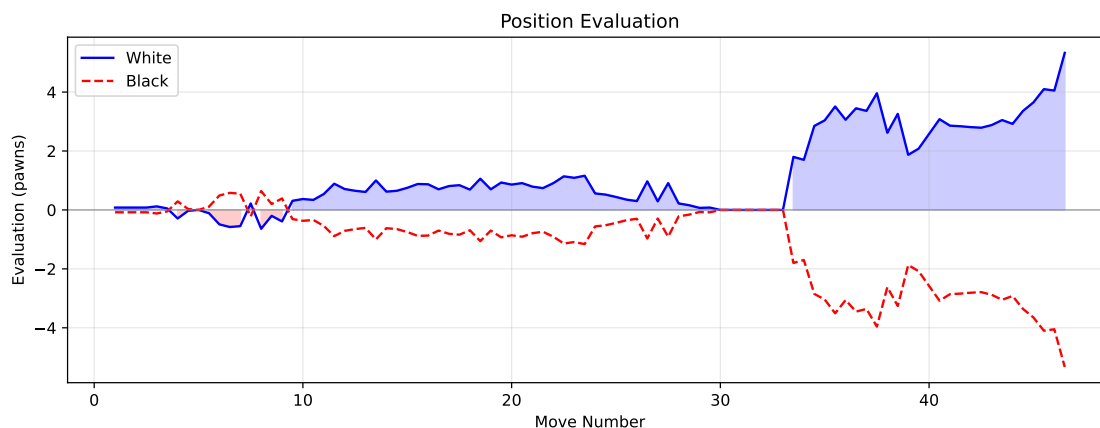
14.3 Positional Analysis

This section shows how key positional factors evolved throughout the game.

14.3.1 Position Evaluation Over Time

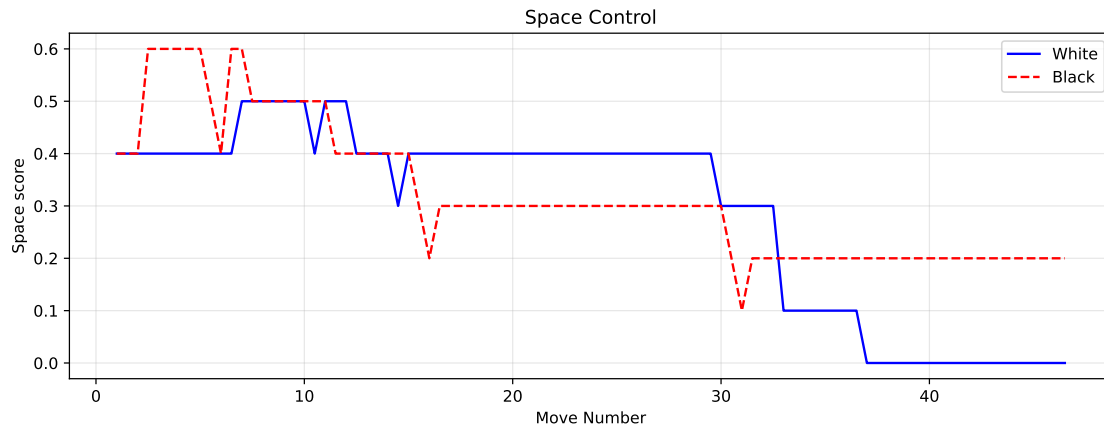
Game Character Classification. Evaluation range: minimum $m_1 = -0.64$, maximum $m_2 = +5.33$, spread $d = m_2 - m_1 = 5.97$.

This game is classified as **tactical** (normal). A tactical game with moderate swings.



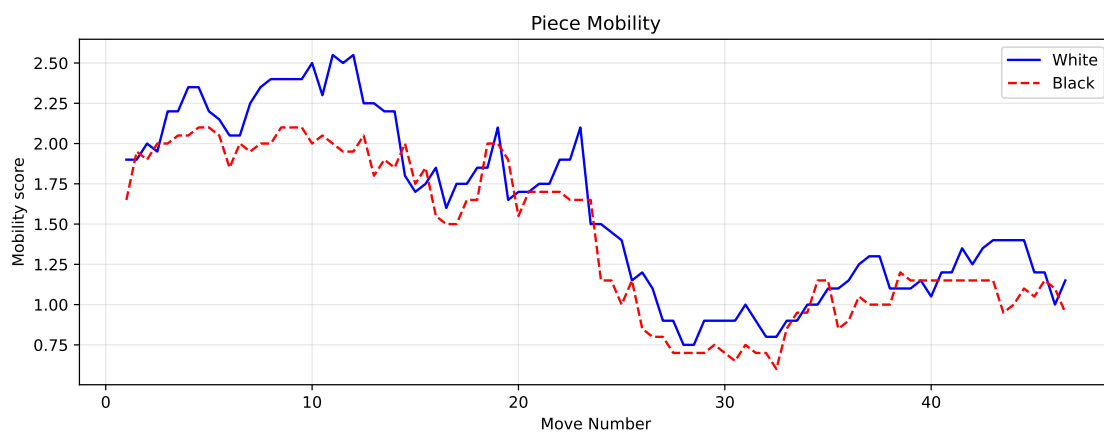
14.3.2 Space Control

- White average: 0.29 (range: 0.00 to 0.50)
- Black average: 0.32 (range: 0.10 to 0.60)



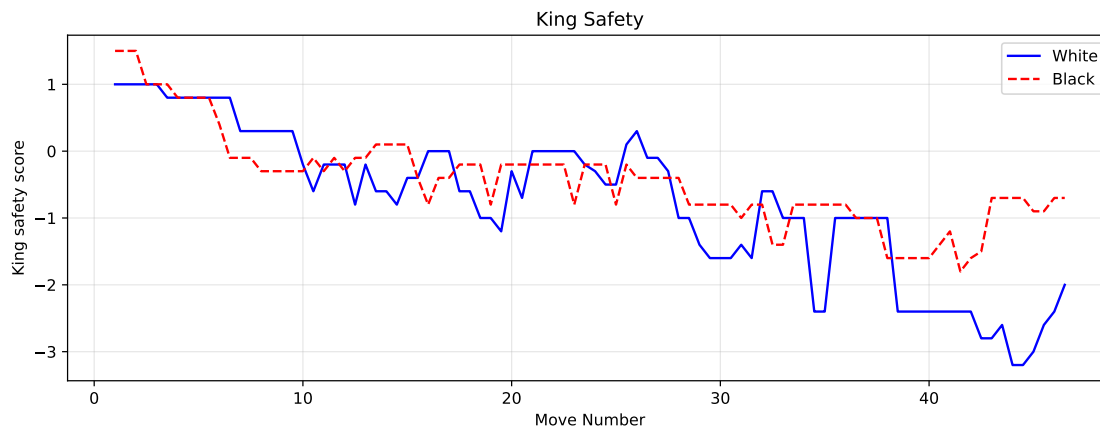
14.3.3 Mobility (Piece Activity)

- White average: 1.59 (range: 0.75 to 2.55)
- Black average: 1.42 (range: 0.60 to 2.10)



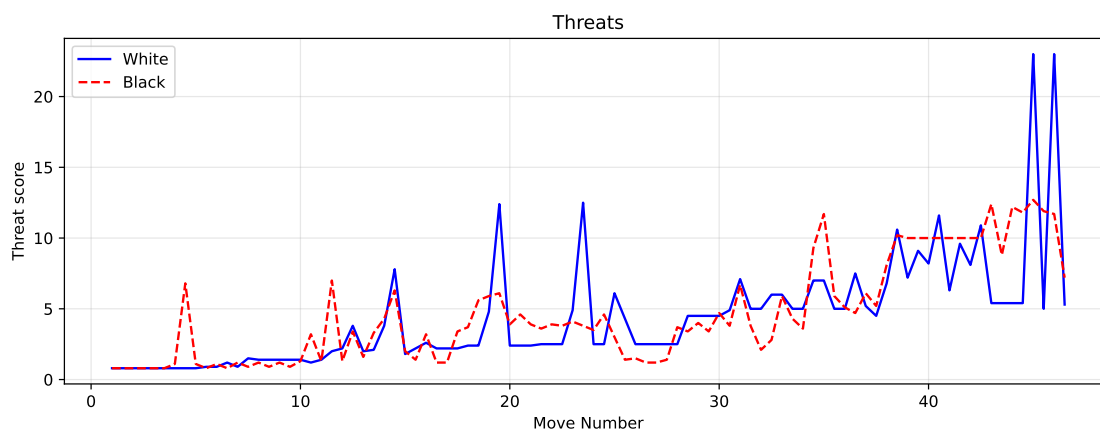
14.3.4 King Safety

- White average: -0.76
- Black average: -0.43



14.3.5 Threats

- White average: 4.46 (range: 0.80 to 23.00)
- Black average: 4.63 (range: 0.80 to 12.70)
- Average advantage: -0.17 (Black maintained more threats)



14.4 Annotated Game

1. d4 Nf6 2. c4 e6 3. Nc3 Bb4 4. Bg5?! h6 5. Bh4 c5 6. d5?! d6 7. e3 e5? *Black loses 77cp. Consider instead: Bxc3+, g5, Qe7.*

8. Qc2? *White loses 85cp. Consider instead: Ne2.*

8...Nbd7?! 9. Bd3?! Qe7? *Black loses 68cp. Consider instead: Bxc3+, Qe7.*

10. f3! e4 11. Bxe4 g5 12. Bg3 Nxe4! 13. fxe4 Ne5?! 14. a3?! Bxc3+! 15. Qxc3! f6 16. Bxe5! fxe5! 17. Ne2 O-O! 18. Ng3 Bg4?! 19. Rf1?! Rxf1+ 20. Kxf1! Rf8+ 21. Kg1! h5! 22. Qc2! Qh7 23. Rf1! Rxf1+! 24. Kxf1? *White loses 60cp. Consider instead: Nxf1, Kxf1.*

24...h4! 25. Nf5 Bxf5! 26. exf5! g4?! 27. Qf2? *White loses 92cp. Consider instead: Ke1, b4, Ke2.*

27...Qh6? *Black loses 76cp. Consider instead: Qh5, Kf7.*

28. Ke2? *White loses 73cp. Consider instead: h3.*

28...Kf7 29. Kd3 Qg5! 30. b4 b6! 31. bxc5 bxc5! 32. Kc2 Kf6! 33. e4! Kf7?? *Black loses 180cp. Consider instead: g3.*

34. Kb3 Qc1? *Black loses 97cp. Consider instead: Qd8, Qe7, Qc1.*

35. Qe2! Qf4?! 36. h3?! gxh3! 37. gxh3 Ke7! 38. Qh5? *White loses 114cp. Consider instead: Ka4.*

38...Qc1 39. Qxh4+?? *White loses 135cp. Consider instead: Qe2.*

39...Kd7 40. Qh7+! Kd8?! 41. Qg8+ Kc7 42. Qg7+! Kb6 43. Qg2! Qb1+ 44. Kc3! Qc1+ 45. Qc2! Qe3+ 46. Qd3! Qf4?? *Black loses 140cp. Consider instead: Qe1+.*

1-0

14.5 Critical Positions

This section highlights critical moments where the evaluation shifted substantially—either due to errors or missed opportunities. Each diagram shows the position *after* the move was played.

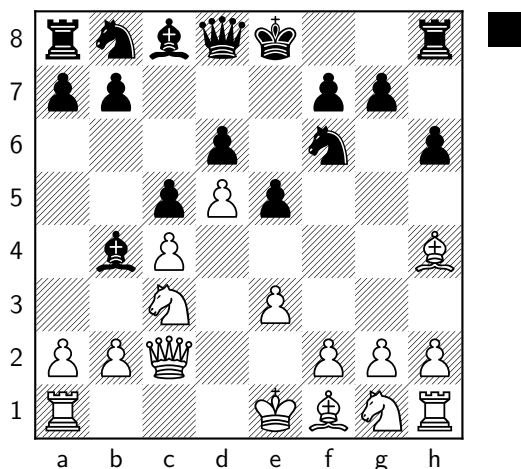
- **Instead of [move]:** The move(s) that should have been played instead.
- **Best continuation:** The optimal sequence of moves from the diagrammed position.

Positions marked with ★ represent the biggest evaluation swings in the game. The positional metrics table summarizes the spatial control, piece activity, king safety, and tactical threats for each side.

Position 1: After 8. Qc2 ★

Biggest evaluation swing(s) (86cp)

Evaluation: -0.64



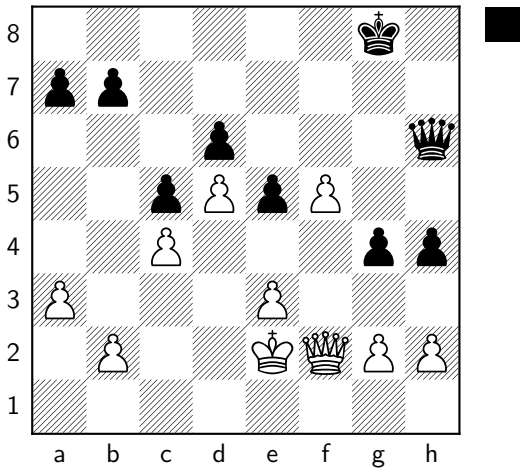
Metric	White	Black
Space	0.50	0.50
Mobility	2.40	2.00
King Safety	0.30	-0.30
Threats	1.40	1.20

Instead of Qc2: Ne2 Best continuation: Bxc3+ bxc3 g5

Position 2: After 28. Ke2 ★

Biggest evaluation swing(s) (69cp)

Evaluation: +0.22



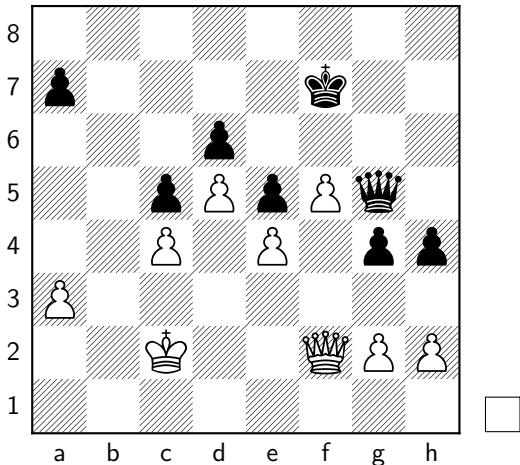
Metric	White	Black
Space	0.40	0.30
Mobility	0.75	0.70
King Safety	-1.00	-0.40
Threats	2.50	3.70

Instead of Ke2: h3 Best continuation: Qg5 Qf1 Kf7

Position 3: After 33...Kf7

Black blunders (+180cp)

Evaluation: +1.80



Metric	White	Black
Space	0.10	0.20
Mobility	0.90	0.95
King Safety	-1.00	-0.80
Threats	5.00	4.30

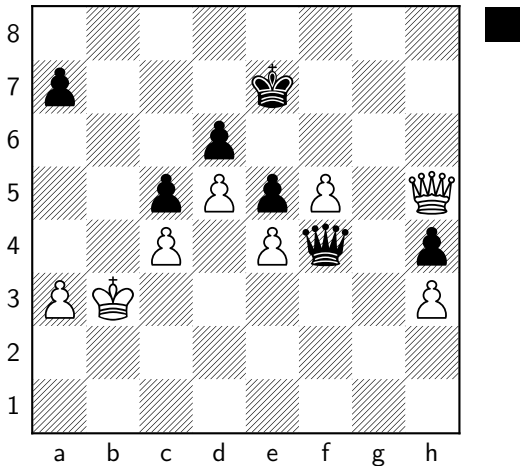
Instead of Kf7: g3

Best continuation: Kb3 Qd8 Qe1

Position 4: After 38. Qh5

Critical moment

Evaluation: +2.62



Metric	White	Black
Space	0.00	0.20
Mobility	1.10	1.00
King Safety	-1.00	-1.60
Threats	6.80	8.10

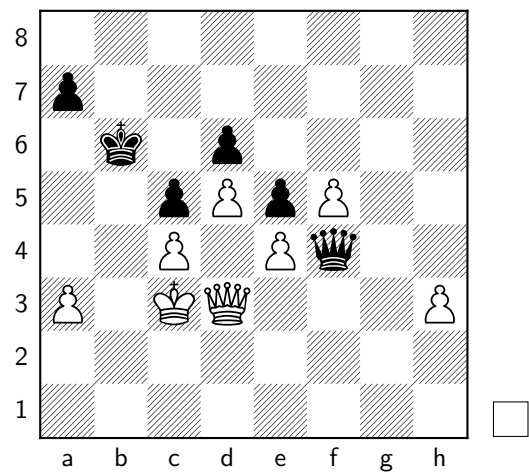
Instead of Qh5: Ka4

Best continuation: Qc1 Qe2 Qa1

Position 5: After 46...Qf4

Black blunders (+128cp)

Evaluation: +5.33



Metric	White	Black
Space	0.00	0.20
Mobility	1.15	0.95
King Safety	-2.00	-0.70
Threats	5.30	7.20

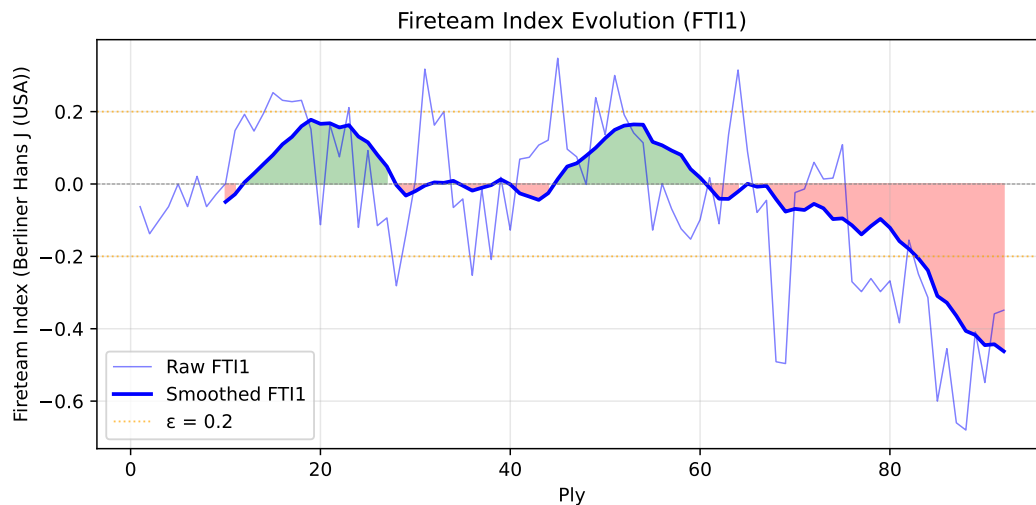
Instead of Qf4: Qe1+ Best continuation: Qb1+ Kc7 Kb3

14.6 Fireteam Index Prediction

Tracking: **Berliner Hans J (USA)** (White)

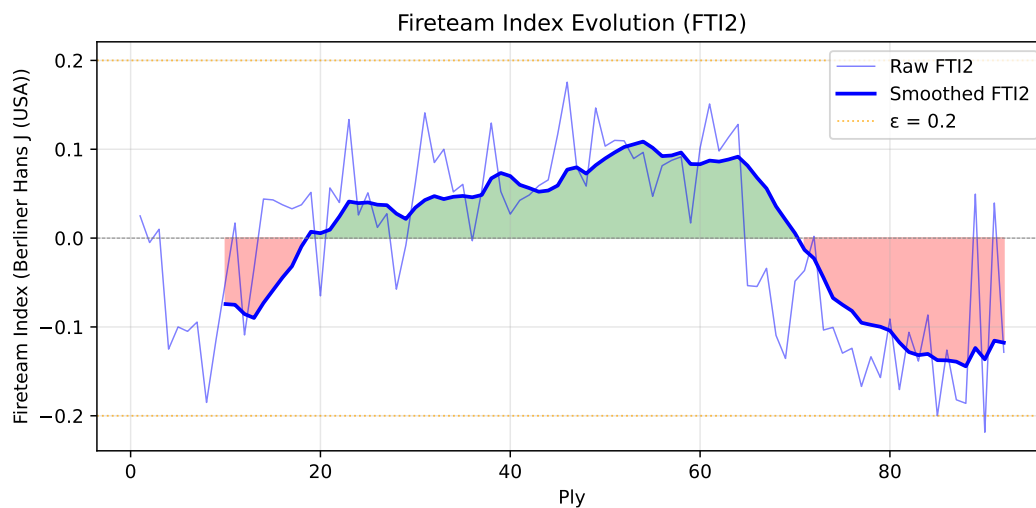
FTI1: Harmonious ($\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}$)

Algorithm	Prediction	Max Streak / Confidence
Per-ply	DRAW	1 plies
Windowed	DRAW	0 plies
Balance	DRAW	$B_n=-0.16$, $C=0.11$



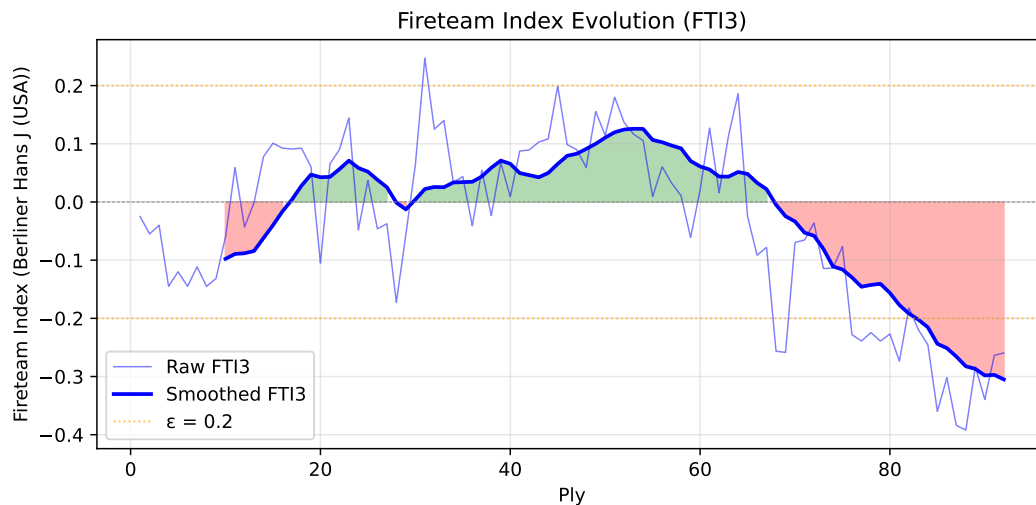
FTI2: Tactical (0.6, 0.1, 0, 0.3)

Algorithm	Prediction	Max Streak / Confidence
Per-ply	DRAW	0 plies
Windowed	DRAW	0 plies
Balance	DRAW	$B_n = -0.01$, $C = 0.01$

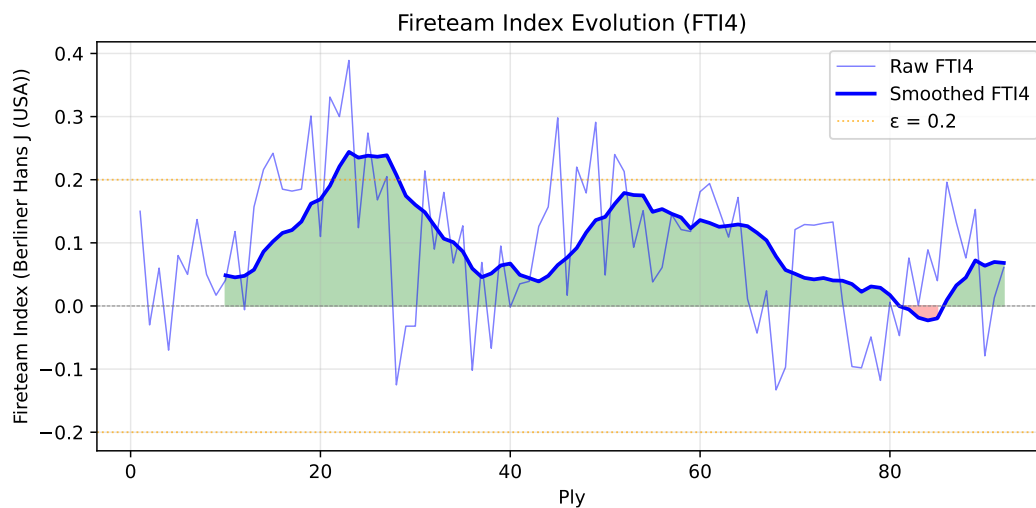


FTI3: Strategic (0.7, 0.1, 0.1, 0.1)

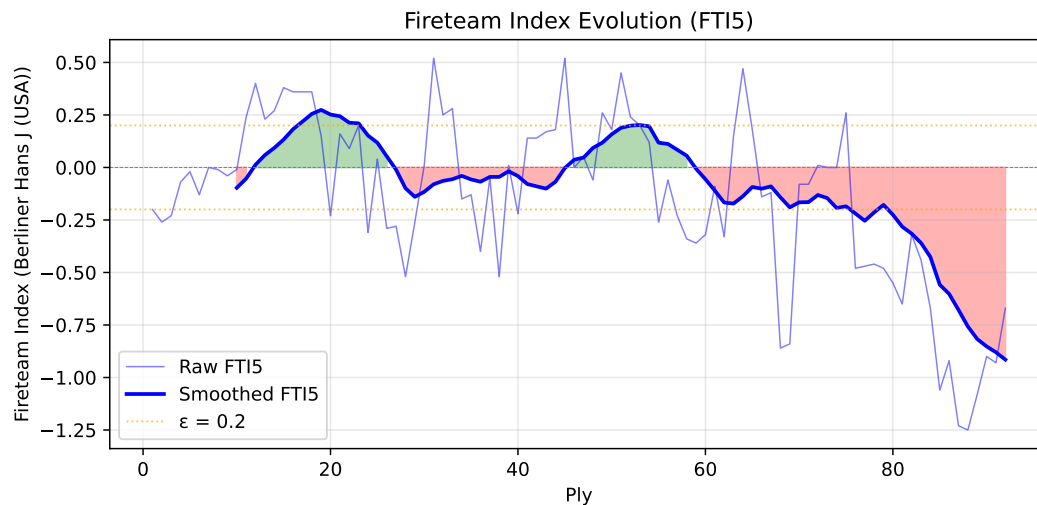
Algorithm	Prediction	Max Streak / Confidence
Per-ply	DRAW	0 plies
Windowed	DRAW	0 plies
Balance	LOSS	$B_n = -0.22$, $C = 0.21$

**FTI4: Dynamic (0.2, 0.6, 0, 0.2)**

Algorithm	Prediction	Max Streak / Confidence
Per-ply	DRAW	3 plies
Windowed	DRAW	0 plies
Balance	WIN	$B_n = +0.16$, $C = 0.16$

**FTI5: Solid (0.3, 0.2, 0.5, 0)**

Algorithm	Prediction	Max Streak / Confidence
Per-ply	DRAW	3 plies
Windowed	DRAW	3 plies
Balance	DRAW	$B_n = -0.30$, $C = 0.12$



Actual result: 1-0 (Berliner Hans J (USA)'s outcome: WIN)

FTI1:	Per-ply	×	Windowed	×	Balance	×
FTI2:	Per-ply	×	Windowed	×	Balance	×
FTI3:	Per-ply	×	Windowed	×	Balance	×
FTI4:	Per-ply	×	Windowed	×	Balance	✓
FTI5:	Per-ply	×	Windowed	×	Balance	×

14.7 Analysis Information

- Engine: Stockfish 17.1
- Depth: 22
- Analysis time: 611.6 seconds

Chapter 15

Werner Stern vs Hans Berliner

15.1 Game Information

Event	5th Correspondence World Championship
Site	ICCF
Date	1965.07.01
Round	11
White	Werner Stern
Black	Hans Berliner
Result	0-1
Opening	B04 – Unknown

15.2 Player Statistics

15.2.1 Werner Stern (White)

- Total moves: 20
- Accuracy: 80.6%
- Average centipawn loss: 43.2
- Best/Excellent moves: 3 / 2
- Good moves: 8
- Inaccuracies: 3
- Mistakes: 2
- Blunders: 2

15.2.2 Hans Berliner (Black)

- Total moves: 20
- Accuracy: 92.8%
- Average centipawn loss: 15.0

- Best/Excellent moves: 5 / 5
- Good moves: 8
- Inaccuracies: 1
- Mistakes: 1
- Blunders: 0

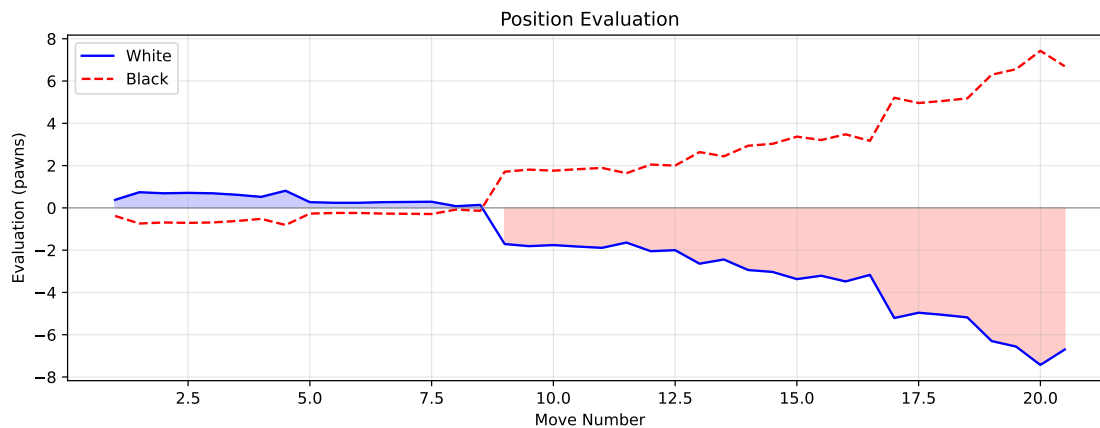
15.3 Positional Analysis

This section shows how key positional factors evolved throughout the game.

15.3.1 Position Evaluation Over Time

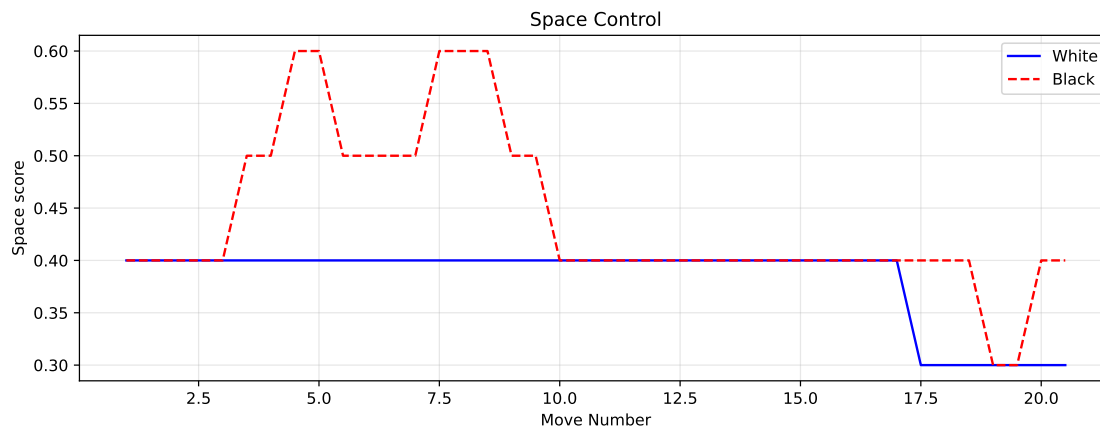
Game Character Classification. Evaluation range: minimum $m_1 = -7.43$, maximum $m_2 = +0.81$, spread $d = m_2 - m_1 = 8.24$.

This game is classified as **chaotic** (normal). A chaotic game with moderate swings.



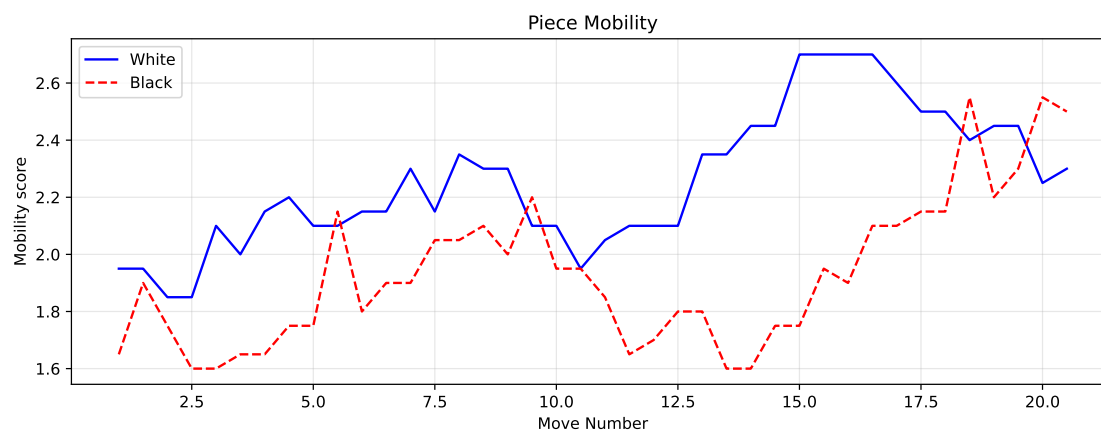
15.3.2 Space Control

- White average: 0.38 (range: 0.30 to 0.40)
- Black average: 0.44 (range: 0.30 to 0.60)



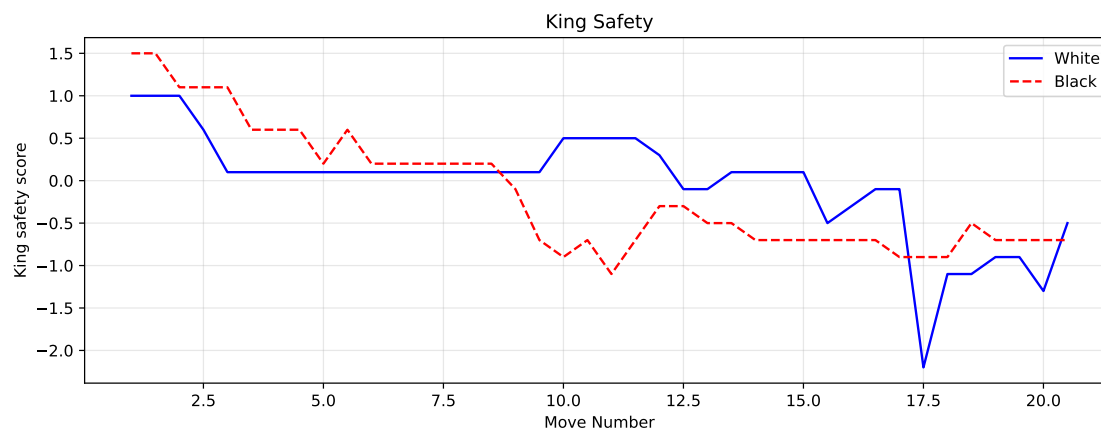
15.3.3 Mobility (Piece Activity)

- White average: 2.26 (range: 1.85 to 2.70)
- Black average: 1.93 (range: 1.60 to 2.55)



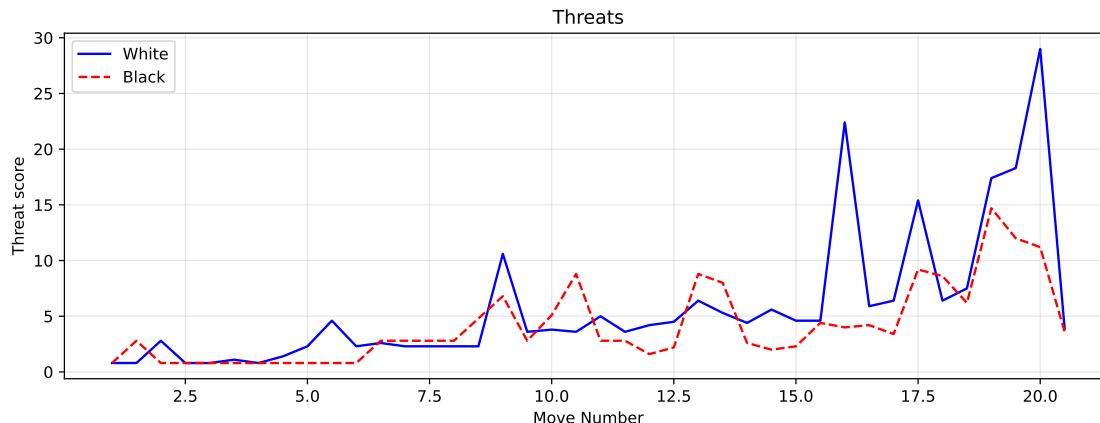
15.3.4 King Safety

- White average: -0.04
- Black average: -0.15



15.3.5 Threats

- White average: 5.82 (range: 0.80 to 29.00)
- Black average: 4.12 (range: 0.80 to 14.70)
- Average advantage: 1.69 (White maintained more threats)



15.4 Annotated Game

1. e4 Nf6?! 2. e5 Nd5 3. d4 d6 4. Nf3 g6 5. Ng5?! dxe5 6. dxe5 Bg7 7. Bc4! c6 8. Nc3 h6! 9. Nxf7?? *White loses 186cp. Consider instead: Nge4, Nf3.*

9...Kxf7! 10. Nxd5 cxd5! 11. Bxd5+! Ke8 12. e6 Rf8 13. Qg4?! Rf6 14. Qc4?! Na6! 15. Be3 Qa5+ 16. Bd2 Qb6 17. Bc3?? *White loses 223cp. Consider instead: O-O, O-O-O.*

17...Qxf2+ 18. Kd1! Bxe6! 19. Bxf6? *White loses 109cp. Consider instead: Bxe6, Kc1, Qb5+.*

19...Rd8 20. Bh4? *White loses 78cp. Consider instead: Bd4, Qb5+.*

20...Qb6? *Black loses 107cp. Consider instead: Qe3.*

0-1

15.5 Critical Positions

This section highlights critical moments where the evaluation shifted substantially—either due to errors or missed opportunities. Each diagram shows the position *after* the move was played.

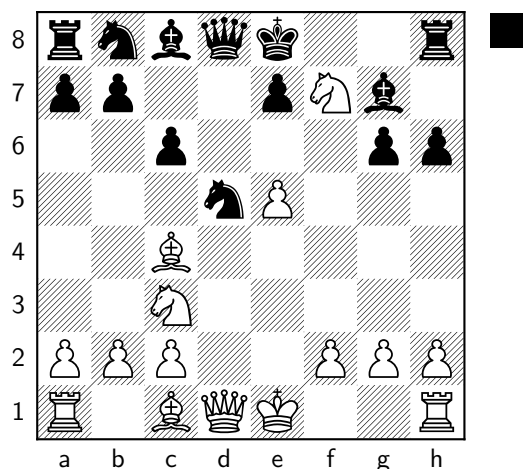
- **Instead of [move]:** The move(s) that should have been played instead.
- **Best continuation:** The optimal sequence of moves from the diagrammed position.

Positions marked with ★ represent the biggest evaluation swings in the game. The positional metrics table summarizes the spatial control, piece activity, king safety, and tactical threats for each side.

Position 1: After 9. Nxf7

White blunders (-185cp)

Evaluation: -1.71



Metric	White	Black
Space	0.40	0.50
Mobility	2.30	2.00
King Safety	0.10	-0.10
Threats	10.60	6.80

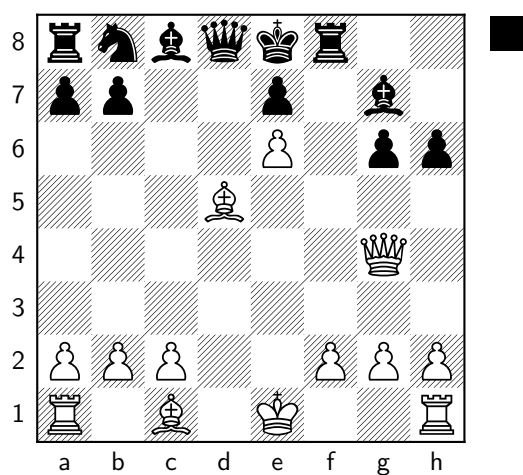
Instead of Nxf7: Nge4, Nf3

Best continuation: Kxf7 Nxd5 cxd5

Position 2: After 13. Qg4 ★

Biggest evaluation swing(s) (64cp)

Evaluation: -2.64



Metric	White	Black
Space	0.40	0.40
Mobility	2.35	1.80
King Safety	-0.10	-0.50
Threats	6.40	8.80

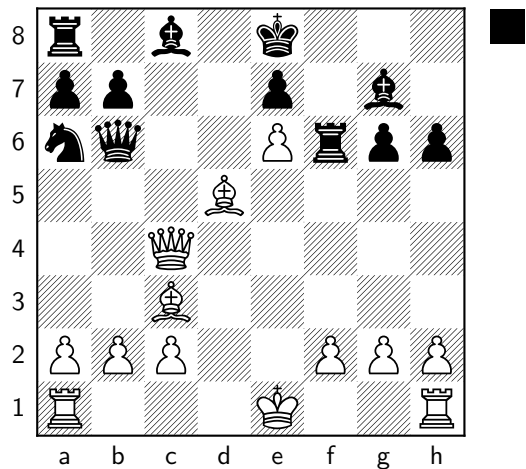
Instead of Qg4: c4, O-O

Best continuation: Rf6 Qa4+ Kf8

Position 3: After 17. Bc3

White blunders (-204cp)

Evaluation: -5.21



Metric	White	Black
Space	0.40	0.40
Mobility	2.60	2.10
King Safety	-0.10	-0.90
Threats	6.40	3.40

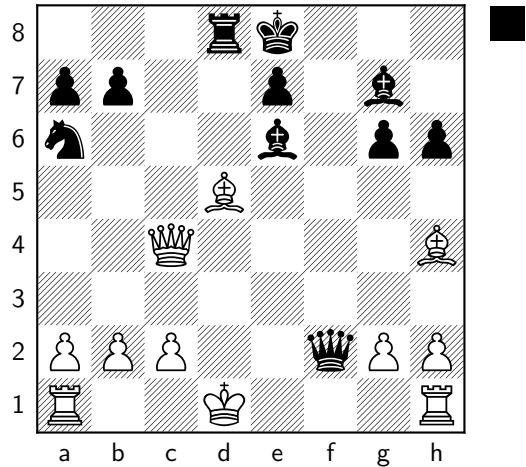
Instead of Bc3: O-O, O-O-O

Best continuation: Qxf2+ Kd1 Bxe6

Position 4: After 20. Bh4 ★

Biggest evaluation swing(s) (87cp)

Evaluation: -7.43



Metric	White	Black
Space	0.30	0.40
Mobility	2.25	2.55
King Safety	-1.30	-0.70
Threats	29.00	11.20

Instead of Bh4: Bd4, Qb5+

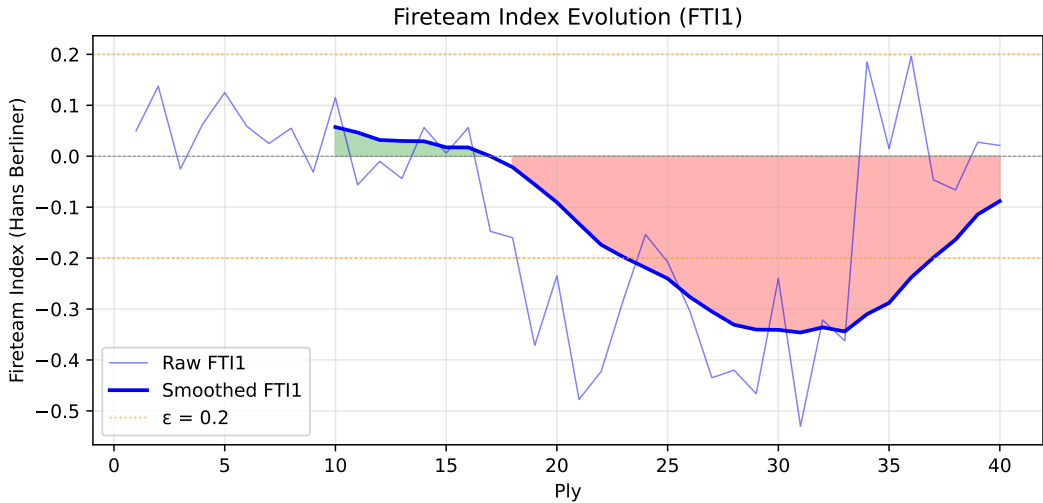
Best continuation: Qe3 Qb5+ Rd7

15.6 Fireteam Index Prediction

Tracking: **Hans Berliner** (Black)

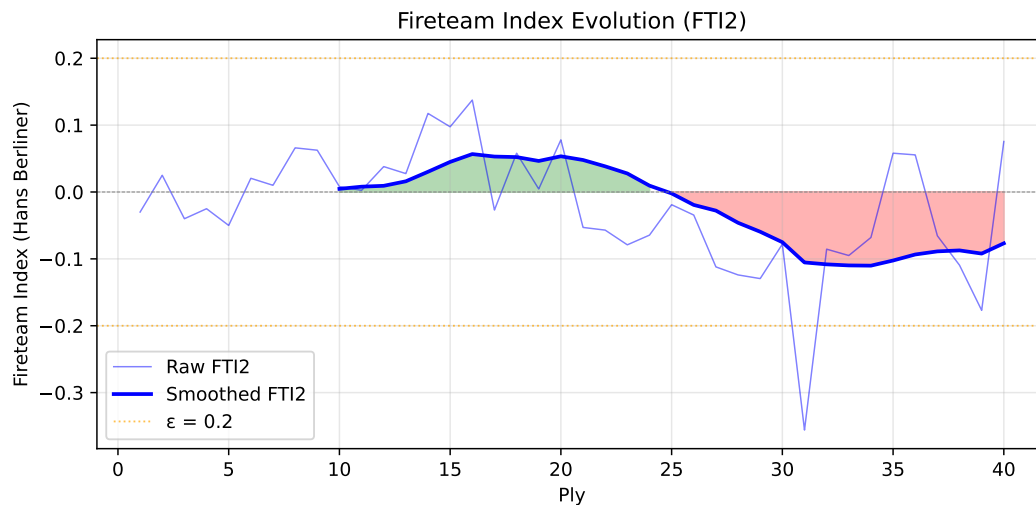
FTI1: Harmonious ($\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}$)

Algorithm	Prediction	Max Streak / Confidence
Per-ply	DRAW	0 plies
Windowed	DRAW	0 plies
Balance	LOSS	$B_n=-0.58$, $C=0.58$

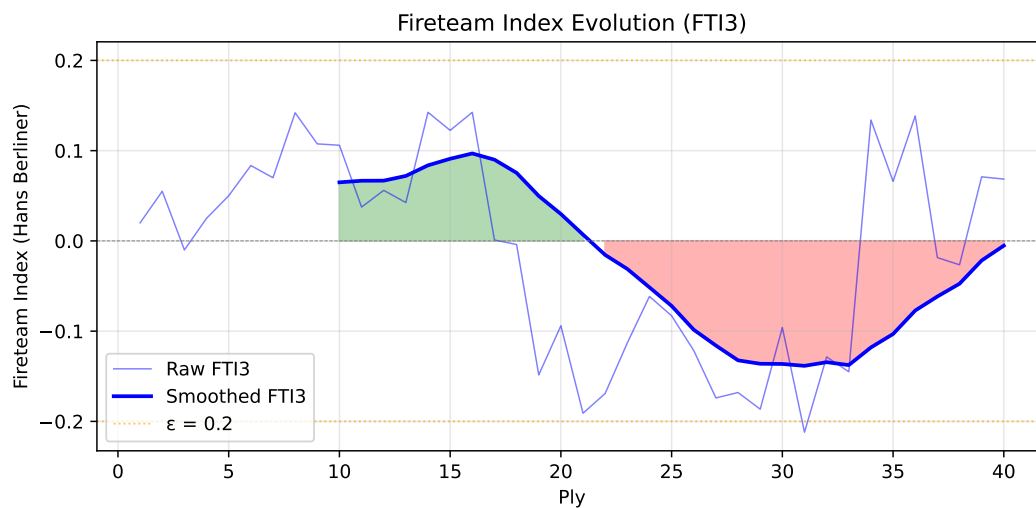


FTI2: Tactical (0.6, 0.1, 0, 0.3)

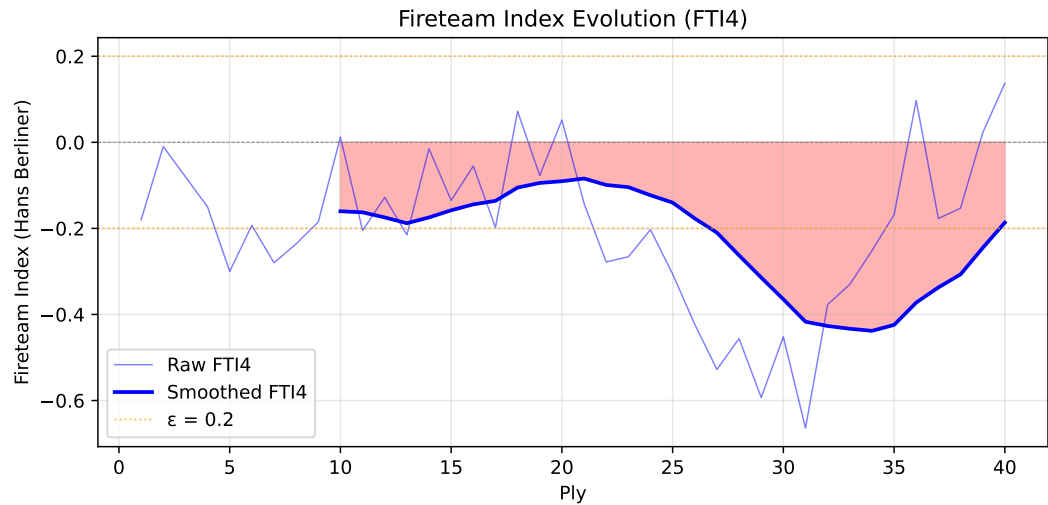
Algorithm	Prediction	Max Streak / Confidence
Per-ply	DRAW	0 plies
Windowed	DRAW	0 plies
Balance	DRAW	$B_n=-0.04$, $C=0.04$


FTI3: Strategic (0.7, 0.1, 0.1, 0.1)

Algorithm	Prediction	Max Streak / Confidence
Per-ply	DRAW	0 plies
Windowed	DRAW	0 plies
Balance	DRAW	$B_n = -0.04$, $C = 0.04$

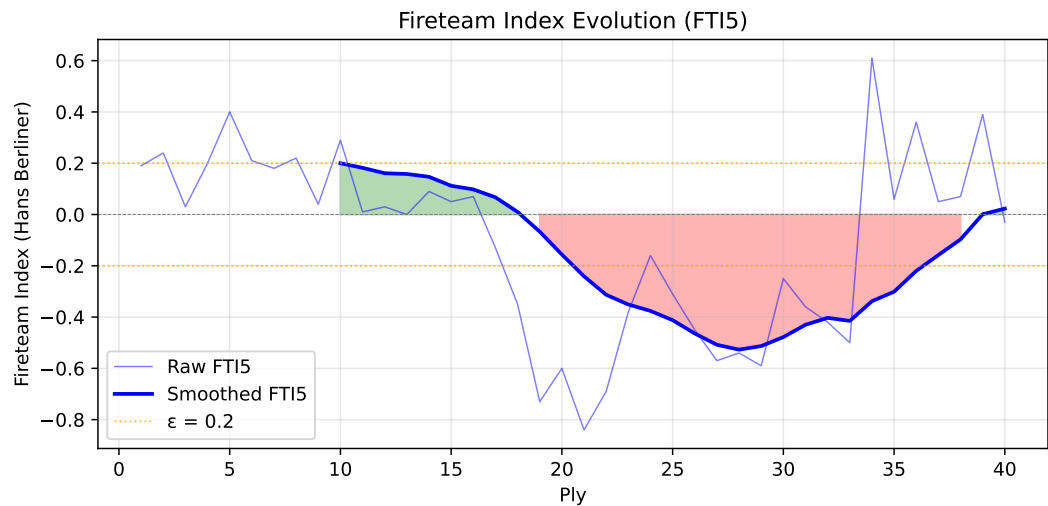

FTI4: Dynamic (0.2, 0.6, 0, 0.2)

Algorithm	Prediction	Max Streak / Confidence
Per-ply	DRAW	0 plies
Windowed	DRAW	0 plies
Balance	LOSS	$B_n = -0.54$, $C = 0.54$



FTI5: Solid (0.3,0.2,0.5,0)

Algorithm	Prediction	Max Streak / Confidence
Per-ply	DRAW	1 plies
Windowed	DRAW	0 plies
Balance	LOSS	$B_n=-0.50$, $C=0.40$



Actual result: 0-1 (Hans Berliner's outcome: WIN)

- FTI1: Per-ply ✗ Windowed ✗ Balance ✗
- FTI2: Per-ply ✗ Windowed ✗ Balance ✗
- FTI3: Per-ply ✗ Windowed ✗ Balance ✗
- FTI4: Per-ply ✗ Windowed ✗ Balance ✗
- FTI5: Per-ply ✗ Windowed ✗ Balance ✗

15.7 Analysis Information

- Engine: Stockfish 17.1
- Depth: 22
- Analysis time: 260.9 seconds

Chapter 16

Hans Berliner vs Vladimir Zagorovsky

16.1 Game Information

Event	5th Correspondence World Championship
Site	ICCF
Date	1965.07.01
Round	3
White	Hans Berliner
Black	Vladimir Zagorovsky
Result	1/2-1/2
Opening	D35 – Unknown

16.2 Player Statistics

16.2.1 Hans Berliner (White)

- Total moves: 38
- Accuracy: 94.5%
- Average centipawn loss: 11.3
- Best/Excellent moves: 18 / 6
- Good moves: 9
- Inaccuracies: 4
- Mistakes: 1
- Blunders: 0

16.2.2 Vladimir Zagorovsky (Black)

- Total moves: 38
- Accuracy: 94.4%
- Average centipawn loss: 11.6

- Best/Excellent moves: 18 / 8
- Good moves: 8
- Inaccuracies: 3
- Mistakes: 0
- Blunders: 1

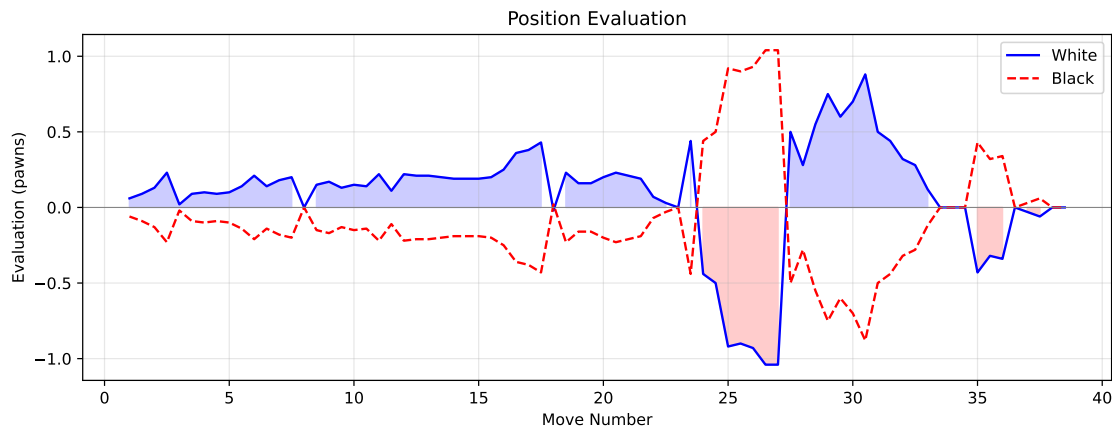
16.3 Positional Analysis

This section shows how key positional factors evolved throughout the game.

16.3.1 Position Evaluation Over Time

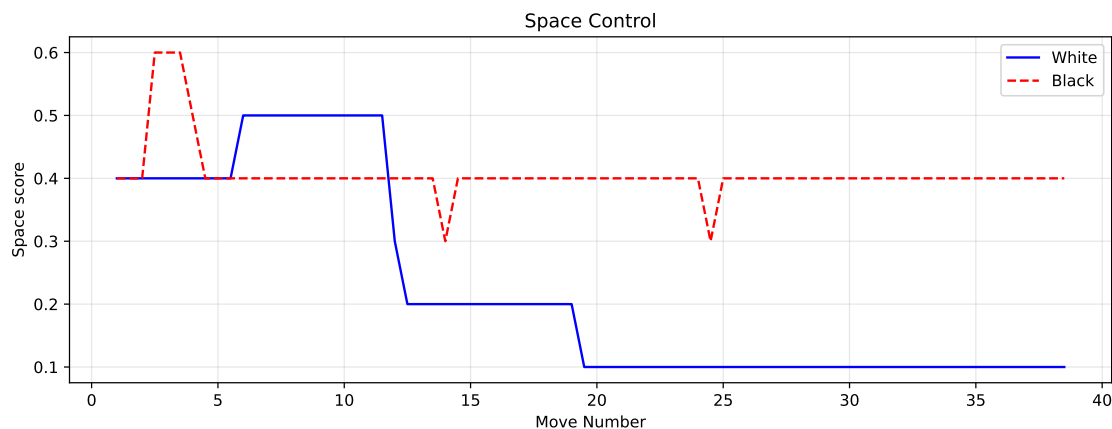
Game Character Classification. Evaluation range: minimum $m_1 = -1.04$, maximum $m_2 = +0.88$, spread $d = m_2 - m_1 = 1.92$.

This game is classified as **tense** (normal). A tense game with moderate swings.



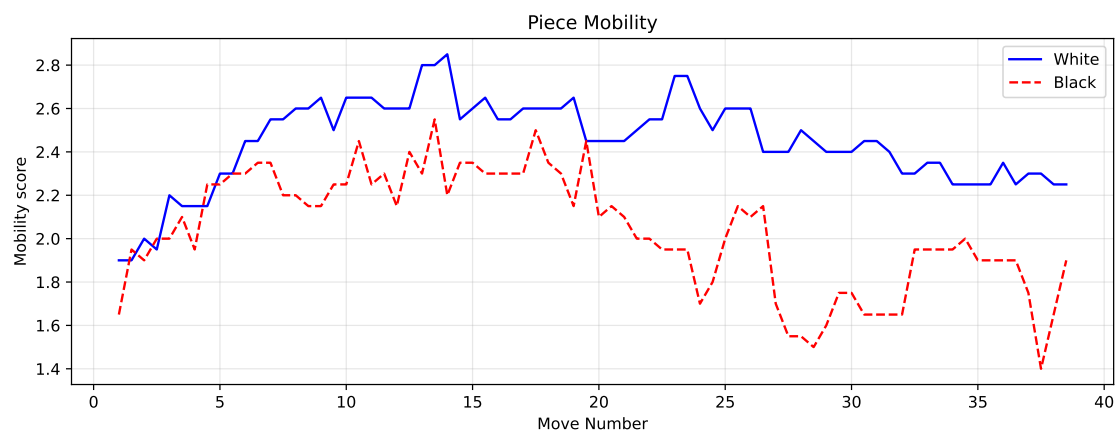
16.3.2 Space Control

- White average: 0.22 (range: 0.10 to 0.50)
- Black average: 0.41 (range: 0.30 to 0.60)



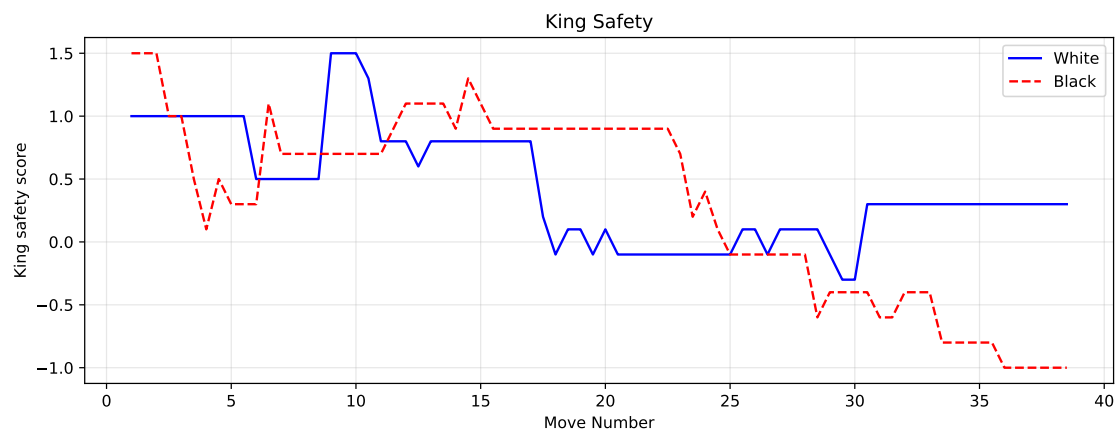
16.3.3 Mobility (Piece Activity)

- White average: 2.45 (range: 1.90 to 2.85)
- Black average: 2.04 (range: 1.40 to 2.55)



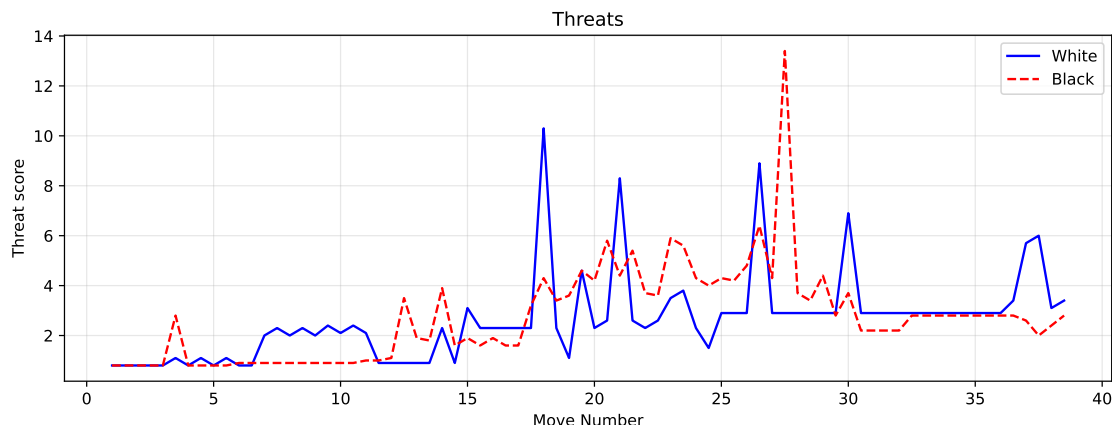
16.3.4 King Safety

- White average: 0.44
- Black average: 0.32



16.3.5 Threats

- White average: 2.64 (range: 0.80 to 10.30)
- Black average: 2.74 (range: 0.80 to 13.40)
- Average advantage: -0.10 (Black maintained more threats)



16.4 Annotated Game

1. d4 Nf6 2. c4 e6 3. Nc3 d5 4. cxd5 exd5 5. Bg5 Be7 6. e3 O-O 7. Bd3! Nbd7 8. Nge2 Re8 9. O-O! c6! 10. Qc2! Nf8! 11. f3! Ng6! 12. e4 dxe4! 13. fxe4! Be6! 14. Bxf6! Bxf6 15. e5! Be7! 16. Rad1 Bf8 17. Kh1! Qh4 18. g3?! Qh6?! 19. Nf4 Nxf4! 20. gxf4! Bg4! 21. Be2! Bh3 22. Rf3! Rad8! 23. Rg3 Kh8?! 24. f5? *White loses 86cp. Consider instead: Bc4, Bg4.*

24...g6! 25. f6?! Be6 26. Bg4 Bxg4! 27. Rxc4! Qh5?? *Black loses 168cp. Consider instead: c5.*

28. Qe2 h6 29. Re4! Qf5! 30. Rf1! Qe6 31. Qg2?! h5! 32. Qf3 Bh6! 33. Rd1 Kh7! 34. Reel Rd7! 35. b3?! Red8 36. Ne4 b6?! 37. Nd6! Bf8 38. Ne4! Bh6!

1/2-1/2

16.5 Critical Positions

This section highlights critical moments where the evaluation shifted substantially—either due to errors or missed opportunities. Each diagram shows the position *after* the move was played.

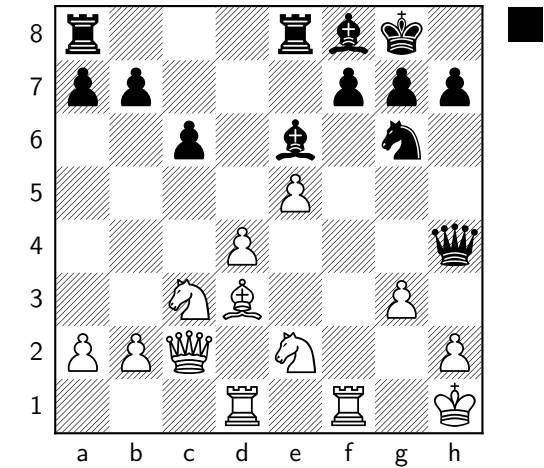
- **Instead of [move]:** The move(s) that should have been played instead.
- **Best continuation:** The optimal sequence of moves from the diagrammed position.

Positions marked with ★ represent the biggest evaluation swings in the game. The positional metrics table summarizes the spatial control, piece activity, king safety, and tactical threats for each side.

Position 1: After 18. g3 ★

Biggest evaluation swing(s) (45cp)

Evaluation: -0.02



Metric	White	Black
Space	0.20	0.40
Mobility	2.60	2.35
King Safety	-0.10	0.90
Threats	10.30	4.30

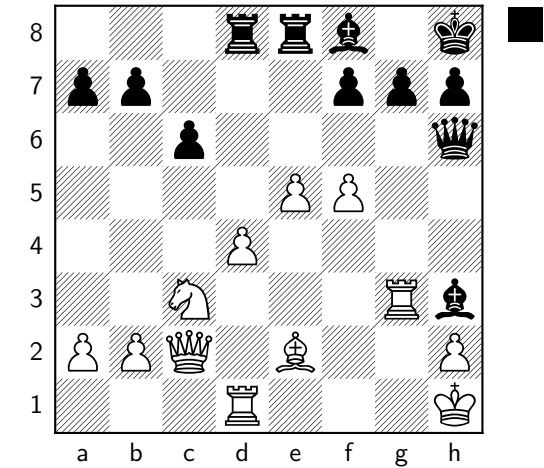
Instead of g3: Bf5, a3, Ne4

Best continuation: Qh5 Bf5 Rad8

Position 2: After 24. f5 ★

Biggest evaluation swing(s) (88cp)

Evaluation: -0.44



Metric	White	Black
Space	0.10	0.40
Mobility	2.60	1.70
King Safety	-0.10	0.40
Threats	2.30	4.30

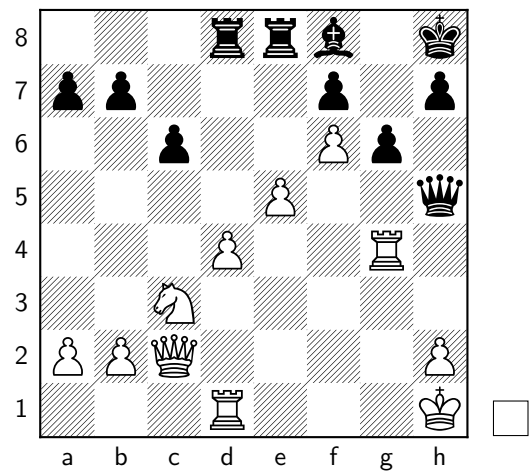
Instead of f5: Bc4, Bg4

Best continuation: g6 Bg4 Bxg4

Position 3: After 27...Qh5

Black blunders (+154cp)

Evaluation: +0.50



Metric	White	Black
Space	0.10	0.40
Mobility	2.40	1.55
King Safety	0.10	-0.10
Threats	2.90	13.40

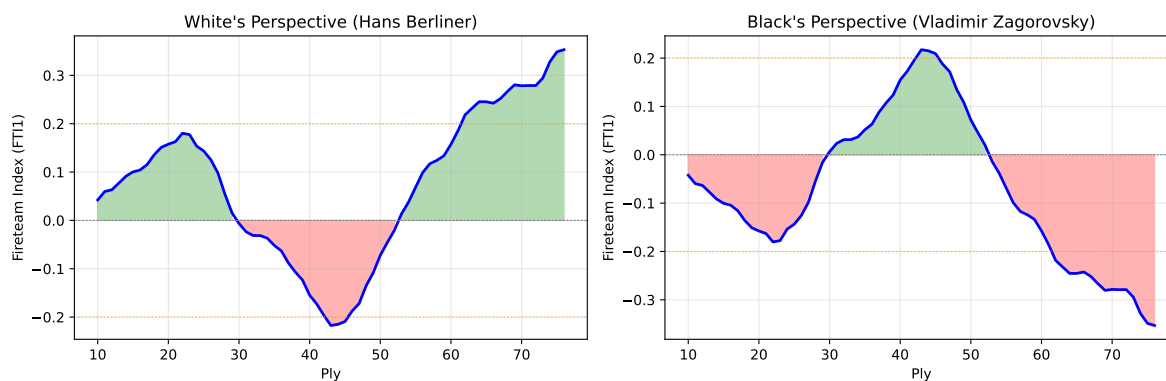
Instead of Qh5: c5 Best continuation: Re4 b5 Qg2

16.6 Fireteam Index Prediction

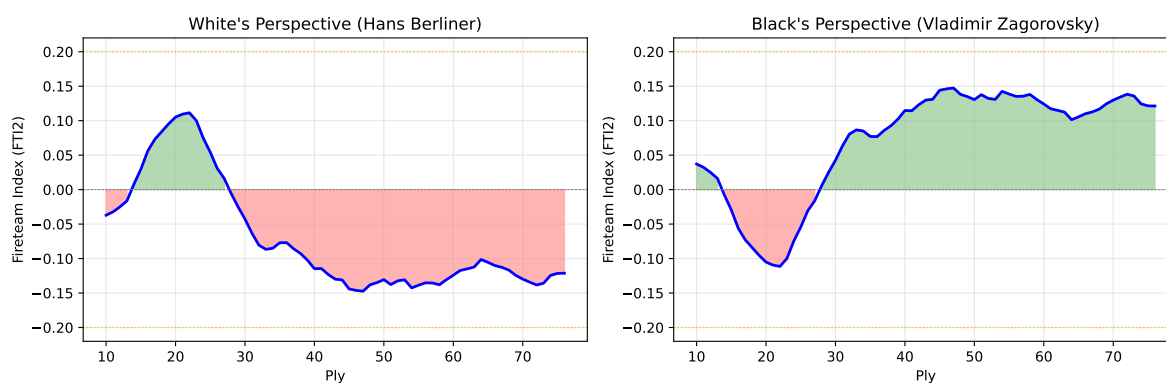
This game ended in a draw. The Fireteam Index is analyzed from both players' perspectives to determine whether either side achieved sustained positional dominance that might have predicted a win.

FTI1: Harmonious ($\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}$)

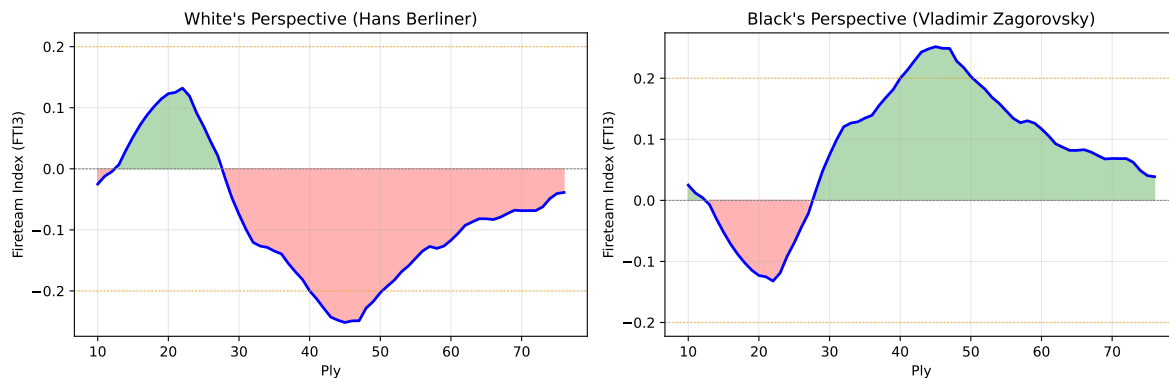
Player	Algorithm	Prediction	Max Streak
Hans Berliner	Per-ply	WIN	12 plies
Hans Berliner	Windowed	WIN	15 plies
Vladimir Zagorovsky	Per-ply	DRAW	3 plies
Vladimir Zagorovsky	Windowed	DRAW	3 plies

**FTI2: Tactical** (0.6, 0.1, 0, 0.3)

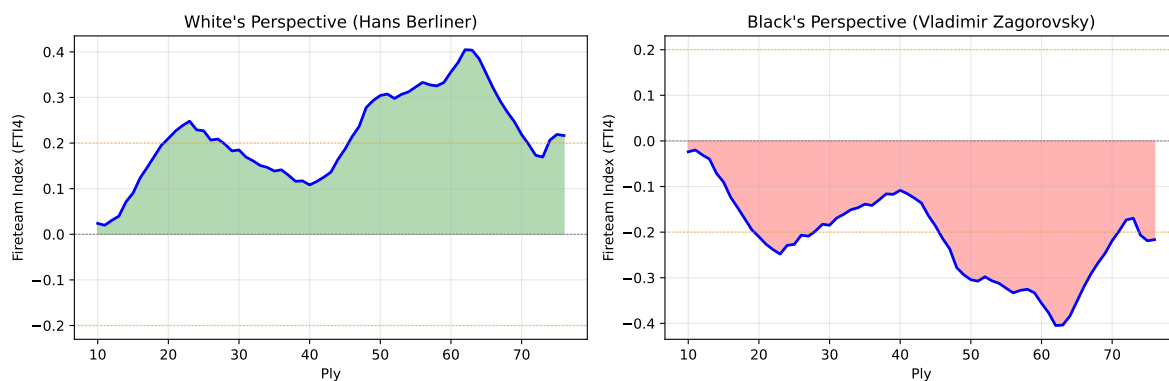
Player	Algorithm	Prediction	Max Streak
Hans Berliner	Per-ply	DRAW	0 plies
Hans Berliner	Windowed	DRAW	0 plies
Vladimir Zagorovsky	Per-ply	DRAW	1 plies
Vladimir Zagorovsky	Windowed	DRAW	0 plies

**FTI3: Strategic** (0.7, 0.1, 0.1, 0.1)

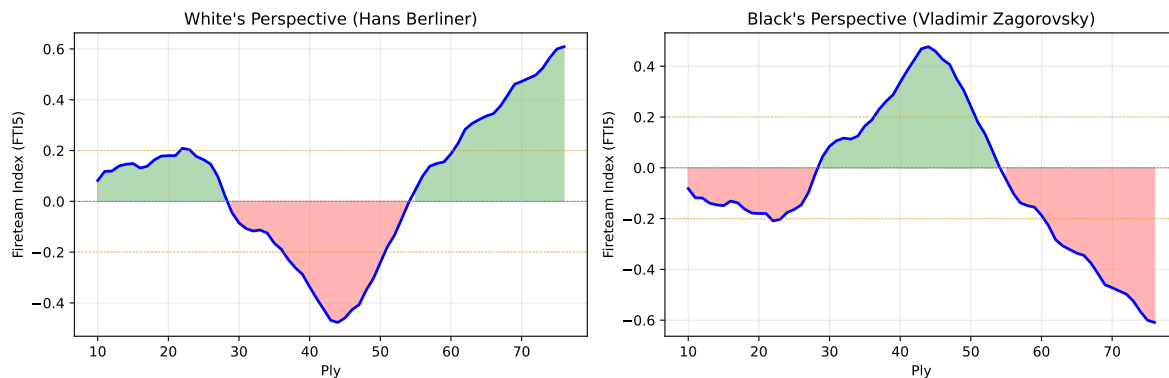
Player	Algorithm	Prediction	Max Streak
Hans Berliner	Per-ply	DRAW	1 plies
Hans Berliner	Windowed	DRAW	0 plies
Vladimir Zagorovsky	Per-ply	DRAW	8 plies
Vladimir Zagorovsky	Windowed	WIN	10 plies

**FTI4: Dynamic (0.2, 0.6, 0, 0.2)**

Player	Algorithm	Prediction	Max Streak
Hans Berliner	Per-ply	WIN	11 plies
Hans Berliner	Windowed	WIN	25 plies
Vladimir Zagorovsky	Per-ply	DRAW	0 plies
Vladimir Zagorovsky	Windowed	DRAW	0 plies

**FTI5: Solid (0.3, 0.2, 0.5, 0)**

Player	Algorithm	Prediction	Max Streak
Hans Berliner	Per-ply	WIN	17 plies
Hans Berliner	Windowed	WIN	16 plies
Vladimir Zagorovsky	Per-ply	WIN	12 plies
Vladimir Zagorovsky	Windowed	WIN	14 plies



Interpretation. White (Hans Berliner) achieved sustained positional dominance, with FTI1 predicting a win. However, the game ended in a draw, suggesting Black successfully defended or White failed to convert the advantage.

16.7 Analysis Information

- Engine: Stockfish 17.1
- Depth: 22
- Analysis time: 526.2 seconds

Chapter 17

Fireteam Index Prediction Summary

This chapter summarizes the accuracy of the Fireteam Index prediction algorithms across all games in this collection.

17.1 Overall Results

Metric	Value
Total games	16
Decisive games	12
Draws (both sides analyzed)	4
Games with prediction	16

17.1.1 FTI1: Harmonious ($\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}$)

Algorithm	Correct	Accuracy
Per-ply (raw)	5/16	31.2%
Windowed (smoothed)	6/16	37.5%
Balance	5/16	31.2%

17.1.2 FTI2: Tactical (0.6, 0.1, 0, 0.3)

Algorithm	Correct	Accuracy
Per-ply (raw)	3/16	18.8%
Windowed (smoothed)	3/16	18.8%
Balance	3/16	18.8%

17.1.3 FTI3: Strategic (0.7, 0.1, 0.1, 0.1)

Algorithm	Correct	Accuracy
Per-ply (raw)	4/16	25.0%
Windowed (smoothed)	4/16	25.0%
Balance	5/16	31.2%

17.1.4 FTI4: Dynamic (0.2, 0.6, 0, 0.2)

Algorithm	Correct	Accuracy
Per-ply (raw)	4/16	25.0%
Windowed (smoothed)	6/16	37.5%
Balance	8/16	50.0%

17.1.5 FTI5: Solid (0.3, 0.2, 0.5, 0)

Algorithm	Correct	Accuracy
Per-ply (raw)	4/16	25.0%
Windowed (smoothed)	7/16	43.8%
Balance	3/16	18.8%

17.1.6 Comparison

FTI4 (Dynamic) shows the best predictive accuracy at 50.0%.

17.1.7 Empirically Optimal Weights

Using linear regression on the per-ply positional data from all games, the following weights best predict game outcomes (win=1, draw=0.5, loss=0):

$$\text{FTI}_{\text{opt}} = 0.32\Delta S + 0.68\Delta M + 0.00\Delta K + 0.00\Delta T' \quad (17.1)$$

Weights: $w_S = 0.319$, $w_M = 0.675$, $w_K = 0.003$, $w_T = 0.003$ ($R^2 = -12.525$)

Prediction Accuracy with Optimal Weights

Algorithm	Correct	Accuracy
Per-ply	7/16	43.8%
Windowed	9/16	56.2%
Balance	4/16	25.0%

For comparison, the predefined signatures are:

- FTI1 (Harmonious): (0.25, 0.25, 0.25, 0.25)
- FTI2 (Tactical): (0.60, 0.10, 0.00, 0.30)
- FTI3 (Strategic): (0.70, 0.10, 0.10, 0.10)
- FTI4 (Dynamic): (0.20, 0.60, 0.00, 0.20)
- FTI5 (Solid): (0.30, 0.20, 0.50, 0.00)

17.2 Interpretation

These predictions tracked the eventual winner in each decisive game. Three prediction algorithms are compared:

- **Per-ply:** Predicts WIN if $FTI > \epsilon$ for 10 consecutive plies (raw values)
- **Windowed:** Same logic but applied to a 10-ply rolling average
- **Balance:** Uses normalized balance B_n (plies ahead – plies behind) and volatility V (sign changes). Predicts DRAW if $|B_n| < 0.2$ or $V \geq 10$

The Balance algorithm explicitly models draws by detecting games where neither side maintains a consistent advantage, or where the advantage changes hands frequently.

The different FTI signatures may perform better or worse depending on the playing style of the players being analyzed. FTI2 (Tactical) tends to predict well for players who emphasize territorial control and tactical threats, while FTI3 (Strategic) may better capture the style of players who prioritize space and piece activity.

Appendix A

Methodology: How Metrics Are Computed

This appendix explains how the positional metrics are computed.

This section explains how Stockfish’s classical evaluation computes each positional factor. Note that modern Stockfish primarily uses NNUE (neural network) evaluation, but the classical evaluation terms provide interpretable metrics for educational purposes.

A.1 Space

Space evaluation counts squares in the center (files c-f, ranks 2-4 for White, 5-7 for Black) that are:

1. Attacked by at least one friendly pawn, AND
2. Not occupied by any enemy piece

The computation is weighted by the number of pieces behind the pawn chain—this rewards positions where you control territory AND have pieces to exploit that space. The space term is scaled by total piece count; it matters more in closed positions with many pieces.

A.2 Mobility

Mobility counts legal moves for each piece type, with different weights:

- **Knights:** Safe squares attacked (excluding squares defended by enemy pawns)
- **Bishops:** Safe squares on diagonals (bonus for long, unobstructed diagonals)
- **Rooks:** Squares on files/ranks (bonus for open files, 7th rank penetration)
- **Queens:** Combination of bishop and rook mobility patterns

“Safe squares” exclude those defended by enemy pawns or where exchanges would lose material. Mobility in the opponent’s territory is valued higher than mobility in your own.

A.3 King Safety

King safety combines several factors:

- **Pawn shield:** Pawns on 2nd/3rd rank directly in front of the castled king
- **King tropism:** Distance of enemy pieces to your king (closer = more dangerous)
- **Attack units:** Count of enemy pieces attacking squares adjacent to your king
- **Safe checks:** Number of safe checking squares available to the opponent

A.4 Threats

Threat evaluation considers:

- Hanging pieces (attacked but not adequately defended)
- Safe pawn advances that create threats
- Minor pieces attacking major pieces
- Weak squares in the pawn structure

A.5 Middlegame vs Endgame (MG/EG)

Each term has separate middlegame (MG) and endgame (EG) values. Stockfish blends these based on remaining material (the “phase”). With all pieces on the board, MG dominates; as pieces trade, EG values become more important. For example, space matters greatly in the middlegame but is nearly irrelevant in endgames.

A.6 Game Character Classification

Games are classified based on evaluation volatility and directionality. Let m_2 be the maximum evaluation for White during the game (in pawns) and m_1 be the minimum. The *spread* $d = m_2 - m_1$ measures the total evaluation range.

Spread Classifications. Based on the magnitude of d :

- **Balanced** ($d < 1$): Minimal advantage shifts; controlled, technical play with neither side gaining a significant edge.
- **Tense** ($1 \leq d < 3$): Normal competitive tension; typical of well-played games where small advantages are contested.
- **Tactical** ($3 \leq d < 6$): Significant evaluation swings; tactical complications, missed opportunities, or speculative play.
- **Chaotic** ($d \geq 6$): Wild swings in evaluation; likely includes major blunders, speculative sacrifices, or complex tactical melees.

Directionality Classifications. Based on whether the advantage changed hands:

- **One-sided:** Either $m_1 > -\frac{1}{2}$ (White always comfortable) or $m_2 < \frac{1}{2}$ (Black always comfortable). The leading side maintained control throughout.
- **Seesaw:** Both $m_1 < -1$ and $m_2 > 1$. The advantage genuinely changed hands, with each side holding a significant advantage at some point.
- **Normal:** Neither one-sided nor seesaw; moderate swings without complete reversals.

These classifications combine to describe the overall game character. For example, a “tactical seesaw battle” indicates significant evaluation swings ($3 \leq d < 6$) with the advantage changing hands multiple times.

A.7 Fireteam Index (FTI) and Win Prediction

The Fireteam Index is a composite metric inspired by the ancient mathematical board game *Rithmo-machia*, where a “progression victory” is achieved when a coordinated group of pieces (a “fireteam”) establishes control deep in enemy territory. Similarly, in chess, sustained positional dominance across multiple factors often precedes victory.

The FTI weights can be thought of as *signatures* that capture different playing styles. By adjusting which positional factors receive emphasis, the FTI can be tuned to better predict outcomes for players with different strategic preferences.

Threats Scaling. The raw Threats metric (T) produces values typically ranging from 5–20, while Space, Mobility, and King Safety range from approximately 0.1–0.5. To make these comparable, Threats is divided by a scaling factor of 20 before weighting. The scaled value $T' = T/20$ brings Threats into a similar range as the other metrics.

FTI1: Harmonious Formula. The harmonious Fireteam Index weights all four positional metrics equally (with Threats pre-scaled):

$$\text{FTI}_1 = \frac{1}{4}\Delta S + \frac{1}{4}\Delta M + \frac{1}{4}\Delta K + \frac{1}{4}\Delta T' \quad (\text{A.1})$$

where ΔS = Space, ΔM = Mobility, ΔK = King Safety, and $\Delta T' = T/20$ = scaled Threats, each computed as (Player’s value – Opponent’s value). Positive FTI indicates the player has overall positional superiority; negative indicates the opponent is better.

FTI2: Tactical. This variant emphasizes territorial control and tactical pressure through Space and Threats:

$$\text{FTI}_2 = 0.6\Delta S + 0.1\Delta M + 0.3\Delta T' \quad (\text{A.2})$$

FTI3: Strategic. This variant emphasizes long-term spatial dominance:

$$\text{FTI}_3 = 0.7\Delta S + 0.1\Delta M + 0.1\Delta K + 0.1\Delta T' \quad (\text{A.3})$$

FTI4: Dynamic. This variant emphasizes piece activity and maneuvering:

$$\text{FTI}_4 = 0.2\Delta S + 0.6\Delta M + 0.2\Delta T' \quad (\text{A.4})$$

FTI5: Solid. This variant emphasizes king safety and defensive play:

$$\text{FTI}_5 = 0.3\Delta S + 0.2\Delta M + 0.5\Delta K \quad (\text{A.5})$$

Prediction Algorithms. Three algorithms are used to predict game outcomes based on FTI (applied independently to each of the five variants):

1. **Per-Ply (Raw):** Examines the raw FTI value at each ply after the opening phase (ply 16, approximately 8 moves). If $\text{FTI} > \epsilon$ (default $\epsilon = 0.25$) for 10 consecutive plies (5 full moves), a WIN is predicted for that player.
2. **Windowed (Smoothed):** Computes a 10-ply rolling average of FTI before applying the same streak detection. This smooths out tactical noise and captures sustained trends rather than momentary fluctuations.
3. **Balance (Confidence-based):** Predicts outcomes using a confidence score that combines balance and volatility:

- *Normalized Balance* $B_n = \frac{(\text{plies with FTI} > \epsilon) - (\text{plies with FTI} < -\epsilon)}{\text{total plies}}$
- *Volatility* $V = \text{number of times FTI crosses zero (sign changes)}$
- *Volatility threshold* $V_{\max} = 0.2 \times \text{total plies}$ (scales with game length)
- *Confidence* $C = |B_n| \times \max(0, 1 - V/V_{\max})$

The prediction is: WIN if $C > 0.15$ and $B_n > 0$; LOSS if $C > 0.15$ and $B_n < 0$; DRAW otherwise. Low confidence (due to balanced play or high volatility) predicts a draw.

The Per-Ply and Windowed algorithms predict DRAW only when no streak is achieved, whereas the Balance algorithm explicitly detects drawish characteristics (balanced plies, back-and-forth play).

Interpretation. The FTI prediction measures *positional* dominance, which is a necessary but not sufficient condition for winning. A player may dominate positionally yet fail to convert due to:

- Tactical errors (blunders, missed combinations)
- Time pressure in rapid/blitz games
- Tenacious defense by the opponent
- Drawing tendencies in certain structures (opposite-colored bishops, fortresses)

Conversely, a player may win without ever achieving sustained FTI dominance through a single tactical strike or opponent error. The FTI metric is most predictive in games where positional understanding—rather than pure calculation—determines the outcome.

Choosing a Signature. Different FTI signatures may perform better for different players or playing styles. FTI2 (Tactical) tends to predict well for players who build up territorial control and convert through direct threats. FTI3 (Strategic) may better capture the style of players who emphasize piece activity and spatial domination. Comparing prediction accuracy across signatures can reveal which positional factors most strongly correlate with a player’s wins.

For Draws. When analyzing drawn games, the FTI is computed from both players’ perspectives. This reveals whether:

- Neither side achieved dominance (a “true draw” with balanced play)
- One side dominated but failed to convert (a “missed win”)
- Both sides dominated at different points (a “mutual chances draw”)